Strategic Deindustrialization and Industrial Policy Games*

David Andolfatto
University of Miami

Roozbeh Hosseini
University of Georgia

Preliminary draft, October 11, 2026

Abstract

This paper asks whether a country can use trade to deindustrialize a geopolitical rival. In our model, imports reduce domestic employment and learning by doing in the industrial goods sector. If imports remain high long enough, industrial knowledge can fall below a critical threshold, leaving the domestic economy in a low-productivity state even after imports cease. We show that a geopolitical rival motivated by commercial or security concerns may optimally induce this outcome. We then study domestic policy responses. A sufficiently high tariff deters the exporter, while an employment subsidy can preserve domestic industrial knowledge without restricting trade. We characterize the conditions under which the threat of a subsidy alone deters deindustrialization.

Keywords: learning by doing; multiple steady states; hysteresis; deindustrialization; trade and national security; industrial policy; tariffs; sectoral export surplus.

JEL codes: F12, F13, F52, L52, O14.

*The authors thank participants at the "Recent Advances in Money & Macro" workshop in Paris and the 2026 SED meetings in Athens for helpful comments and discussion.

1 Introduction

This paper examines the extent to which trade policy can be used to reduce a geopolitical rival's industrial capacity. Specifically, we ask whether a surge of imported manufactured goods can cause a lasting loss of domestic productive capacity, whether an exporter would choose to cause that loss, and the ways in which an importer can protect itself.

These questions matter when activity in production sustains knowledge through, for example, learning by doing. While an importing country gains access to cheaper goods, the resulting decline in domestic employment can reduce its future productivity. In sectors such as semiconductors and shipbuilding, industrial capacity also supports military production. A geopolitical rival may therefore value the loss of that capacity for security reasons. Recent policy reflects both concerns: the U.S. CHIPS and Science Act, the European Chips Act, and the export-control measures of 2024–26 treat the capacity to manufacture advanced goods as a matter of national security rather than ordinary commerce.

We develop a dynamic model of two countries, East and West, in which the East exports manufactured goods to the West in exchange for consumption goods, either currently or through claims on future consumption.[1] Productivity in each country depends on industrial knowledge shared among its domestic producers. Sectoral knowledge accumulates through sectoral learning by doing; that is, individuals do not internalize the contribution of their work to future domestic productivity. The West's government is initially passive: it does not correct the learning externality or act on national security concerns. The East's government chooses exports, taking into account both their commercial return and the security benefit of reducing Western industrial capability. The weight it places on that benefit, its *security weight*, is the model's central parameter.

Our first result is that a temporary import surge can cause a permanent decline in domestic industrial capacity (the reverse is true for the exporting nation). Productivity is S-shaped in knowledge. When the feedback between productivity and employment is sufficiently strong, the economy has two stable steady states separated by an unstable knowledge threshold. Imports displace domestic production and reduce employment. Lower employment slows learning, and knowledge falls. A sufficiently large and persistent surge pushes the West below the threshold. Once there, it converges to the low-industrialization steady state even after imports return to their original level.

Our second result characterizes the exporter's incentive to cause this decline. When commercial incentives alone are not strong enough to justify deindustrializing the West, a sufficiently strong security concern can optimally induce the East to do so. The export path has two phases. The East first ships enough manufactured goods to push Western knowledge down to just below the critical threshold. Once the West is trapped, the East cuts exports back to the commercial level appropriate for a low-productivity customer. The West's own employment and knowledge dynamics then sustain the decline.

A concern for security is not necessary for this outcome. East and West have the same

[1] While we model trade as being balanced in an intertemporal sense, this is by no means necessary. The predation and deterrence results below hold qualitatively if imports appear in the form of seigniorage or even as outright gifts (Appendix D).

preferences and technology, but play different roles in the trading game. The West is competitive, while the East is a large exporter that recognizes how its exports affect the price of manufactured goods. By reducing Western productive capacity, the East can raise the price the West pays for future imports. This creates a purely commercial motive for deindustrialization. The strategy requires a temporary sacrifice, since the East must first ship enough manufactured goods to reduce Western employment and knowledge. When future gains matter enough, that sacrifice is worthwhile; otherwise, absent a security motive, the equilibrium is autarky.

Our third result concerns the West's response. A sufficiently high tariff deters deindustrialization. Below the deterrent level, the East absorbs the tariff and continues to export enough to push Western knowledge across the threshold; above it, the East abandons the attempt. The deterrent tariff rises with the East's security motive. But the West need not observe that motive if it is willing to prohibit trade altogether. In the benchmark, a prohibitive tariff protects against every security motive and carries no static cost. Two industrialized countries would not trade absent a strategic motive. The informational problem matters only if the West wants to deter the East without shutting down trade.

An employment subsidy addresses the domestic learning externality directly. The policy supports employment when imports threaten to push Western knowledge below the critical threshold. We give conditions under which the West is willing to provide enough support to prevent that crossing. With lump-sum financing and a subsidy rate that can respond to the size of the East's exports, the policy deters deindustrialization from the high steady state at any security weight. The East anticipates that additional exports cannot push the West into the low-productivity state and therefore does not initiate the surge. Along the equilibrium path, the subsidy is not paid. Unlike a prohibitive tariff, the policy does not restrict imports. This distinction has no welfare value in our analytical benchmark because two industrialized countries do not trade in the absence of the strategic motive.

The analytical deterrence result relies on a sharp threshold in productivity and security losses. With smooth schedules, numerical solutions continue to exhibit deterrence for security weights well above the benchmark's critical security weight.[2] In these computations, the West pays a positive subsidy along the equilibrium path, equal to the subsidy that corrects its domestic learning externality.

The analysis makes the importer's domestic knowledge stock central to industrial policy. The West does not need to know why the East exports. It only needs to know how much domestic employment is required to preserve productive capacity and how imports affect that employment. The relevant objects are import flow, domestic employment, and the knowledge sustained by that employment. The aggregate trade balance says little about how close an industry is to losing the capacity to sustain itself.

Related literature. The paper relates to three strands of research. The first studies learning by doing, industrial development, and persistent trade effects. Bahk and Gort (1993), Benkard (2000), and Irwin and Klenow (1994) provide evidence on the accumulation, depre-

[2]At higher weights, we do not find an equilibrium; see Appendix E.

ciation, and diffusion of production knowledge. Baldwin (1969) and Dasgupta and Stiglitz (1988) study the implications of learning for infant-industry protection. Our multiplicity of steady states descends from the literature on increasing returns and development traps (Murphy et al., 1989; Matsuyama, 1991); here an international flow of manufactured goods, rather than domestic spillovers, moves the economy between them. Krugman (1987) shows how learning can make specialization persistent, while Baldwin and Krugman (1989) show how sunk costs can give temporary shocks lasting trade effects. We study a knowledge threshold below which an established industry cannot recover on its own and a foreign government's incentive to push it across that threshold.

The second strand studies strategic trade policy. Brander and Spencer (1985) analyze export subsidies that shift profits towards domestic producers. Costinot et al. (2014) study capital controls as a means of manipulating intertemporal terms of trade. Recent work studies trade policy that responds to security risk (Traiberman and Rotemberg, 2023) and how geopolitical conflict reshapes trade (Adolph et al., 2025). Our exporter can improve its future terms of trade by reducing the productive capacity of its customer. A security motive provides an additional return from the same policy. We characterize how the response of the importing country changes the incentive to pursue it.

The third strand connects international imbalances, industrial policy, and geopolitical competition. Pflueger and Yared (2024) study the relationship between financing advantages and hegemony. The closest contemporaneous paper is Cesa-Bianchi et al. (2026), which shows how industrial policies combined with weak domestic demand generate surpluses that reduce innovation and productivity abroad. Our contribution is to characterize a threshold beyond which the importer's loss persists after the surplus ends, an exporter's choice to cause that loss, and an equilibrium response that can deter it. The mechanism operates through a sectoral flow of goods and does not require a particular current-account balance.

Section 2 presents the motivating episodes. Section 3 develops the learning mechanism and the conditions for lasting deindustrialization. Section 4 studies the exporter's choices, and Section 5 analyzes the importer's policies. Section 6 provides the numerical illustration. Sections 7 and 8 discuss implications and conclude. The appendices contain proofs, extensions, and the numerical analysis of the smooth economy.

2 Motivating Facts

This section describes three episodes that motivate the analysis: the concentration of solar manufacturing in China, the concentration of commercial shipbuilding in East Asia, and the declining share of the U.S. in semiconductor fabrication. Each raises a question about the relationship between access to imported goods and the capacity to produce them domestically. We focus on the location of production, changes in industrial leadership, and efforts to expand domestic capacity.

In solar photovoltaics, manufacturing became concentrated throughout the production process. The International Energy Agency (IEA) places China's share above 80 percent in each of the five main stages: polysilicon, ingots, wafers, cells, and modules (Interna-

tional Energy Agency, 2022). China's manufacturing share was more than twice its share of global demand. During the preceding decade, manufacturing had shifted toward China from Europe, Japan, and the United States. This expansion also lowered production costs and made solar power more affordable. The same development therefore increased access to inexpensive equipment and concentrated the capacity to supply it. The IEA identifies cost competitiveness, with China the lowest-cost location for every stage, as a key challenge to diversifying supply.

Commercial shipbuilding illustrates concentration among several exporting countries. In 2023, China delivered more than half of the world's new ship tonnage. China, South Korea, and Japan together represented approximately 95 percent, leaving the rest of the world, including the United States and Europe, about a twentieth of the total (United Nations Conference on Trade and Development, 2024). The episode also shows that our model's two-country structure is an abstraction: the concentration of production can involve several competing exporters.

Semiconductors illustrate a decline in relative manufacturing capacity alongside efforts to expand domestic production. The U.S. share of global fabrication capacity fell from 37 percent in 1990 to 10 percent in 2022 (Semiconductor Industry Association and Boston Consulting Group, 2020, 2024). The 1990 figure should be read with care. The source counts only fabs that use wafers of eight inches or more, and it notes that leaving out the smaller wafers common in some regions in the 1990s significantly affects its shares for that decade (Semiconductor Industry Association and Boston Consulting Group, 2024). Mainland China accounted for 24 percent in 2022, and for a third of logic capacity at 28 nanometers and older, while China, Taiwan, South Korea, and Japan together held about three quarters. These shares describe the location of fabrication rather than the entire semiconductor industry. The U.S. decline is relative, not absolute: U.S. fabrication capacity grew by 11 percent between 2012 and 2022. The outcome of the policy response is not settled. The industry's 2024 projections have the U.S. share rising to 14 percent by 2032 with the investments supported by the CHIPS and Science Act, and falling to 8 percent without them (Semiconductor Industry Association and Boston Consulting Group, 2024). These are conditional projections.

These episodes motivate a theory of how trade changes the capacity to produce. They do not establish that imports caused a permanent loss through learning by doing or that an exporting government intended such a loss. Production shares alone cannot identify either claim. Employment sustains industrial knowledge, and a contraction in production can reduce the knowledge available to future producers. We ask when this feedback gives a temporary import surge lasting effects, when an exporter benefits from causing those effects, and when domestic policy prevents them. Section 7 returns to historical precedents, including the decline of India's cotton-textile industry.

3 A Baseline Model

This section studies one economy, the West, that receives an exogenous flow of manufactured goods. We describe its households and its learning-by-doing technology, reduce the

equilibrium to a law of motion for industrial knowledge, and characterize the steady states. When the feedback between productivity and employment is strong enough, the economy has a low and a high steady state separated by an unstable one, and a large enough flow of manufactured goods eliminates the high steady state. We then show that a temporary flow can have a permanent effect, using a resource windfall as an example, and derive the subsidy that corrects the learning externality.

Time, denoted t, is discrete, and the horizon is infinite: $t = 0, 1, 2, \ldots$. There are two countries: East and West. We ignore the East for the moment. The baseline is intended to isolate the basic economic mechanisms that carry over into the more complicated setting in which the East behaves strategically.

A representative household living in the West has preferences defined over the streams of consumption goods (c_t) and labor (n_t) given by:

$$\max_{\{c_t, n_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \left[c_t - v(n_t)\right] \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor. Consumption goods are produced with manufactured goods (k_t) according to:

$$c_t = f(k_t) \tag{2}$$

Assumption 1 (Preferences and technology)**.** *The labor-disutility function $v : \mathbb{R}_+ \to \mathbb{R}_+$ is twice continuously differentiable, satisfies $v(0) = v'(0) = 0$ and $v''(n) > 0$ for all $n \geq 0$, and has $\lim_{n\to\infty} v'(n) = \infty$. The production function $f : \mathbb{R}_+ \to \mathbb{R}_+$ is continuous on $\mathbb{R}_+$, twice continuously differentiable on $\mathbb{R}_{++}$, satisfies $f(0) = 0$, $f'(k) > 0$, and $f''(k) < 0$ for all $k > 0$, and has $\lim_{k\downarrow 0} f'(k) = \infty$ and $-kf''(k)/f'(k) < 1$ for all $k > 0$.*

Manufactured goods are produced by individual households using their labor (n_t) and an accumulation of *collective* domestic knowledge (h_t) that determines domestic labor productivity $\alpha(h_t)$. The latter technology is given by:

$$\alpha(h_t) = \alpha_L + \frac{\alpha_H - \alpha_L}{1 + \exp(-\gamma(h_t - \chi))} \tag{3}$$

where $\alpha_L > 0$ is the baseline productivity, $\alpha_H > \alpha_L$ is the productivity frontier, χ is the inflection point, and $\gamma > 0$ governs the steepness of the learning externalities.

There is also an exogenous flow of manufactured goods (m_t) that can be interpreted as arising from a natural resource or another external transfer. In the sequel, the East is explicitly modeled and the same flow is interpreted as imports of manufactured goods exchanged for a claim on the West. For now, we refer to m_t as imports of manufactured goods. The supply of manufactured goods, the sum of domestic production and imports, is given by:

$$k_t = \alpha(h_t)n_t + m_t \tag{4}$$

Finally, the body of domestic collective knowledge evolves according to

$$h_{t+1} = (1 - \delta)h_t + \lambda \mathbf{n}_t \tag{5}$$

where $\mathbf{n}_t$ denotes aggregate (or per capita) employment, $\delta \in (0,1)$ captures depreciation (organizational forgetting), and $\lambda > 0$ governs the intensity of learning by doing.

The manufactured good represents the output of any industry in which production builds knowledge: consumer goods, components, or equipment. It does not need to be durable. The supply k_t is used up within the period, and the only stock that carries over is knowledge h_t. The mechanism requires that imported and domestic units are perfect substitutes and that only domestic production adds to domestic knowledge. Autor et al. (2013) document a decline in U.S. manufacturing employment associated with import competition from China. In the model, lower employment also reduces the accumulation of industrial knowledge.

3.1 Equilibrium Conditions

An individual household maximizes utility (1) subject to (2) and (4) taking the evolution of h_t and m_t as given. Because an individual household has a negligible impact on the evolution of collective knowledge, each household's choice problem is static in nature. Substituting (2) into the objective (1), the desired labor supply of a household at each date t is characterized by:

$$v'(n_t) = \alpha(h_t)\, f'(\alpha(h_t) n_t + m_t) \tag{6}$$

In equilibrium, $n_t = \mathbf{n}_t$ for all t. With this latter condition imposed, equations (5) and (6) characterize the model's equilibrium dynamics for a given initial condition, h_0 and an exogenous sequence $\{m_t\}$.

In a steady state with $m_t = m$, employment and the stock of knowledge satisfy:

$$v'(n^*) = \alpha(h^*)\, f'(\alpha(h^*)\, n^* + m) \tag{7}$$

Setting $h_{t+1} = h_t = h^*$, $n_t = \mathbf{n}_t = n^*$ and using (5), we have:

$$h^* = (\lambda/\delta)\, n^* \tag{8}$$

Substituting (8) into (7) reduces the steady state to a single equation in n^*,

$$v'(n^*) = \alpha\big(\tfrac{\lambda}{\delta}\, n^*\big)\, f'\big(\alpha\big(\tfrac{\lambda}{\delta}\, n^*\big)\; n^* + m\big) \tag{9}$$

Multiple steady states are possible here because of the S-shaped property of $\alpha(\cdot)$. The remainder of this section makes that statement precise: we reduce the equilibrium to a scalar dynamical system, characterize its steady states and their stability, and show how a permanent change in the exogenous flow of manufactured goods m affects the steady state.

3.2 Equilibrium Dynamics

Equation (6) determines the equilibrium level of employment as a function of the levels of domestic knowledge and imports. Write the solution of (6) as the *employment function*

$$n_t = \phi(h_t; m) \tag{10}$$

Lemma 1 (Employment function)**.** *Under Assumption 1, for every* $(h, m) \in \mathbb{R}^2_+$*, equation* (6) *has a unique solution* $\phi(h; m) > 0$*. The function* ϕ *is continuously differentiable and satisfies* $\phi_h(h; m) > 0$ *and* $\phi_m(h; m) < 0$*. The supply of manufactured goods* $k(h; m) \equiv \alpha(h)\phi(h; m) + m$ *is strictly increasing in both arguments, with*

$$k_m \;=\; 1 + \alpha\phi_m \;=\; \frac{v''}{v'' - \alpha^2 f''} \in (0, 1), \qquad k_h \;=\; \alpha'\phi + \alpha\phi_h \;=\; \frac{\alpha'\left[\phi\, v'' + \alpha f'\right]}{v'' - \alpha^2 f''} > 0.$$

Proof. Fix (h, m) and let $a = \alpha(h) > 0$. The left-hand side of equation (6), $v'(n)$, is continuous and strictly increasing from zero to infinity. The right-hand side, $af'(an+m)$, is strictly positive and strictly decreasing in n, with a possibly infinite limiting value as $n \downarrow 0$. Hence, equation (6) has a unique solution $\phi(h; m) > 0$. At the solution, let $k = \alpha\phi + m$. The comparative statics are

$$\phi_h = \frac{\alpha'\left[f' + \alpha\phi f''\right]}{v'' - \alpha^2 f''} \qquad \text{and} \qquad \phi_m = \frac{\alpha f''}{v'' - \alpha^2 f''}$$

Since $f'' < 0$, it immediately follows that $\phi_m < 0$. For ϕ_h, observe that

$$f' + \alpha\phi f'' = f'\left[1 - \left(-\frac{kf''}{f'}\right)\frac{\alpha\phi}{k}\right]$$

Since $\alpha\phi/k \leq 1$ and $-kf''/f' < 1$, the bracketed term is strictly positive. Because $\alpha' > 0$, it follows that $\phi_h > 0$. For the last claim, substitute the expressions for ϕ_m and ϕ_h: $1 + \alpha\phi_m = (v'' - \alpha^2 f'' + \alpha^2 f'')/(v'' - \alpha^2 f'')$, and $\alpha'\phi + \alpha\phi_h = \alpha'[\phi(v'' - \alpha^2 f'') + \alpha f' + \alpha^2\phi f'']/(v'' - \alpha^2 f'')$. Both denominators are positive, and both numerators are positive. An inflow therefore raises the supply of manufactured goods, but by less than one for one: part of it crowds out domestic production. □

Condition (6) equates the marginal cost of employment, $v'(n_t)$, with its marginal benefit, the product $\alpha(h_t)f'(k_t)$ of productivity and the marginal product of manufactured goods. For any given level of domestic production $\alpha(h_t)n_t$, an exogenous increase in imported manufactured goods m_t lowers their marginal product and therefore lowers the marginal benefit of employment. As the return to domestic employment declines, households work less.

Substituting the employment function into the law of motion for collective knowledge gives

$$h_{t+1} = G(h_t; m) \equiv (1 - \delta)h_t + \lambda\phi(h_t; m) \tag{11}$$

A steady state is a fixed point $G(h^*; m) = h^*$. Define the steady-state residual $D(h; m) \equiv G(h; m) - h = \lambda\phi(h; m) - \delta h$. Thus, h^* is a steady state if and only if $D(h^*; m) = 0$. Moreover, $G_h(h; m) = 1 - \delta + \lambda\phi_h(h; m) > 0$ and $D_h(h; m) = G_h(h; m) - 1$. Therefore, a steady state is locally asymptotically stable if $D_h(h^*; m) < 0$ and unstable if $D_h(h^*; m) > 0$.

The multiplicity mechanism requires the equilibrium law of motion $G(\cdot; m)$ to be S-shaped relative to the 45-degree line. When the stock of knowledge is low or high, the productivity schedule $\alpha(h)$ is relatively flat, so an increase in knowledge produces only a modest response of employment. However, in the intermediate range, $\alpha(h)$ is steep, and the resulting increase

in employment reinforces the accumulation of knowledge through learning by doing. If this feedback is sufficiently strong, the slope of $G(\cdot; m)$ rises above one over an intermediate interval before falling below one again.

This feedback can produce two stable fixed points separated by an unstable threshold. However, because the employment response $\phi(h; m)$ is endogenous, the S-shape of α alone does not guarantee the required shape of G. Therefore, we impose the following restriction directly on the induced steady-state residual $D(h; m) = G(h; m) - h$.

Assumption 2 (Shape restriction)**.** *There exists $m_1 > 0$ such that D is twice continuously differentiable on $\mathbb{R}_{++} \times [0, m_1]$. For every $m \in [0, m_1]$, $D_h(\cdot; m)$ has exactly two interior zeros $h_1(m) < h_2(m)$, with*

- *$D_h(h; m) < 0$ for $h < h_1(m)$,*
- *$D_h(h; m) > 0$ for $h_1(m) < h < h_2(m)$,*
- *$D_h(h; m) < 0$ for $h > h_2(m)$.*

Moreover,

$$D_{hh}(h_1(m); m) > 0 \qquad \text{and} \qquad D_{hh}(h_2(m); m) < 0,$$

and the functions h_1 and h_2 are continuously differentiable on $[0, m_1]$.

Proposition 1 (Steady-state multiplicity)**.** *Suppose Assumption 2 holds and*

$$D(h_1(0); 0) < 0 < D(h_2(0); 0), \qquad D(h_2(m_1); m_1) < 0.$$

Then there exists a unique import threshold $\bar{m} \in (0, m_1)$ such that:

(i) For $0 \leq m < \bar{m}$, the economy has exactly three distinct steady states,

$$h_L^*(m) < h_M^*(m) < h_H^*(m).$$

(ii) At $m = \bar{m}$, the economy has exactly two distinct steady states. The middle and high steady states coalesce at $h_2(\bar{m})$ in a saddle-node bifurcation, characterized by the tangency

$$D(h_2(\bar{m}); \bar{m}) = 0 \qquad \text{and} \qquad D_h(h_2(\bar{m}); \bar{m}) = 0, \tag{12}$$

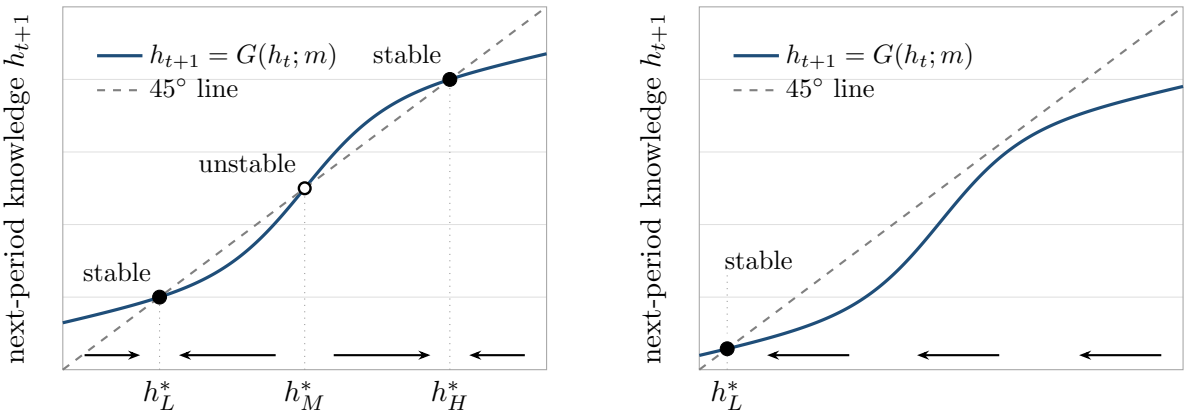
equivalently $G(h_2(\bar{m}); \bar{m}) = h_2(\bar{m})$ and $G_h(h_2(\bar{m}); \bar{m}) = 1$.

(iii) For $\bar{m} < m \leq m_1$, the economy has a unique steady state, namely the low-level steady state $h_L^*(m)$.

(iv) On the three-steady-state region $0 \leq m < \bar{m}$, the two outer steady states are strictly decreasing in imports while the middle steady state is strictly increasing:

$$\frac{dh_L^*}{dm} < 0, \qquad \frac{dh_M^*}{dm} > 0, \qquad \frac{dh_H^*}{dm} < 0.$$

The surviving low-level steady state is likewise strictly decreasing in m for $\bar{m} < m \leq m_1$.



(a) $m < \bar{m}$ (b) $\bar{m} < m \leq m_1$

Figure 1: The law of motion $G(\cdot; m)$ against the 45° line. Panel (a): for $m < \bar{m}$ the S-shaped map crosses the line three times; the low and high steady states are stable and the middle one is unstable, and the arrows show the direction of motion. Panel (b): for $\bar{m} < m \leq m_1$ the map crosses once, the surviving low steady state is stable, and knowledge converges to it from every starting point. Both panels are drawn for a stylized map.

The proof is given in Appendix A.

For $m < \bar{m}$, the low and high steady states are locally asymptotically stable, while the middle steady state is unstable. For $\bar{m} < m \leq m_1$, the surviving low steady state is locally asymptotically stable. Both statements follow from the sign of the residual $D(\cdot; m)$ between its zeros, because $G(\cdot; m)$ is increasing. D is positive just below the low and high steady states and negative just above them. Therefore, nearby knowledge moves monotonically toward each of them, and the pattern is reversed around the middle steady state.

When three steady states coexist, the unstable one divides the state space in two: since $G(\cdot; m)$ is increasing, an economy starting above $h_M^*(m)$ converges to the high steady state, and one starting below it converges to the low steady state. We therefore call $h_M^*(m)$ the *threshold*, and the two intervals it separates the *basins of attraction* of $h_H^*(m)$ and $h_L^*(m)$. We call the low steady state the *trap* and say that an economy is *trapped* when its knowledge lies in the basin of attraction of the low steady state.

Figure 2 demonstrates Proposition 1 in the space of steady states. As m increases, the two stable branches fall and the threshold rises (part (iv)), so the high steady state and the threshold move toward each other. At $\bar{m}$ they meet and then cease to exist for higher levels of m. An increase in imports moves the economy to the right in this diagram, and a large enough increase leaves it with no high-industrialization steady state to return to.

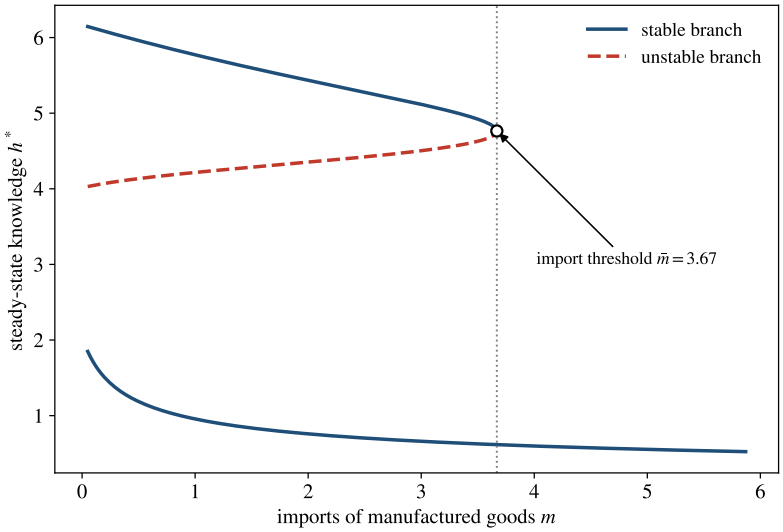


Figure 2: Steady-state knowledge h^* as a function of imports m of manufactured goods. Of the two stable branches (h^*_L falling, h^*_H falling), the high one meets the unstable branch (h^*_M rising) at the import threshold $\bar{m}$, beyond which only the low branch remains.

3.3 A Resource Windfall: Dutch Disease with Hysteresis

So far, nothing in the analysis requires that the flow m come from an adversary, or indeed from abroad. As noted above, m_t is simply an exogenous flow of manufactured goods, and a natural resource is one reading of it. Suppose that the economy is at the high steady state $h^*_H(m_0)$ and a resource is discovered: a Lucas tree appears whose dividend is paid in manufactured goods, which enter the supply k_t through (4) exactly as any other source of m does. If the dividend is large enough that the flow rises to some $m' > \bar{m}$, Proposition 1(iii) applies, and the high steady state ceases to exist. The map falls to the configuration of panel (b) of Figure 1. The only remaining steady state is the low-industrialization state, and knowledge declines toward it. A windfall thus contracts tradable manufacturing, which is the familiar Dutch-disease outcome, though it arrives here through the return to industrial employment rather than through a real appreciation.[3]

With an S-shaped map, the loss of knowledge can persist after the flow ends. A resource boom is typically temporary: the dividend eventually stops and the flow reverts to m_0. Whether the economy recovers depends only on which side of the threshold $h^*_M(m_0)$ it has reached by the time the dividend is stopped. In stating this, we write $G^T(h; m)$ for the T-fold composition of the map $G(\cdot; m)$ with itself, that is, the knowledge stock reached after T

[3]In the canonical account (Corden and Neary, 1982) a windfall raises domestic demand and the real exchange rate, draws labor out of tradable manufacturing, and reverses when the windfall ends. Here there is no real appreciation and no reallocation across sectors: the additional manufactured goods lower the return to industrial employment, and because the knowledge dynamics have an unstable threshold, a boom that outlasts the critical duration of Proposition 2 leaves a permanent loss. Krugman (1987) embeds learning by doing in a Dutch-disease setting. We add the multiplicity and the hysteresis.

periods starting from h when the flow is held fixed at m, with the convention $G^0(h;m) = h$.

Proposition 2 (Hysteresis: temporary booms, permanent losses)**.** *Let $m_0 < \bar{m}$, so that three steady states coexist, and let the economy begin at $h_0 = h^*_H(m_0)$. Suppose $m_t = m' > \bar{m}$ for $t = 0, \ldots, T-1$ and $m_t = m_0$ for all $t \geq T$, so that $h_T = G^T(h_0; m')$. Define the critical duration*

$$\bar{T}(m') \;\equiv\; \min\left\{T \geq 1 \;:\; G^T(h^*_H(m_0); m') < h^*_M(m_0)\right\}. \tag{13}$$

Then

(i) h_t is strictly decreasing for $t = 0, \ldots, T$. If the boom never ended, that is, if $m_t = m'$ at every date, then h_t would decrease monotonically and converge to $h^*_L(m')$;

(ii) $\bar{T}(m')$ is finite. If $T \geq \bar{T}(m')$ the economy converges to $h^*_L(m_0)$. If $T < \bar{T}(m')$ it converges to $h^*_H(m_0)$, with one exception: if $T = \bar{T}(m') - 1$ and $h_T = h^*_M(m_0)$ exactly, the economy remains at the threshold. The exception arises for at most countably many boom sizes m';

(iii) $\bar{T}(m')$ is nonincreasing in m'.

The proof is given in Appendix A.

The path in Proposition 2 does not depend on what agents expect. The preferences are quasilinear in consumption and the household does not have assets, so its choice of labor solves the static condition (6). Learning has a spillover externality. No household takes account of the fact that its own employment raises tomorrow's productivity. Equilibrium employment is therefore the function $\phi(h_t; m_t)$ of the current state alone, and the transition path is identical whether the duration T of the boom is perfectly foreseen at date 0 or arrives as a complete surprise. This is a property of competitive equilibrium, not of learning by doing in general. The trap is therefore an externality problem, not a problem of coordinating expectations, and the corrective policy below is a wedge on the labor margin, not a device for selecting among self-fulfilling paths.

That policy is easy to characterize here, where there is one country and no strategic interaction. A planner who values the same household preferences but internalizes (5) solves

$$\max_{\{n_t\}} \sum_{t=0}^{\infty} \beta^t \left[f(\alpha(h_t)n_t + m_t) - v(n_t)\right] \quad \text{subject to} \quad h_{t+1} = (1-\delta)h_t + \lambda n_t. \tag{14}$$

Let $\beta^t \nu_t$ be the Lagrange multiplier on the law of motion, with $\nu_t \geq 0$ the social value of the knowledge created at date t. The planner's first-order condition for employment is

$$v'(n_t) = \alpha(h_t)\, f'(k_t) + \lambda\, \nu_t, \tag{15}$$

which differs from the competitive condition (6) by exactly the term $\lambda\nu_t$, the social value of the knowledge created by an extra unit of work. The competitive economy therefore under-employs at every date, weakly, and strictly at every date at which $\nu_t > 0$. Too little employment means too little learning, so the whole law of motion sits below its efficient counterpart.

Proposition 3 (The corrective subsidy)**.** *Let* $\{n_t^P, h_t^P\}$ *be a solution to* (14)*, and let* $\nu_t \geq 0$ *be the multiplier on the law of motion for knowledge along it, so that* (15) *and* (17) *hold. Suppose that labor income is subsidized at rate* τ_t *and the subsidy is financed by a lump-sum tax, so the household's condition becomes* $v'(n_t) = (1+\tau_t)\,\alpha(h_t)f'(k_t)$*. The competitive allocation from the same* h_0 *coincides with* $\{n_t^P, h_t^P\}$ *if and only if*

$$\tau_t = \frac{\lambda\,\nu_t}{\alpha(h_t^P)\,f'(k_t^P)} \qquad \text{for every } t. \tag{16}$$

The subsidy is strictly positive whenever $\nu_t > 0$*, and at given* (h_t, ν_t) *it increases in the flow* m_t*.*

Proof. If the two allocations coincide, the household's condition and (15) both hold at (h_t^P, n_t^P), and subtracting gives $\tau_t\,\alpha(h_t^P)f'(k_t^P) = \lambda\nu_t$, which is (16). Conversely, suppose the rates are set by (16) and argue by induction on t. Suppose $h_t = h_t^P$, which holds at $t = 0$. At the stated rate, the household's first-order condition reads $v'(n) = \alpha(h_t^P)f'(\alpha(h_t^P)n + m_t) + \lambda\nu_t$, whose left side is strictly increasing in n and right side strictly decreasing, so it has a unique solution; n_t^P solves it by (15). Hence $n_t = n_t^P$ and $h_{t+1} = h_{t+1}^P$. Positivity is immediate. For monotonicity, at given (h_t, ν_t) employment solves (15), and differentiating gives $dk_t/dm_t = v''/(v'' - \alpha^2 f'') > 0$ as in Lemma 1, so k_t increases with m_t, $f'(k_t)$ falls, and the ratio in (16) increases. □

The proposition takes a solution of the planner's problem as given. The problem is not concave, so its first-order conditions (15) and (17) are necessary but not sufficient. Moreover, the solution need not be unique. Appendix B shows that at low discount factors the problem has two basins of attraction, the history dependence of Skiba (1978) and Dechert and Nishimura (1983). The proposition gives the form of the wedge that decentralizes any solution of the planner's problem.[4]

The subsidy (16) is forward-looking. It depends on the multiplier ν_t, which must be computed along the planner's path. The first-order condition of the problem (14) for h_{t+1} is

$$\nu_t = \beta\Big[\,\alpha'(h_{t+1})\,n_{t+1}\,f'(k_{t+1}) + (1-\delta)\,\nu_{t+1}\,\Big], \tag{17}$$

which says that the value of the knowledge created today is the extra output it produces tomorrow, plus its discounted undepreciated value thereafter. At a steady state, $\nu_t = \nu_{t+1}$, so the multiplier is $\beta\alpha'(h)\,n\,f'(k)/[1-\beta(1-\delta)]$ and the subsidy of Proposition 3 takes the closed form $\beta\lambda\alpha'(h)n/\big(\alpha(h)[1-\beta(1-\delta)]\big)$.

The subsidy therefore does not depend on the source of the flow. It depends on m_t only through its effect on employment, so the same formula applies to a resource dividend and to a rival's export surplus. Two further implications follow from Proposition 2. Since only

[4] A solution exists: flows are bounded above and v is unbounded, so employment can be confined to a compact interval without loss, and the discounted objective is continuous on the resulting product space. Condition (17) is the first-order condition for a perturbation of h_{t+1} alone, with n_t and n_{t+1} adjusting, and is therefore necessary at any interior optimum; its bounded solution, the discounted sum of future marginal products of knowledge, is nonnegative because $\alpha' > 0$.

$m' > \bar{m}$ eliminates the high steady state, a policy that keeps the dividend out of the supply of manufactured goods (saving it abroad rather than converting it into manufactured goods) preserves multiplicity however large the discovery. And since the loss becomes permanent only after $\bar{T}(m')$ periods, the policy must also act in time: keeping the dividend out of that supply works before the threshold is crossed and not after.

4 Trade Between Two Countries

Section 3 treated the flow of manufactured goods as exogenous. In this section, a second economy, the East, produces the manufactured goods, sells them to the West at a market price, and chooses how much to ship. We formalize the intensity of the East's security concern with a parameter ψ. That is, the East's welfare falls with the West's industrial capability at a rate governed by ψ, where $\psi = 0$ denotes the case in which countries are motivated solely by commerce. We derive the equilibrium conditions for general ψ and solve the problem in a special case where exact results are available. There is a critical security weight. Above it, the East exports manufactured goods to the West and the West is trapped. Below it, commerce between the two industrialized countries is autarky, given that the special case assumes no static gain from trade between them. We call an export plan *predation* if it traps the West, in the sense of Section 3, whatever the East's motive.

Throughout, the two countries share the same technology and the same quasilinear preferences over consumption and leisure.

4.1 Environment

Each country $j \in \{E, W\}$ has the technology of Section 3: output $c_t^j = f(k_t^j)$, the supply of manufactured goods $k_t^j = \alpha(h_t^j)n_t^j + m_t^j$ with m_t^j the quantity of manufactured goods the country imports, or minus the quantity it exports, productivity (3), and knowledge that evolves by (5).

West. The West is a decentralized market economy. Consumption is the numeraire and imported manufactured goods trade at the relative price q_t (in consumption units). Western households maximize $\sum_{t=0}^{\infty} \beta^t [c_t^W - v(n_t^W)]$ by choosing employment n_t^W, the quantity of imports $x_t \geq 0$, and the issuance of one-period real bonds b_t^W, subject to the budget

$$c_t^W + R_{t-1}\, b_{t-1}^W \;=\; f\big(k_t^W\big) \;-\; q_t\, x_t \;+\; b_t^W, \qquad k_t^W = \alpha(h_t^W)\, n_t^W + x_t, \tag{18}$$

taking (q_t, R_t) as given. Optimality gives two conditions each period, a labor margin and an import margin.

$$v'(n_t^W) = \alpha(h_t^W)\, f'\big(\alpha(h_t^W)n_t^W + x_t\big), \qquad f'\big(\alpha(h_t^W)n_t^W + x_t\big) = q_t. \tag{19}$$

Because utility is linear in consumption, the bond condition pins the safe rate at the rate of time preference:

$$1 = \beta R_t \quad \Longrightarrow \quad R_t = 1/\beta. \tag{20}$$

The import margin says that the West buys manufactured goods until their marginal product equals the price. Combined with the labor margin, it reproduces the employment function of Section 3, $n_t^W = \phi(h_t^W; x_t)$, with the flow now equal to the traded quantity. The only change is that the flow now has a market price, $q_t = f'(k_t^W)$, rather than being an unexplained transfer. Everything Section 3 established about ϕ applies unchanged. A larger inflow depresses Western employment (Lemma 1), and Western knowledge follows the law of motion (11) at $m = x_t$.

East. The East is governed by a social planner. It produces manufactured goods, ships $x_t \geq 0$ of them to the West, and is paid the world price q_t per unit, which it holds as a claim on the West, its reserves b_t^E. Shipping x_t reduces the East's own supply of manufactured goods one-for-one, so domestic output is $f(\alpha(h_t^E)n_t^E - x_t)$ and the flow budget is

$$c_t^E + \left(b_t^E - R_{t-1}b_{t-1}^E\right) \;=\; f\left(\alpha(h_t^E)n_t^E - x_t\right) + q_t\, x_t, \tag{21}$$

with $m_t^E = -x_t$ and $m_t^W = x_t$: consumption plus net reserve accumulation equals domestic output plus export earnings. Feasibility requires $0 \leq x_t \leq \alpha(h_t^E)n_t^E$.

The East's objective. The Eastern planner's per-period welfare adds one term to the common consumption–leisure payoff:

$$W_t^E \;=\; c_t^E \;-\; v(n_t^E) \;-\; \psi\, g(h_t^W), \tag{22}$$

with the security-loss function

$$g(h^W) \;\equiv\; \mathcal{D}\big(\alpha(h^W)\big)\,, \tag{23}$$

where $\mathcal{D}$ is strictly increasing. Here, $\psi \geq 0$ is the weight on the security term. One reading is that ψ is the per-period probability of a geopolitical contingency, a war, a sanctions episode, or a technological embargo, and $\mathcal{D}$ is the damage the West could inflict in that contingency with industrial capability $\alpha(h^W)$. The damage depends on how much of that capability converts to military use, which is represented by the function $\mathcal{D}$. By (23), the West's knowledge enters the East's payoff only through $\alpha(h^W)$, so g is increasing in h^W and inherits the S-shape of α. Consumption enters linearly, as in the Western household problem (1): the two countries share the same preferences over consumption and leisure and differ only in this term, which is absent from the West's objective. At $\psi = 0$ the term vanishes.

The reserve asset and market clearing. The West pays for its imports with consumption goods, now or later. The asset market clears at $b_t^E = b_t^W$, and the market for the manufactured good clears at the traded quantity x_t. The asset is real, denominated in consumption, and safe; by (20) it earns $R_t = 1/\beta$. The East exports manufactured goods and imports consumption goods, and the bond moves only the timing of the second flow. In a steady state with a constant shipment x, price q, and bond positions $\bar{b}^E = \bar{b}^W$, the West ships $qx + (R-1)\bar{b}^W$ units of consumption to the East each period, the import bill plus the

interest on its debt, and the East's trade balance is a deficit of $(R-1)\bar{b}^E$ matched by interest income of the same size. The surplus we study is a surplus in the manufactured good, not a current-account surplus. Along a path, the split between current consumption goods and future claims is indeterminate. Utility is linear in consumption on both sides, and the return is equal to the rate of time preference. Therefore, the West is indifferent between paying and borrowing, and the East between consuming and lending. What is pinned down is the present value. We impose the no-Ponzi condition $\lim_{t\to\infty} \beta^t b_t^W = 0$ on the debt of the West, under which the present value of the West's consumption exports equals the present value of its imports of the manufactured good plus its initial debt. Nothing below depends on the size of the East's claim.

4.2 The East's Problem

The East chooses its exports anticipating how the West will respond. It cannot choose Western employment directly. Its only instrument is the export flow x_t, and Western employment then follows from the West's own employment function (10). The problem is therefore a Ramsey problem. The East optimizes over allocations that can be implemented in the Western economy as a competitive equilibrium.

The East faces two prices and treats them differently. In the bond market, it is a price-taker. The safe return on the bond is $R_t = 1/\beta$, determined by the intertemporal optimality condition of the West (20). In the goods market, it is large. The West's employment function (10) determines Western employment, $n_t^W = \phi(h_t^W; x_t)$, and the West's import margin (19) ties the price the West pays to the quantity it receives. The East chooses its price and quantity subject to the West's demand schedule.

The East chooses sequences of its consumption, domestic employment, exports, reserves, price, and the two knowledge stocks, $\{ c_t^E,\ n_t^E,\ x_t,\ b_t^E,\ q_t,\ h_{t+1}^E,\ h_{t+1}^W \}_{t=0}^{\infty}$, to solve

$$V^E\big(h_0^E, h_0^W, b_{-1}^E\big) \;=\; \max \sum_{t=0}^{\infty} \beta^t \Big[c_t^E - v(n_t^E) - \psi\, g(h_t^W) \Big] \tag{24}$$

subject to, for all $t \geq 0$, the East's budget, the two laws of motion for knowledge, the West's import demand, and feasibility:

$$c_t^E + b_t^E - R_{t-1} b_{t-1}^E = f\big(\alpha(h_t^E)\, n_t^E - x_t\big) + q_t\, x_t, \tag{25}$$

$$h_{t+1}^E = (1-\delta)\, h_t^E + \lambda\, n_t^E, \tag{26}$$

$$h_{t+1}^W = (1-\delta)\, h_t^W + \lambda\, \phi(h_t^W; x_t), \tag{27}$$

$$q_t = f'\big(\alpha(h_t^W)\, \phi(h_t^W; x_t) + x_t\big)\,, \tag{28}$$

$$x_t \geq 0, \quad b_t^E \geq 0, \quad x_t \leq \alpha(h_t^E)\, n_t^E, \tag{29}$$

with initial conditions (h_0^E, h_0^W, b_{-1}^E) given and the safe return $R_t = 1/\beta$ from (20).

The constraint (27) is the law of motion for knowledge in the West, with Western employment replaced by the employment function $n_t^W = \phi(h_t^W; x_t)$. The constraint (28) is

the West's import demand at that employment. Together, (27) and (28) are the implementability constraints of the East's Ramsey problem. They are the only channels through which the East's choices affect the Western economy. The budget (25) determines the East's consumption c_t^E.[5]

The first-order conditions are stated here for interpretation; the analysis of Section 4.3 argues from plans. Let $\beta^t \mu_t^E$ and $\beta^t \mu_t^W$ be the Lagrange multipliers on the two laws of motion (26) and (27), and let $\beta^t \vartheta_t$ be the multiplier on the demand constraint (28). Because the objective is linear in consumption, the multiplier on the budget (25) is β^t. We sign μ_t^W so that raising h_{t+1}^W is costly: $\mu_t^W \geq 0$ is the value to the East of a one-unit reduction in West knowledge. The first-order condition for q_t gives $\vartheta_t = x_t$. Relaxing the demand constraint by one unit of price is worth the x_t units sold at that price. At an interior optimum, the conditions for n_t^E and x_t are

$$v'(n_t^E) = \alpha(h_t^E)\, f'\big(\alpha(h_t^E) n_t^E - x_t\big) + \lambda\, \mu_t^E, \tag{30}$$

$$f'\big(\alpha(h_t^E) n_t^E - x_t\big) = \underbrace{q_t + x_t \frac{\partial q_t}{\partial x_t}}_{\text{marginal export revenue}} + \underbrace{\lambda\, \mu_t^W \left(-\frac{\partial \phi(h_t^W; x_t)}{\partial x_t}\right)}_{\text{marginal deindustrialization benefit}}, \tag{31}$$

and the conditions for h_{t+1}^E and h_{t+1}^W are

$$\mu_t^E = \beta\Big[\, \alpha'(h_{t+1}^E)\, n_{t+1}^E\, f'\big(\alpha(h_{t+1}^E) n_{t+1}^E - x_{t+1}\big) + (1-\delta)\, \mu_{t+1}^E \,\Big], \tag{32}$$

$$\mu_t^W = \beta\Big[\, \psi\, g'(h_{t+1}^W) \;+\; x_{t+1}\left(-\frac{\partial q_{t+1}}{\partial h_{t+1}^W}\right) \;+\; \mu_{t+1}^W \left((1-\delta) + \lambda\, \tfrac{\partial \phi(h_{t+1}^W; x_{t+1})}{\partial h_{t+1}^W}\right)\Big], \tag{33}$$

where $\partial q_t / \partial x_t < 0$ and $\partial q_t / \partial h_t^W < 0$ by (28) and Lemma 1. At a corner with $x_t = 0$, (31) holds with $\geq$ replacing $=$.

Condition (33) lists the two reasons why the East values a reduction in h_{t+1}^W. The first term is the security loss $\psi g'$. The second is a terms-of-trade gain: $\partial q / \partial h^W < 0$, so a less industrialized West pays a higher price, on each of the x_{t+1} units it buys. The security weight enters the system only through the first term. Conditions (30) and (32) do not involve ψ. The security motive enters the optimality condition of the East's employment only through the shipment x_t.

Condition (31) equates the marginal cost and marginal benefit of one shipped unit. The left-hand side is the marginal cost: one unit less of the manufactured good at home, worth $f'(\alpha(h_t^E) n_t^E - x_t)$ in consumption. The first term on the right is marginal export revenue: the price q_t, net of the terms-of-trade loss $x_t\, \partial q_t / \partial x_t < 0$ on the units already sold. The second term is the deindustrialization benefit. Shipping more depresses Western employment ($\partial \phi / \partial x < 0$, Lemma 1), which lowers h_{t+1}^W, valued by the East at μ_t^W.

The East's financial claim is payoff-irrelevant. With the safe rate pinned at $R_t = 1/\beta$ by (20), the East's intertemporal margin holds with equality for any path of b_t^E, so its level

[5] The West's budget is not a separate constraint on the East's problem. West consumption does not enter the East's objective, and by Walras's law the East's budget, asset-market clearing $b_t^E = b_t^W$, and the West's budget together imply the world resource constraint $c_t^E + c_t^W = f(k_t^E) + f(k_t^W)$, so any two imply the third. The West's budget only pins down c_t^W, which we assume stays nonnegative.

is indeterminate. The East effectively consumes its export earnings. In the benchmark of Section 4.3, the East also stays in the high regime, and its problem reduces to one state, the West's knowledge h_t^W.

4.3 Predation and Welfare

This subsection contains the main result of the section. There is a critical security weight above which the East's optimal plan drives the West across its threshold. The plan is a finite *flood*, a temporary expansion of exports that carries the West's knowledge down to the threshold, followed by the commercial export to a West whose knowledge is below it. The East's problem is not concave, so we do not argue from first-order conditions. Instead, we let the steepness γ of (3) grow without bound, so that the productivity schedule becomes a step, α_L at and below χ and α_H above it. In this *sharp-threshold benchmark*, productivity and security loss are constant within each regime, so the East gains from lowering Western knowledge only by moving it to or below χ. Using this assumption, we prove the results analytically. The subsection first reduces the East's problem to a problem with a single state, the West's knowledge. It then derives the critical security weight, first in an economy in which the East receives nothing for its shipments and then with paid shipments. Next, it asks whether commerce alone, without a security motive, can produce predation. Finally, it characterizes the optimal export path above the critical security weight and compares the East's gain from predation with the West's loss.

An export plan is predation if it traps the West, whatever the East's motive. In the benchmark a West whose knowledge reaches χ never recovers (Lemma 2), so a plan traps the West exactly when it carries the West's knowledge to or below χ at some date. We measure its size by the exports in excess of the East's *commercial export*, the shipment that maximizes its current payoff given the West's current state. In the following benchmark, the commercial export between two industrialized countries is zero, so every unit shipped to an industrialized West exceeds it.

In the benchmark, a country's productivity depends only on which side of χ its knowledge lies. Call the side its *regime*, $i \in \{H, L\}$. Write $\varphi_i(x)$ for the West's employment when its regime is i and it imports x, the solution of $v'(n) = \alpha_i f'(\alpha_i n + x)$, and $q_i(x) = f'(\alpha_i \varphi_i(x) + x)$ for the price it pays. Write $n^E(x)$ for the solution of $v'(n) = \alpha_H f'(\alpha_H n - x)$, and define

$$\Pi_i(x) \;\equiv\; f\big(\alpha_H\, n^E(x) - x\big) - v\big(n^E(x)\big) + q_i(x)\, x. \tag{34}$$

Assumption 3 (Sharp-threshold benchmark)**.** *The productivity schedule is the step*

$$\alpha(h) = \begin{cases} \alpha_L & h \le \chi, \\ \alpha_H & h > \chi, \end{cases} \qquad 0 < \alpha_L < \alpha_H, \tag{35}$$

and:

(i) $\lambda\varphi_L(0) < \delta\chi < \lambda\varphi_H(0)$*: in autarky the benchmark has a low steady state below* χ *and a high steady state* $h_H = \lambda\varphi_H(0)/\delta$ *above it.*

(ii) Both countries begin at h_H.

(iii) Π_H attains its maximum at $x = 0$, and the ratio $-\Pi'_H(x)/\big(-\lambda\varphi'_H(x)\big)$ is strictly increasing on $(0, \infty)$.

(iv) There is a feasible shipment x with $\lambda\varphi_H(x) < \delta\chi$. If the East ships this x at every date, the West's knowledge falls below χ in finite time, so the East can move the West from regime H to regime L.

Part (i) is a condition on the decentralized West. At low productivity, its employment is too low to sustain a knowledge stock of χ, and at high productivity, it sustains more than χ. The decentralized West therefore has an autarky steady state in each regime. Part (ii) starts both countries industrialized, which is the case we study. A West that starts at or below χ is already trapped. An East that starts at or below χ cannot trap the West and remain pre-industrial. A flood requires labor, and at low productivity, the employment needed to produce it raises the East's own knowledge above χ (Appendix C). This assumption also removes the East's own knowledge as a state variable. The first half of part (iii) says that autarky maximizes the East's current payoff against an industrialized West, so pushing the West down always costs flow payoff. It does not rule out a commercial motive for predation. A less productive West pays more for future imports and Proposition 6 characterizes when that gain justifies the cost. Its second half is a sufficient condition for the two-phase shape of the optimal export path. We need this sufficient condition because the problem is not concave. Part (iv) says that a large enough import flow pulls the West's knowledge below χ. The smallest such flow is the import threshold of the benchmark $\bar{m}_\infty$, the counterpart of the import threshold $\bar{m}$ of Section 3. Without part (iv), no export path can trap the West. Each object of the benchmark has a counterpart in Section 3. The employment function $\varphi_i(x)$ is $\phi(h; x)$ evaluated on the plateau i; the high autarky stock h_H is $h^*_H(0)$; the import threshold $\bar{m}_\infty$ is $\bar{m}$; and the discontinuity at χ replaces the unstable steady state h^*_M as the boundary between the two regimes.

Assumption 3 simplifies the problems of both countries. The East starts above the threshold, and the next lemma shows that it stays there whatever it ships, so its productivity is α_H at every date. Employment today therefore adds nothing to the East's productivity tomorrow, and its employment at date t is $n^E(x_t)$, which depends only on that date's shipment. Its payoff at date t from shipping x to a West in regime i is $\Pi_i(x)$. Its reserves do not enter this payoff. The gross return of the bond $1/\beta$ is the inverse of the East's discount factor, so the reserve terms cancel from its present value. On the other hand, West employment and the price it pays depend only on its regime and shipment, through $\varphi_i(x)$ and $q_i(x)$, and the security loss takes two values, $g_i = \mathcal{D}(\alpha_i)$, with $g_L < g_H$. Write x^{c}_i for the shipment that maximizes Π_i, the East's commercial export to a West in regime i. Between two industrialized countries, the autarky prices are equal, so $x = 0$ is a critical point of Π_H, and part (iii) of the assumption makes it the global maximum: $x^{\mathrm{c}}_H = 0$. A trapped West pays a higher price, so $x^{\mathrm{c}}_L > 0$.

Lemma 2 (Reduction to a one-state problem)**.** *Under Assumption 3, for every $\psi \geq 0$:*

(i) an export path $\{x_t \geq 0\}$ is optimal for the East if and only if it solves

$$\max_{\{x_t \geq 0\}} \sum_{t=0}^{\infty} \beta^t \left[\Pi_{i_t}(x_t) - \psi\, g_{i_t}\right] \qquad \text{subject to} \quad h_{t+1}^W = (1-\delta)h_t^W + \lambda\, \varphi_{i_t}(x_t), \quad h_0^W = h_H, \tag{36}$$

where $i_t = H$ if $h_t^W > \chi$ and $i_t = L$ otherwise;

(ii) regime L is absorbing: once $h_t^W \leq \chi$, no export path returns the West above χ;

(iii) among plans that never send the West's knowledge to or below χ, the best ships $x_t = x_H^{\mathrm{c}} = 0$ at every date; and once the West's knowledge is at or below χ, the optimal continuation ships $x_t = x_L^{\mathrm{c}}$ at every later date.

Proof. (i) With $R_t = 1/\beta$, the reserve term cancels out of the East's present-value budget for any bounded reserve path, so its period payoff is consumption net of labor disutility plus the security term, $f(\alpha(h_t^E)n_t^E - x_t) - v(n_t^E) + q_t x_t - \psi g_{i_t}$. Fix any shipment path. Because f is increasing and $\alpha(h) \leq \alpha_H$ for every h, the date-t payoff is at most $\max_n[f(\alpha_H n - x_t) - v(n)] + q_t x_t - \psi g_{i_t}$, and the price q_t depends only on the shipment and on the West. The static employment $n^E(x_t)$ attains this bound at every date provided that the East is in regime H throughout, and it is: $dn^E/dx > 0$ by the argument of Lemma 1 applied at $m = -x$, so $n^E(x_t) \geq n^E(0)$, and since the East starts at $h_H = \lambda n^E(0)/\delta$, induction gives $h_t^E \geq h_H > \chi$ at every date. Any other employment path earns strictly less at some date, because $f(\alpha_H n - x) - v(n)$ is strictly concave in n, and the static path is feasible, since $n^E(x)$ leaves the East with a positive supply of manufactured goods. Whatever the shipments, the East's payoff at date t is therefore $\Pi_{i_t}(x_t) - \psi g_{i_t}$ on any optimal path, and the only constraint linking periods is the law of motion for Western knowledge with φ_{i_t}.

(ii) Let $h_t^W \leq \chi$. Then $h_{t+1}^W = (1-\delta)h_t^W + \lambda\varphi_L(x_t) \leq (1-\delta)\chi + \lambda\varphi_L(0) < (1-\delta)\chi + \delta\chi = \chi$, using φ_L decreasing in x and Assumption 3(i). Induction completes the claim.

(iii) In regime L the regime is permanent by (ii), and the flow payoff $\Pi_L(x_t) - \psi g_L$ involves no state, so it is maximized date by date at x_L^{c}. In regime H, conditional on never crossing, the flow payoff is $\Pi_H(x_t) - \psi g_H$; its date-by-date maximum is $x_t = 0$ by Assumption 3(iii), and under $x \equiv 0$ the West stays at $h_H > \chi$, so the no-crossing restriction does not bind. □

By the lemma, the East's problem reduces to one decision: whether to move the West's knowledge to or below χ, and if so, at which date. If the East never moves the West below χ, its best policy is to ship nothing, by Lemma 2(iii), and it earns $\Pi_H(0) - \psi g_H$ in every period. If it moves the West below χ, it gives up flow payoff in each period before the crossing, because shipping to an industrialized West pays less than autarky. After crossing, it earns more in each period for two reasons. First, a trapped West produces few manufactured goods of its own, so their marginal product is high, and it pays a higher price for imports. Second, a trapped West is less of a threat: $g_L < g_H$. Write $K(x) \equiv \Pi_H(0) - \Pi_H(x)$ for the flow payoff that the East gives up by shipping x rather than nothing to an industrialized West; it is nonnegative by Assumption 3(iii). An export path $P = \{x_t\}_{t \geq 0}$ is a *crossing plan* if it carries the West's knowledge to or below χ at some finite date. Write T_P for the first such

date and define the cost of the plan as

$$C_P \equiv \sum_{t<T_P} \beta^t K(x_t),$$

the discounted flow payoff the plan gives up before the crossing. The subscript records that T_P and C_P are functions of the path. Both can be computed from the path alone. While the West is in regime H, its employment and the price it pays at date t depend only on that date's shipment: employment is $\varphi_H(x_t)$, and the price is $q_H(x_t) = f'(\alpha_H \varphi_H(x_t) + x_t)$, the West's marginal product of the manufactured good, as its import margin (19) requires. The path therefore determines the West's knowledge at every date, and hence T_P. It also determines the East's flow payoff $\Pi_H(x_t)$ on each date, through the East's employment $n^E(x_t)$ and the price $q_H(x_t)$, and hence C_P. Neither computation requires solving the East's problem. Write $\Delta\Pi \equiv \Pi_L(x_L^c) - \Pi_H(0) > 0$ for the commercial gain and $\Delta g \equiv g_H - g_L > 0$ for the security gain.

4.3.1 Predation with free shipments

We first find the critical security weight in a simpler economy in which the East receives nothing for its shipments. This case isolates the security motive and is a step toward the main result. The West absorbs the shipments as the inflow m of Section 3. At a price of zero, the West's import margin has no finite solution. We assume that the West does not choose the quantity; the East does. Write $\Pi^0(x) \equiv \Pi_i(x) - q_i(x)x = f(\alpha_H n^E(x) - x) - v(n^E(x))$ for the East's flow payoff from a gift. This object does not depend on the West's regime and falls strictly in x, by the envelope theorem. Let $K^0(x) \equiv \Pi^0(0) - \Pi^0(x)$ be the flow cost of a gift, and $C_P^0 \equiv \sum_{t<T_P} \beta^t K^0(x_t)$ be the discounted cost of a crossing plan. Since Π^0 does not depend on the West's regime, trapping the West does not have a commercial benefit for the East. Therefore, the gift economy's counterpart of $\Delta\Pi$ is zero.

Proposition 4 (The critical security weight with free shipments)**.** *Let the East receive nothing for its shipments, and let parts (i), (ii), and (iv) of Assumption 3 hold. Then the crossing is strictly optimal for $\psi > \psi_0^*$ and strictly suboptimal for $\psi < \psi_0^*$, where*

$$\psi_0^* = \frac{1-\beta}{\Delta g} \inf_P \beta^{-T_P} C_P^0,$$

the infimum over crossing plans.

Proof. The reduction of Lemma 2 applies because its proof does not use the price. The one difference is that after a crossing the East's best continuation ships nothing, because Π^0 is decreasing. The value of never crossing is $[\Pi^0(0) - \psi g_H]/(1-\beta)$, and the value of a crossing plan P is $\sum_{t<T_P} \beta^t [\Pi^0(x_t) - \psi g_H] + \beta^{T_P}[\Pi^0(0) - \psi g_L]/(1-\beta)$. Subtracting the first from the second, the gain from plan P over never crossing is

$$-C_P^0 + \beta^{T_P} \frac{\psi \Delta g}{1-\beta}.$$

This gain is affine in ψ with positive slope $\beta^{T_P}\Delta g/(1-\beta)$. It is zero at $\psi = z_P^0 \equiv (1-\beta)\beta^{-T_P}C_P^0/\Delta g$, positive for $\psi > z_P^0$, and negative for $\psi < z_P^0$. If $\psi > \inf_P z_P^0$, then $\psi > z_P^0$ for some crossing plan, and crossing is strictly optimal. If $\psi < \inf_P z_P^0$, then every crossing plan loses, and the crossing is strictly suboptimal. The infimum is taken over a nonempty set: by Assumption 3(iv), shipping a constant x with $\lambda\varphi_H(x) < \delta\chi$ carries the West's knowledge below χ in finite time, so crossing plans exist. □

4.3.2 Predation with paid shipments

With the price restored, each crossing plan's break-even weight is similar to the above, adjusted for what the price adds: the revenue on the active shipments and the commercial gain once the customer is trapped. Write $z_P \equiv [(1-\beta)\beta^{-T_P}C_P - \Delta\Pi]/\Delta g$ for the weight at which plan P breaks even when the East is paid for shipments. Because $K(x) = K^0(x) - q_H(x)x$,

$$z_P \;=\; z_P^0 \;-\; \frac{(1-\beta)\,\beta^{-T_P}}{\Delta g}\sum_{t<T_P}\beta^t\,q_H(x_t)\,x_t \;-\; \frac{\Delta\Pi}{\Delta g}, \tag{37}$$

where $\Delta\Pi = \Pi_L(x_L^{\mathrm{c}}) - \Pi_H(0)$ is the commercial gain per period (over autarky) for the East, from shipping x_L^{c} to the West in regime L. Both subtracted terms are nonnegative, so $z_P \le z_P^0$ for every crossing plan. Taking the infimum over plans and truncating at zero gives $\psi^* \le \psi_0^*$. In other words, the critical security weight is lower when the East is paid than when it gives its shipments away. Payment lowers the critical security weight in two ways (Figure 3(a)). One, revenue from shipments offsets part of their cost. Two, after crossing, a West in regime L pays a higher price.

Proposition 5 (The critical security weight)**.** *Under Assumption 3:*

(i) There is a unique critical security weight

$$\psi^* \;=\; \max\left\{0,\; \inf_P\;\frac{(1-\beta)\,\beta^{-T_P}\,C_P \;-\; \Delta\Pi}{\Delta g}\right\}, \tag{38}$$

the infimum taken over crossing plans, such that crossing is strictly optimal for $\psi > \psi^*$ and strictly suboptimal for $\psi < \psi^*$.

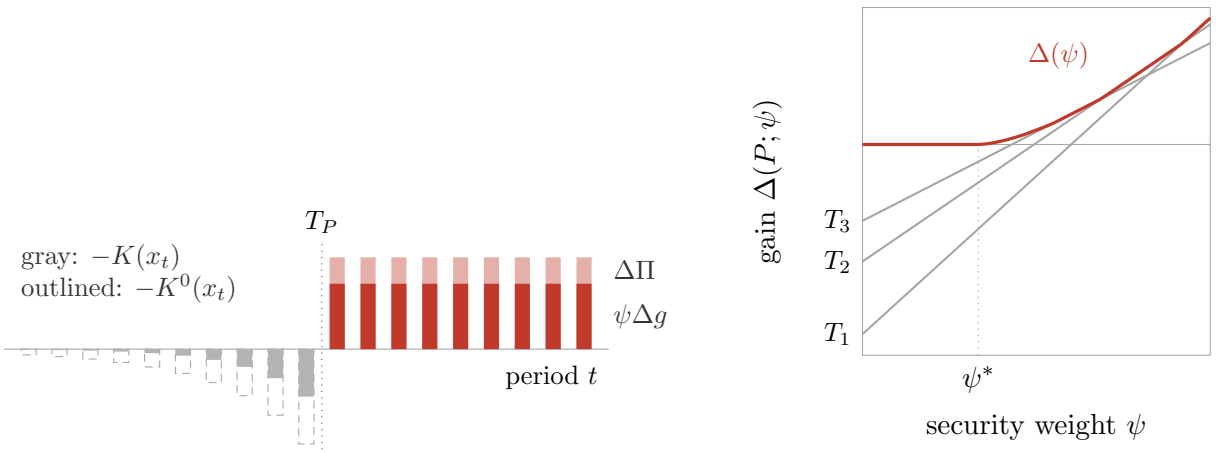
(ii) When $\psi^* > 0$, it is strictly decreasing in the security gap Δg.

Proof. By Lemma 2(iii), the value of never crossing is $V_{\mathrm{stay}}(\psi) = [\Pi_H(0) - \psi g_H]/(1-\beta)$. A crossing plan P with date T_P and active exports $\{x_t\}_{t<T_P}$ has value

$$V(P;\psi) \;=\; \sum_{t<T_P}\beta^t\big[\Pi_H(x_t) - \psi g_H\big] \;+\; \beta^{T_P}\,\frac{\Pi_L(x_L^{\mathrm{c}}) - \psi g_L}{1-\beta},$$

where the continuation uses Lemma 2(ii) and (iii). Subtracting, we get

$$\Delta(P;\psi) \;\equiv\; V(P;\psi) - V_{\mathrm{stay}}(\psi) \;=\; -\,C_P \;+\; \beta^{T_P}\,\frac{\Delta\Pi + \psi\,\Delta g}{1-\beta}. \tag{39}$$



(a) Flow payoff relative to never crossing

(b) Gains of crossing plans

Figure 3: The East's choice of a crossing plan. Panel (a): before the crossing date T_P the plan gives up $K(x_t)$ each period, or $K^0(x_t)$ when shipments are free; from T_P on it gains $\psi\Delta g$, plus $\Delta\Pi$ when shipments are paid. Panel (b): the gain $\Delta(P;\psi)$ of the cheapest plans with crossing dates $T_1 < T_2 < T_3$, and their upper envelope over all crossing plans, $\Delta(\psi)$. Schematic.

Each $\Delta(P;\cdot)$ is affine in ψ (Figure 3(b)) with a strictly positive slope $\beta^{T_P}\Delta g/(1-\beta)$, so it has a unique zero, the break-even weight z_P defined before the proposition, and $\Delta(P;\psi) > 0$ exactly when $\psi > z_P$. Crossing plans exist by Assumption 3(iv): a constant export $\bar{x}$ with $\lambda\varphi_H(\bar{x}) < \delta\chi$ drives h^W monotonically toward $\lambda\varphi_H(\bar{x})/\delta < \chi$ and so crosses in finite time. The infimum in (38) is therefore over a nonempty set, and $\psi^* = \max\{0, \inf_P z_P\}$ is finite. If $\psi > \psi^*$, then $\psi > z_P$ for some crossing plan P, so $\Delta(P;\psi) > 0$: some crossing plan beats never crossing, and crossing is strictly optimal. If $\psi < \psi^*$, then $\psi < z_P$ for every crossing plan, so every crossing plan loses to never crossing, whose best version is autarky by Lemma 2(iii): crossing is strictly suboptimal. This proves (i). The comparison must be made plan by plan, because the supremum of $\Delta(P;\psi)$ over crossing plans is never negative: a plan P preceded by N periods of autarky, which keep the West at h_H, is a crossing plan with gain $\beta^N\Delta(P;\psi)$, and that tends to zero.

For later use, write $\Delta(\psi) \equiv \sup_P \Delta(P;\psi)$ for the East's value gain from predation, the value of its problem less $V_{\text{stay}}(\psi)$. By the argument just given, wherever $\psi^* > 0$ the gain is zero on $[0, \psi^*]$, and it is positive on (ψ^*, ∞). Since $\Delta(P;\psi) \le (\Delta\Pi + \psi\Delta g)/(1-\beta)$ for any fixed crossing plan P, $\Delta(\psi)$ is finite. Moreover, $\Delta(\psi)$ is convex and continuous on $(0,\infty)$, since it is the finite supremum of affine functions. Since each $\Delta(P;\cdot)$ is nondecreasing, so is $\Delta(\psi)$. Finally, for any fixed crossing plan P, $\Delta(P;\psi) \to \infty$ as $\psi \to \infty$. Therefore, $\Delta(\psi)$ is unbounded above.

For (ii): in (38) the numerator is free of Δg and strictly positive wherever $\psi^* > 0$, so ψ^* falls strictly as Δg rises. □

4.3.3 Predation without a security motive

We now ask whether the East would trap the West with no security motive at all, that is, whether the critical security weight of Proposition 5 can be zero. It is zero when some crossing plan is profitable at $\psi = 0$, so that the East traps the West only for commercial reasons. By (38), the critical security weight is strictly positive if and only if

$$\inf_P \beta^{-T_P} C_P \;>\; \frac{\Delta\Pi}{1-\beta}.$$

The left side is the lowest cost of crossing, with each plan's cost measured at its own crossing date. The right side is the present value, at the crossing date, of the commercial gain $\Delta\Pi$ that the East earns in every period from the crossing onward.[6]

Whether the inequality holds depends on how patient the East is. The left side can be written as

$$\beta^{-T_P} C_P \;=\; \sum_{t<T_P} \beta^{-(T_P-t)} K(x_t),$$

which carries each period's forgone payoff forward to the crossing date with interest. It falls as β increases, while the right side increases with β. Nothing else in the comparison depends on β. The flow payoffs, the West's law of motion, and hence the set of crossing plans do not depend on β. The inequality therefore holds when the East is impatient and fails when it is patient enough.

The inequality also has a sufficient condition that does not require finding the cheapest plan. Every period of forgone payoff comes at least one period before the crossing. Therefore, each term of the sum is at least $K(x_t)/\beta$, and the cost of any crossing plan, measured at its crossing date, is at least C_0/β, where

$$C_0 \;\equiv\; \inf\Big\{ \textstyle\sum_{t=0}^{T-1} K(x_t) \;:\; h_0 = h_H,\; h_{t+1} = (1-\delta)h_t + \lambda\varphi_H(x_t),\; h_T \le \chi \Big\}. \tag{40}$$

The infimum is taken over export paths of any finite length T. C_0 is the smallest total flow payoff, summed without discounting, that any export path gives up while it carries the West's knowledge from h_H to χ or below. It is computed from the West's law of motion and the flow payoff Π_H alone; neither β, nor ψ, nor the East's problem enters it. Comparing C_0/β with $\Delta\Pi/(1-\beta)$ gives the condition

$$C_0 \;>\; \frac{\beta\,\Delta\Pi}{1-\beta}, \tag{D}$$

which is equivalent to $\beta < C_0/(C_0 + \Delta\Pi)$.

Proposition 6 (Commerce and patience)**.** *Under Assumption 3 there is a discount factor $\hat\beta$ with $C_0/(C_0+\Delta\Pi) \le \hat\beta < 1$ such that:*

[6] The inequality concerns the infimum. If every crossing plan loses money at $\psi = 0$ but the losses, measured at the crossing date, approach zero along a sequence of plans, then any positive security weight makes some plan profitable and $\psi^* = 0$. This happens only when the two sides of the inequality are equal.

(i) *If $\beta > \hat{\beta}$, then $\psi^* = 0$, and at $\psi = 0$ crossing is strictly optimal: the East traps the West for commercial gain alone.*

(ii) *If $\beta < \hat{\beta}$, then $\psi^* > 0$, and at $\psi = 0$ the optimal path is the commercial equilibrium: the East does not ship anything and the West stays at its high steady state forever.*

(iii) *Condition (D) is sufficient for case (ii): if it holds, then $\beta < C_0/(C_0 + \Delta\Pi) \leq \hat{\beta}$. The critical security weight then satisfies the explicit bound*

$$\psi^* \;\geq\; \frac{(1-\beta)\, C_0/\beta - \Delta\Pi}{\Delta g} \;>\; 0.$$

The proof is given in Appendix A. Parts (i) and (ii) say that a purely commercial East traps an industrialized customer when it is patient enough and ships nothing when it is not. Part (iii) gives a condition that can be checked without solving the East's problem. The condition is sufficient, but not necessary. It bounds the cost of each crossing plan from below by C_0/β, which carries every period's forgone payoff forward by one period only, while the actual cost carries each payoff forward to the crossing date. The cutoff $\hat{\beta}$ can therefore exceed $C_0/(C_0 + \Delta\Pi)$. The difference comes only from compounding within the active phase, and the compounding factors $\beta^{-(T_P - t)}$ approach one as β approaches one. Section 6 computes $\hat{\beta}$ and both bounds.

4.3.4 The two-phase path of predation

Propositions 5 and 6 say when the East crosses, but not how. We now describe the East's optimal export path when $\psi > \psi^*$.

Proposition 7 (Two-phase predation)**.** *Under Assumption 3, let $\psi > \psi^*$. An optimal export path exists, and every optimal export path has exactly two phases.*

(i) Active phase *($0 \leq t < T^*$, with T^* finite): the export is positive at every date, is strictly increasing, and reaches its maximum in the period when the West's knowledge reaches χ; the West's knowledge falls strictly throughout.*

(ii) Passive phase *($t \geq T^*$): the West's regime is L and remains so, and the East ships its commercial export x_L^{c}.*

T^* is the first date at which $h_t^W \leq \chi$, and at that date $h_{T^*}^W = \chi$.

The proof is given in Appendix A. When $\psi^* > 0$, the East's optimal plan at $\psi = 0$ ships nothing, by Proposition 5 and Lemma 2(iii), so the active phase is due to the security motive alone.

The crossing date. Predation costs the East commercial payoff before the crossing and yields commercial and security gains after it. Once the West is in regime L, it pays a higher price for imports, and the East's security loss is lower in every period from then on. A slower flood is cheaper, because shipping a little to an industrialized West costs the East

almost nothing at the margin. However, the gains begin only at the crossing, and delay discounts them. For late enough crossing dates, discounting dominates. Whatever the cost of the shipments, a plan that crosses at date T gains at most β^T times the value of the gains measured at the crossing date, and that bound goes to zero as T grows. When $\psi > \psi^*$, some plan that crosses at a finite date gains a positive amount, so crossing ever later cannot be optimal, and the East crosses at a finite date. Appendix A also shows that the optimal crossing date does not increase with the security weight. When $\psi^* > 0$, it also shows that the crossing date grows without bound as the weight falls to the critical security weight, so just above it the East crosses late, and that no crossing plan breaks even exactly at the critical security weight, which is the limit of the break-even weights of ever slower plans (Figure 3(b)). Section 6 reports the computed dates at our parametrization.

The rising export. Once the crossing date is fixed, the East's problem is to bring the West's knowledge down to χ by that date as cheaply as possible. A late shipment lowers the West's knowledge at the crossing date at a lower present cost than an early one, for two reasons. It is paid for later, so it costs less in present value. Also, more of its effect survives to the crossing date. A shipment lowers the knowledge the West creates in the period it arrives. Knowledge that is not created early would have been partly depreciated by the crossing date anyway. The East therefore concentrates its shipments near the crossing date. It does not ship everything in the last period, because a shipment's cost rises faster than its effect on Western employment as the shipment grows, by Assumption 3(iii). Balancing the two, the East ships more in each period than in the one before, and ships the most in the last period before the crossing (Figure 4).

The rising profile is a property of the step function. On the plateau, nothing but the shipments lowers the West's employment. In the smooth economy, the plan has the same two phases, but the peak comes before the crossing rather than at it. As the West's knowledge enters the steep part of the S-curve, its productivity and employment fall on their own, and a smaller shipment is enough to keep its knowledge falling. Section 6 reports both plans at illustrative parameters.

4.3.5 The welfare cost of predation

We now ask whether predation raises or lowers world welfare, that is, whether the East gains more or less than the West loses. Throughout, we assume $\psi^* > 0$, and the welfare of both countries is measured relative to the commercial equilibrium, which is then autarky: the East ships nothing and the West stays at h_H.

Write $u_i^W(x) = f(\alpha_i\varphi_i(x) + x) - q_i(x)\,x - v(\varphi_i(x))$ for the West's welfare in a period in which it is in regime i and imports x: output, less the import bill, less the disutility of work. On an export path $P = \{x_t\}$ with regimes $\{i_t\}$, the West's lifetime loss relative to autarky is

$$L^W(P) \;=\; \sum_t \beta^t \big[u_H^W(0) - u_{i_t}^W(x_t)\big].$$

For $\psi > \psi^*$, write $\mathcal{P}^*(\psi)$ for the set of the East's optimal plans, which is nonempty by

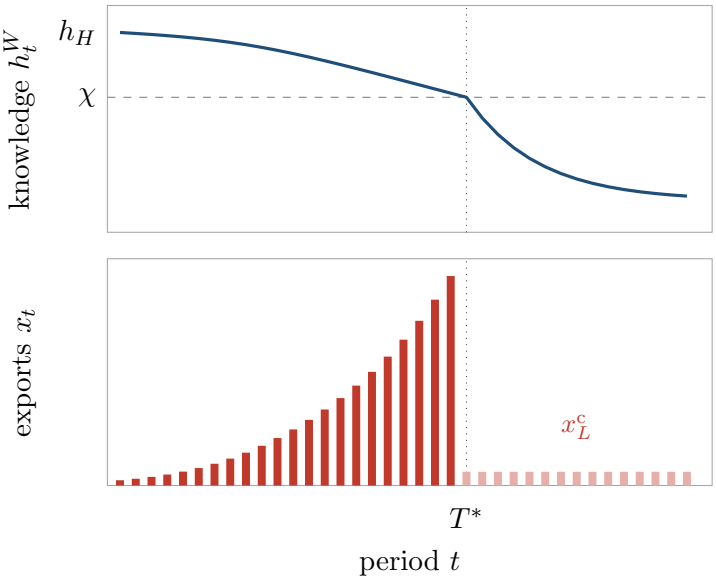


Figure 4: Two-phase predation (Proposition 7). Top: the West's knowledge falls to χ at the crossing date T^* and then toward its low steady state under the commercial export. Bottom: the East's export rises until the last period before the crossing and then falls to the commercial export x_L^{c}. Drawn from the optimal plan at the parameters of Section 6.

Proposition 7, and write

$$\Delta W^E(\psi) \;=\; \sum_t \beta^t \big[\Pi_{i_t}(x_t) - \psi\, g_{i_t}\big] \;-\; \frac{\Pi_H(0) - \psi\, g_H}{1-\beta} \qquad \text{for } \{x_t\} \in \mathcal{P}^*(\psi)$$

for the East's lifetime gain relative to autarky. Every optimal plan gives the East the same value, so its gain does not depend on which optimal plan it follows. However, optimal plans with different crossing dates impose different losses on the West. The result in the following covers every optimal plan.

Near the critical security weight, the East's gain and the West's loss, measured at date zero, are both close to zero, for different reasons. The East's gain is close to zero because it is zero at the critical security weight and rises continuously above it. The West's loss is small because just above it the East crosses late, so the trap begins far in the future and is heavily discounted. Both are discounted by the same factor β^{T_P}, so we compare them at the crossing date. Measured there, the West's loss does not shrink. Most of it is the permanent fall in the West's welfare after the crossing,

$$\Delta u \;\equiv\; u_H^W(0) - u_L^W(x_L^{\mathrm{c}})$$

in every period, and the value of that fall at the crossing date does not depend on when the crossing happens. Against it, the West gains while the shipments last, because it buys them below its autarky price. Under the conditions of the proposition, that gain is at most a third of what the shipments cost the East.

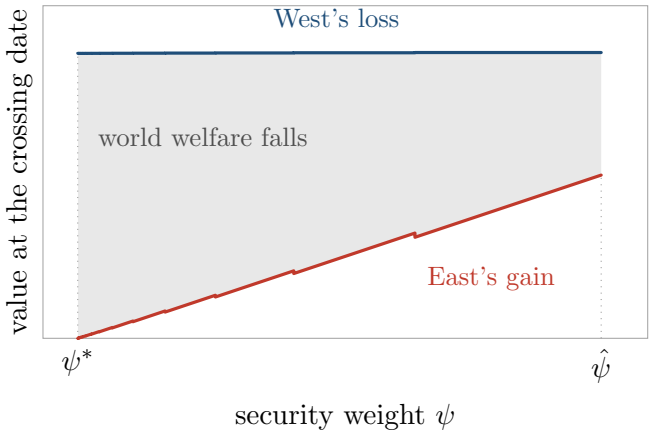


Figure 5: The East's gain and the West's loss from predation for $\psi^* < \psi \leq \hat{\psi}$ (Proposition 8), both measured at the crossing date of the optimal plan. The shaded gap is the fall in world welfare, also measured at the crossing date. The East's gain dips slightly where the optimal crossing date changes. Drawn from the optimal plans at the parameters of Section 6.

Proposition 8 (Predation near the critical security weight lowers world welfare)**.** *Under Assumption 3, suppose that $\psi^* > 0$, v is quadratic, and $f''' \geq 0$. Suppose also that*

$$3\,\Delta u \;>\; \psi^* \Delta g + \Delta\Pi. \tag{N}$$

Define $\hat{\psi}$ by $\hat{\psi}\,\Delta g = \frac{2}{3}\,\psi^* \Delta g + \Delta u - \frac{1}{3}\,\Delta\Pi$, so that $\hat{\psi} > \psi^*$. Then for every weight ψ with $\psi^* < \psi < \hat{\psi}$ and every optimal plan $P \in \mathcal{P}^*(\psi)$,

$$0 \;<\; \Delta W^E(\psi) \;<\; L^W(P).$$

The East gains from predation, the West loses more, and the sum of the two countries' welfare falls: predation is negative-sum.

The proof is given in Appendix A. Condition (N) says that once the West is in regime L, each period it loses more than a third of what the East gains at the critical security weight, its security gain $\psi^* \Delta g$ plus the higher price $\Delta\Pi$. On an optimal plan, the East's gain is $\beta^{T_P} \Delta g(\psi - z_P)/(1-\beta)$, which is positive, so $z_P < \psi$. The West's loss is $\beta^{T_P} \Delta u/(1-\beta)$ minus what it gains while the shipments last. That gain is at most a third of the cost C_P of the shipments to the East (Appendix A). No plan breaks even at the critical security weight, so $z_P > \psi^*$, and the two bounds together put the West's loss above the East's gain at every weight below $\hat{\psi}$ (Figure 5). The conclusion covers every optimal plan at every weight in the interval, including weights at which two crossing dates are both optimal. The security parameters enter only through the product $\psi \Delta g$, so the interval, stated in that product, is the same for every Δg. Section 6 reports Condition (N) at our parameters, the upper end of the interval, and how far above the critical security weight the conclusion extends when the East's gain and the West's loss are compared directly.

Predation lowers world welfare for two reasons. First, shipments waste resources. Each unit shipped is worth more to the East at home, where its value is the East's marginal

product, than to the West, where its value is the price the West pays. Second, the loss of knowledge is permanent. After crossing, the West's labor produces at α_L instead of α_H in every period. The higher price the trapped West pays is a transfer to the East and does not change the sum of the two countries' welfare; the lost productivity does. Near the critical security weight, the East's gain is small, so the fall in world welfare is close to the West's entire loss.

We next allow the Western government to respond with tariffs or employment subsidies.

5 The West's Counterpolicy

Section 4 showed that above the critical security weight ψ^* a security-motivated East temporarily expands its exports to move a passive West's knowledge below the threshold. A passive West is one whose households set employment competitively and whose government does nothing. This section gives the West two instruments, a tariff on the imported manufactured good and a subsidy to industrial employment, and asks which one prevents predation. We work in the benchmark of Section 4.3 and keep its objects: the regimes $i \in \{H, L\}$, the West's competitive employment $\varphi_i(x)$ and the price $q_i(x)$, the employment of the East $n^E(x)$ and the flow payoff $\Pi_i(x)$ of (34), the step security loss g_i with gap Δg, the high autarky stock h_H, the import threshold $\bar{m}_\infty$, and Assumption 3.

The two instruments differ in what the West must know. A tariff deters only if it exceeds a level that depends on the East's motive, which the West does not observe (Section 5.1). The labor subsidy removes the trap at every motive if its rate is not capped, and, capped, deters every motive below a level that rises with the cap; on the equilibrium path it is never paid (Sections 5.2 and 5.3).

5.1 Tariffs

The first instrument is a per-unit tariff $\xi \geq 0$ on imports of manufactured goods, collected from Western households and rebated to them as a lump sum. The West's import margin in (19) becomes $f'(k_t^W) = q_t + \xi$, where q_t is the price the East receives. Households buy until the marginal product equals the price they pay, tariff included. In Section 4.2 the East chose its price and quantity subject to the West's demand. The same is true here, with one change. The demand ties the quantity to the price the West pays, and the East receives that price less the tariff, so when choosing x the East chooses its after-tax price. At a shipment x in regime i, the West's labor margin is unchanged, so its employment is still $\varphi_i(x)$, its marginal product $q_i(x)$ is now the gross price it pays, and the East receives $q_i(x) - \xi$. The tariff shifts the inverse demand that the East faces down by ξ at every quantity and does not change anything else at a given shipment. The East's flow payoff from shipping x to a regime-i West is therefore

$$\Pi_i(x) - \xi\, x, \tag{41}$$

the payoff of (34) less the tariff bill, and the West's flow welfare is $u_i^W(x) + \xi x$, its welfare at that shipment without the tariff plus the receipts. The tariff transfers revenue from the

East to the West and, at a given shipment, leaves Western employment unchanged.

Three objects of Section 4.3 acquire a tariff argument. The commercial export against a trapped West is

$$x_L^{\mathrm{c}}(\xi) \in \arg\max_{x \geq 0} \big[\Pi_L(x) - \xi x\big].$$

The commercial gain is now

$$\Delta\Pi(\xi) \;\equiv\; \max_{x \geq 0} \big[\Pi_L(x) - \xi x\big] - \Pi_H(0) \;\geq\; 0.$$

It is nonnegative because $\Pi_L(0) = \Pi_H(0)$: the East can always ship nothing. The cost of a crossing plan P is

$$C_P(\xi) \;\equiv\; \sum_{t<T_P} \beta^t \big[\Pi_H(0) - \Pi_H(x_t) + \xi x_t\big] \;=\; C_P(0) + \xi \sum_{t<T_P} \beta^t x_t,$$

the cost without the tariff plus the discounted tariff bill of the active phase. The crossing plans themselves are unchanged. They are paths of shipment, and the tariff does not change the West's law of motion. The critical security weight under the tariff is

$$\psi^*(\xi) \;=\; \max\left\{0,\; \inf_P \; \frac{(1-\beta)\,\beta^{-T_P}\,C_P(\xi) \;-\; \Delta\Pi(\xi)}{\Delta g}\right\}, \tag{42}$$

formula (38) with the tariff cost and gain, and $\psi^*(0) = \psi^*$.

Proposition 9 (Tariff absorption)**.** *Under Assumption 3, for every tariff $\xi \geq 0$:*

(i) Lemma 2 holds with (41) in place of Π_i: the East's optimal export paths solve the reduced problem (36) with the tariffed payoff, regime L is absorbing, the best never-crossing plan is autarky, and after a crossing the East ships $x_L^{\mathrm{c}}(\xi)$.

(ii) The crossing is strictly optimal for $\psi > \psi^*(\xi)$ and strictly suboptimal for $\psi < \psi^*(\xi)$. The function $\xi \mapsto \psi^*(\xi)$ is continuous and nondecreasing, and wherever it is positive it rises at least at the rate $(1-\beta)\bar{m}_\infty/\Delta g$:

$$\psi^*(\xi') \;\geq\; \psi^*(\xi) + \frac{(1-\beta)\,\bar{m}_\infty}{\Delta g}\,(\xi' - \xi) \qquad \text{whenever } \xi' > \xi \text{ and } \psi^*(\xi) > 0.$$

In particular $\psi^*(\xi) \to \infty$ as $\xi \to \infty$.

(iii) For $\psi > \psi^*(\xi)$ the East's optimal path crosses: the West's knowledge reaches χ or falls below it at a finite date and stays at or below it; its flow welfare at every date is $u_{i_t}^W(x_t) + \xi x_t$, the welfare it would have at that shipment without the tariff plus the tariff receipts, which are positive at every date with positive shipments. For $\psi < \psi^*(\xi)$ the East ships nothing and the West stays at h_H.

(iv) For every $\psi > \psi^*(0)$ there is a unique $\bar{\xi}(\psi) \in (0, \infty)$ with $\psi^*(\bar{\xi}(\psi)) = \psi$. A tariff below $\bar{\xi}(\psi)$ is absorbed and the West is trapped; a tariff above it deters, and the West stays at h_H. The deterrent tariff $\bar{\xi}(\psi)$ is nondecreasing in ψ.

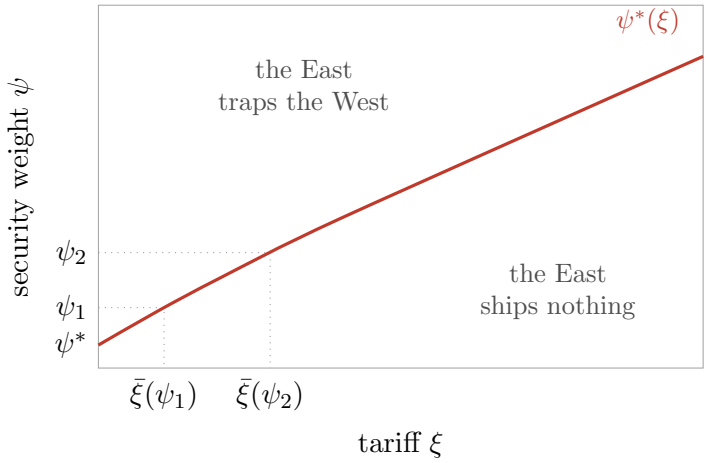


Figure 6: The critical security weight under a tariff, $\psi^*(\xi)$ (Proposition 9). Above the curve the East pays the tariff and traps the West; below it the East ships nothing. A tariff between $\bar{\xi}(\psi_1)$ and $\bar{\xi}(\psi_2)$ deters an East with weight ψ_1 and is absorbed by an East with weight ψ_2. Drawn at the parameters of Section 6.

The proof is given in Appendix A.

By the proposition (Figure 6), a tariff below $\bar{\xi}(\psi)$ raises the cost of predation to the East, but does not prevent it. The East pays ξ on every unit it ships and still ships enough to carry the West's knowledge down to χ. The West collects the revenue while its knowledge falls, and it ends in regime L, as it would under a tariff of zero. If the tariff is higher than $\bar{\xi}(\psi)$, the East does not ship anything. The deterrent tariff $\bar{\xi}(\psi)$ is finite for every $\psi > \psi^*$ and nondecreasing in ψ, so a tariff large enough deters any given motive. Which case applies depends on ψ and on the damage function behind Δg, and the West observes neither ψ nor $\mathcal{D}$. A West that sets a tariff therefore does not know whether it has set it above or below $\bar{\xi}(\psi)$.

Absorption concerns the outcome, not the path. A tariff raises the cost of every unit shipped. The East's optimal plan changes, and so does its crossing date. But a tariff below $\bar{\xi}(\psi)$ does not change the West's final regime. The West still ends in the low regime. A prohibitive tariff has no cost here. Between two industrialized countries, the commercial equilibrium is autarky. A tariff that stops all trade therefore stops no trade that commerce would have produced. It is equivalent to a quota of zero. The deterrent tariffs are also small at our parameters, a fraction of the autarky price (Section 6). A tariff therefore deters at low cost when the East's motive is known. It does not give protection when the motive exceeds the level it was set to deter.

5.2 The Labor Subsidy

The second instrument is a labor subsidy at rate $\tau_t > -1$. It is paid to Western households and is financed by a lump-sum tax, as in Proposition 3. A negative rate is a tax on labor income, rebated lump sum. The household's employment optimality condition, the labor

margin in (19), becomes

$$v'(n_t^W) = (1+\tau_t)\,\alpha(h_t^W)\,f'\big(\alpha(h_t^W)\,n_t^W + x_t\big). \tag{43}$$

The rate multiplies the private return to work by $1+\tau_t$. At $\tau_t = 0$ the condition reduces to that margin. Because the tax is lump sum, the subsidy acts only through the West's employment and through it on Western knowledge and the import price. It does so indirectly. The West is a decentralized economy and its government does not choose employment. It sets the rate, and households choose how much to work at that rate. The next lemma says that the distinction does not matter for what follows.

Lemma 3 (Implementability)**.** *Fix the West's knowledge h and the shipment x, and let the subsidy rate depend on h and x, so that households take it as given. For every rate $\tau > -1$, condition (43) has a unique solution, and every employment level $n > 0$ solves it at exactly one rate,*

$$\tau(n;h,x) \;=\; \frac{v'(n)}{\alpha(h)\,f'\big(\alpha(h)n + x\big)} - 1. \tag{44}$$

This rate is strictly increasing in n and zero at the competitive employment $\phi(h;x)$. It is a subsidy, $\tau \geq 0$, if and only if $n \geq \phi(h;x)$, and a tax otherwise.

Proof. At a given rate, the left side of (43) increases from zero to infinity in n and the right side falls, as in the proof of Lemma 1 with $(1+\tau)\alpha(h)$ in place of $\alpha(h)$, so the solution is unique. Solving (43) for the rate gives (44). Its numerator increases in n and its denominator falls, and it is zero at $\phi(h;x)$ by (6). □

The government can therefore induce any employment level, with a subsidy above the competitive level and a tax below it. From here on, we describe the government as choosing employment. By the lemma, a choice of employment is the choice of one rate, the rate (44), so nothing is lost by the description. Write the West's flow welfare when it works n, imports x, and has knowledge h as

$$u^W(n;h,x) \;\equiv\; f\big(\alpha(h)n + x\big) - f'\big(\alpha(h)n + x\big)\,x - v(n). \tag{45}$$

This is output less the import bill less the disutility of work. The import bill is priced at the marginal product at which the West buys x, as in (28). In the benchmark, the same object is $u_i^W(n;x)$, with α_i in place of $\alpha(h)$. The welfare $u_i^W(x)$ of Section 4.3 is its value at the competitive employment $\varphi_i(x)$.

The government chooses employment to maximize the present value of this flow. Its choice differs from the households' for two reasons. The first is the learning externality of Section 3: work today adds to knowledge tomorrow. The second is the price of imports. More work raises the West's supply of manufactured goods and lowers the marginal product at which it buys the East's shipment, so it lowers the bill for every unit imported. Households take the price as given and ignore both.

In the benchmark, knowledge matters to the government only through its regime. Productivity is flat on each side of χ, so as long as the next period's knowledge stays above χ, it

does not matter by how much. Setting knowledge aside, the best choice of employment by the government against a shipment x in regime i maximizes $u_i^W(n;x)$. Write $n_i^{\mathrm{R}}(x)$ for this optimal employment and call it the *static best response* of the government. It must satisfy the equation

$$v'(n) \;=\; \alpha_i\big[f'(k) - x\,f''(k)\big], \qquad k = \alpha_i n + x. \tag{46}$$

Competitive employment $\varphi_i(x)$, which households choose when the rate is zero, solves the same condition without the term $-xf''(k)$: households take the import price as given, and the government does not. When f is three times differentiable with $f''' \geq 0$, which holds for every f of constant elasticity and in particular for the one of Section 6, u_i^W is strictly concave in n, and $n_i^{\mathrm{R}}(x)$ is unique and nonincreasing in x. The two levels coincide when nothing is shipped, $n_i^{\mathrm{R}}(0) = \varphi_i(0)$, and against every positive shipment, the government's level is higher, $n_i^{\mathrm{R}}(x) > \varphi_i(x)$.

The subsidy. The government implements its choice with the rate (44). To interpret the rate, write

$$u_n^W(n;h,x) \;\equiv\; \alpha(h)\big[f'(k) - x\,f''(k)\big] - v'(n), \qquad k = \alpha(h)n + x,$$

for the derivative of the flow (45) in employment. Using this equation to substitute $v'(n)$ into (44) decomposes the subsidy into two parts:

$$\tau(n;h,x) \;=\; -\frac{x\,f''(k)}{f'(k)} \;-\; \frac{u_n^W(n;h,x)}{\alpha(h)\,f'(k)}. \tag{47}$$

The first part rewards households for the effect of their work on the import bill, which they ignore because they take the price as given. It is zero when nothing is shipped, and it is below one for every shipment, because $x \leq k$ and $-kf''(k)/f'(k) < 1$ by Assumption 1. The second part is zero at the static best response, where $u_n^W = 0$ by (46). The second part is positive when the government employs more than its static best response, which in the benchmark it does only to keep its knowledge in the high regime. It then subsidizes the knowledge that the extra work preserves, as does the learning subsidy of Proposition 3. Neither part depends on the motives of the East. The rate responds to the West's knowledge and to the size of the shipment, not to the motive behind them.

The margin above the threshold. Suppose a shipment is large enough that, at the static best response $n_H^{\mathrm{R}}(x)$, next period's knowledge would fall to χ or below. To stay in the high regime, the government must choose a higher employment level, one that places next period's knowledge strictly above χ, because Assumption 3 puts knowledge of exactly χ in the low regime.[7] Among these levels the government prefers the lowest, because every one of

[7] Putting χ in the high regime would not help. Both players want to be as close to χ as they can, from opposite sides: the East wants to carry the West's knowledge just into the low regime at the least cost, and the West wants to keep it just inside the high regime. A step must assign χ itself to one side, and whichever side it is, the other player's set is open. With χ in the high regime no export path would be the cheapest way to move the West into regime L, the East would have no optimal plan above the critical security weight, and Proposition 7 would lose its existence claim; we therefore put χ in the low regime and close the West's set with a margin.

them keeps the high regime and flow welfare falls as employment rises above $n_H^{\mathrm{R}}(x)$. But the set of these levels is open and has no lowest element, so the government has no best choice. We restrict its choice to a closed set.

Assumption 4 (Margin above the threshold)**.** *Fix a margin $\varepsilon \in \big(0, \delta(h_H - \chi)\big)$. The Western government does not induce an employment level that places next period's knowledge in the interval $(\chi, \chi + \varepsilon)$.*

The restriction is needed only because of the step function. With a smooth productivity schedule, more knowledge is worth more, so the government has a reason to prefer one level of knowledge to another, and no restriction is needed; Appendix E computes that case. The current welfare cost of the restriction is bounded. The employment levels that this assumption rules out lie within ε/λ below the lowest level it allows. Therefore, this restriction lowers flow welfare by less than ε/λ times the marginal disutility of work at that level. Call $\chi + \varepsilon$ the *defended level* and write

$$\underline{n}(h) \;\equiv\; \frac{\chi + \varepsilon - (1-\delta)h}{\lambda}$$

for the employment that places next period's knowledge at the defended level from state h. At $h = \chi + \varepsilon$ it equals $\delta(\chi + \varepsilon)/\lambda$, i.e., the employment that holds the defended level in place. When $\underline{n}(h) > 0$, the subsidy that implements it is (44) evaluated at $\underline{n}(h)$,

$$\tau^{\mathrm{F}}(h, x) \;=\; \frac{v'\big(\underline{n}(h)\big)}{\alpha_H\, f'\big(\alpha_H \underline{n}(h) + x\big)} - 1. \tag{48}$$

The bound $\varepsilon < \delta(h_H - \chi)$ ensures that from any state above χ the autarky employment $\varphi_H(0)$ places next period's knowledge at or above the defended level. The margin is held fixed throughout, and we suppress it in the notation.

Lemma 4 (The employment floor)**.** *Suppose Assumptions 3 and 4 hold and f is three times differentiable with $f''' \geq 0$. Let the West be in regime H, at a state $h > \chi$, and let it receive a shipment x.*

(i) Among the employment levels that place next period's knowledge at or above the defended level, the one with the highest flow welfare is

$$N^{\mathrm{F}}(h, x) \;=\; \max\big\{n_H^{\mathrm{R}}(x),\; \underline{n}(h)\big\}. \tag{49}$$

It is the static best response when that response keeps knowledge at or above the defended level, and the floor $\underline{n}(h)$ otherwise; in the second case, we say that the floor binds. In both cases, it is at least the competitive employment $\varphi_H(x)$.

(ii) At the defended level, $h = \chi + \varepsilon$, the floor binds if and only if $\lambda n_H^{\mathrm{R}}(x) < \delta(\chi + \varepsilon)$. When it binds, the subsidy rate $\tau^{\mathrm{F}}(\chi + \varepsilon, x)$ is strictly increasing in x.

(iii) For each regime i, every shipment x, and all $n' < n$, $u_i^W(n'; x) - u_i^W(n; x) < v(n) - v(n')$.

The proof is given in Appendix A.

By part (i), the government chooses its static best response unless that choice would let its knowledge fall below the defended level. By (47), the subsidy rate is then $-xf''(k)/f'(k)$ when the floor does not bind and $\tau^{\mathrm{F}}(h,x)$ when it binds. By part (ii), the floor binds only against large shipments, and against those the rate that holds it rises with the shipment. By part (iii), holding the floor costs the West at most the floor's disutility $v(\underline{n}(h))$ in current welfare, however large the shipment, and that number depends only on the technology and the margin. Section 6 reports the floor, the bound, and the shipments against which the floor binds.

A capped rate. The subsidy rate that holds the floor increases with the size of the shipment, by part (ii) of the lemma. If the rate cannot exceed a cap $\bar{\tau}$, a large enough flood carries the West's knowledge to χ or below whatever its government does, and an East whose security weight is large enough sends one. A capped subsidy then works like a tariff. It deters every East whose security weight is below a critical security weight $\psi^*_{\mathrm{cap}}(\bar{\tau})$, which rises with the cap, and no East above it.

5.3 Deterrence

We now let the East's planner and the West's government interact. The East chooses its shipments anticipating the West's subsidy response. The West responds to each shipment with its subsidy. The state is the West's knowledge $h \geq 0$. Write $i(h)$ for the West's regime at h: $i(h) = H$ if $h > \chi$ and $i(h) = L$ otherwise. The East's knowledge is not a state. Under Assumption 3 the East stays at or above h_H whatever it ships, by the argument of Lemma 2(i), and that argument does not involve how the West responds. Within each period, the East ships $x \geq 0$. The West observes x and sets its subsidy rate. By Lemma 3 the subsidy rate and the employment it induces determine each other, so we describe the West as choosing its employment n, and back out the subsidy rate from (44). The price clears at the West's marginal product. The West's knowledge then updates by (5).

An *export policy* is a function $X : \mathbb{R}_+ \to \mathbb{R}_+$ from the West's knowledge to the East's shipment: at state h the East ships $x = X(h)$. An *employment policy* is a function $N : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$ from the state and the shipment to the West's employment: at state h, after observing the shipment x, the West works $n = N(h,x)$. By Assumption 4, $N(h,x)$ never places the next period's knowledge in the band $(\chi, \chi+\varepsilon)$. Every other positive level is induced by a rate, by Lemma 3.

A pair of policies (X,N) induces the following objects. From any initial state h_0 the path of the game is

$$x_t = X(h_t), \qquad n_t = N(h_t, x_t), \qquad h_{t+1} = (1-\delta)h_t + \lambda n_t.$$

The price at date t is

$$q_t = f'\big(\alpha_{i(h_t)}\, n_t + x_t\big), \tag{50}$$

the West's marginal product at that date's employment and shipment. Both flow payoffs are evaluated at that price. The East's flow payoff from shipping x to a West that works n in regime i is

$$\hat{\Pi}_i(x,n) \;\equiv\; f\big(\alpha_H n^E(x) - x\big) - v\big(n^E(x)\big) + f'(\alpha_i n + x)\, x.$$

That is the payoff (34) with the import price at the employment level n. Note that $\hat{\Pi}_i(x, \varphi_i(x)) = \Pi_i(x)$. The hat marks the dependence on the West's employment. The West's flow payoff is $u_i^W(n;x)$ of (45).

Given N, the East's problem at state h is a dynamic program. The state is h, the control is x, the transition is $h' = (1-\delta)h + \lambda N(h,x)$, and the flow is $\hat{\Pi}_{i(h)}(x, N(h,x)) - \psi g_{i(h)}$. Write $V^E(h;N)$ for its value, the supremum over the shipment sequences of the discounted payoff from h. Given X, the West's problem at state h after shipment x is also a dynamic program. Each period, the West observes that period's shipment $X(h_t)$ and then chooses its employment. Write $V^W(h,x;X)$ for its value, the supremum over such employment plans of the discounted payoff from (h,x). Let $\mathcal{N}(h) \subseteq \mathbb{R}_+$ be the set of employment levels that the West may choose at state h. By Assumption 4, it is

$$\mathcal{N}(h) \;=\; \big\{n \geq 0 :\; (1-\delta)h + \lambda n \leq \chi\big\} \;\cup\; \big\{n \geq 0 :\; n \geq \underline{n}(h)\big\}.$$

The first set contains the levels that allow next period's knowledge to fall to χ or below; it is empty when $(1-\delta)h > \chi$. The second contains the levels that keep it at or above the defended level. The two value functions satisfy

$$V^E(h;N) = \sup_{x \geq 0}\Big\{\hat{\Pi}_{i(h)}\big(x, N(h,x)\big) - \psi g_{i(h)} + \beta\, V^E(h';N)\Big\}, \quad h' = (1-\delta)h + \lambda N(h,x), \tag{51}$$

$$V^W(h,x;X) = \sup_{n \in \mathcal{N}(h)}\Big\{u^W_{i(h)}(n;x) + \beta\, V^W\big(h', X(h');X\big)\Big\}, \quad h' = (1-\delta)h + \lambda n. \tag{52}$$

We write suprema because nothing guarantees, in general, that a best shipment or a best employment level exists. Conditions (i) and (ii) of Definition 1 below require the policies to attain these suprema at the states the equilibrium covers, and Proposition 10 shows that they do. The feasible set $\mathcal{N}$ in (52) is a primitive of the game. Appendix E computes the equilibrium defined below with a smooth productivity schedule, where $\mathcal{N}$ is the whole employment grid and no margin is needed. Given a pair of policies, call a set of states $S \subseteq \mathbb{R}_+$ *invariant* if the policies do not leave it: $h \in S$ implies $(1-\delta)h + \lambda N(h, X(h)) \in S$.

Definition 1 (Markov perfect equilibrium)**.** *A pair of policies (X, N) is a* Markov perfect equilibrium *on an invariant set S if the following three conditions hold.*

(i) The export policy is optimal for the East given the employment policy. *For every state $h \in S$, the shipment $X(h)$ attains the supremum in (51): the East's payoff from shipping $X(h)$ at h and following X thereafter, with the West responding by N, equals $V^E(h;N)$, and no shipment sequence from h yields more.*

(ii) The employment policy is optimal for the West given the export policy. *For every state $h \in S$ and every shipment $x \geq 0$, the employment $N(h,x)$ attains the supremum in (52) over $\mathcal{N}(h)$: the West's payoff from working $N(h,x)$ at (h,x) and following N thereafter, with the East shipping by X, equals $V^W(h,x;X)$, and no employment plan from (h,x) yields more.*

(iii) The price clears the West's import margin. *At every date the price is (50), the West's marginal product at the realized employment $n_t = N(h_t, x_t)$ and shipment $x_t = X(h_t)$, and the payoffs in (i) and (ii) are evaluated at that price: the East receives q_t per unit shipped, and the West pays q_t per unit received.*

Condition (ii) requires the West's employment to be optimal at every shipment x, not only at the shipment $X(h)$ the East makes in equilibrium. The East's problem in (i) takes as given the West's response $N(h,x)$ to every shipment, including shipments the East does not make. Condition (ii) ensures that each of these responses is one the West would choose if that shipment arrived. In an equilibrium on S, conditions (i) and (ii) are required only at states in S, but the value functions V^E and V^W in those conditions are defined at every state in $\mathbb{R}_+$. This matters because a player that departs from its policy at a state in S can move the West's knowledge outside S, and whether the departure pays then depends on the player's value at that state.

We now construct a pair of policies and show that it is an equilibrium on the states that play from h_H can reach, with or without a departure by either player. The construction has three parts: an employment policy for the West, an export policy for the East, and a condition that makes the West's policy optimal.

The West's policy is the rule of Lemma 4 above χ and the return to the defended level at or below it. Write N^{F} for the employment policy with

$$N^{\mathrm{F}}(h,x) \;=\; \max\bigl\{n_H^{\mathrm{R}}(x),\, \underline{n}(h)\bigr\} \quad \text{for } h > \chi, \qquad N^{\mathrm{F}}(h,x) \;=\; \underline{n}(h) \quad \text{for } h \leq \chi.$$

Under N^{F} the West's knowledge never falls to χ once it is above. From $h > \chi$, the next period's knowledge is at least $\chi + \varepsilon$, because the West works at least $\underline{n}(h)$. From $h \leq \chi$, it is exactly $\chi + \varepsilon$.

The East's policy is autarky above χ and its flow-optimal shipment at or below χ. At or below χ the West returns to the defended level by working $\underline{n}(h)$, so the East's shipment does not change the next period's state. The East's shipment there is then a static choice. Write

$$\tilde{x}(h) \;\in\; \arg\max_{x \geq 0} \hat{\Pi}_L\bigl(x, \underline{n}(h)\bigr)$$

for the shipment that maximizes the East's flow payoff against a West that works $\underline{n}(h)$, at the realized price. Write X^{c} for the export policy with

$$X^{\mathrm{c}}(h) = 0 \quad \text{for } h > \chi, \qquad X^{\mathrm{c}}(h) = \tilde{x}(h) \quad \text{for } h \leq \chi.$$

The condition compares two numbers. The first is the most the floor can cost the West in one period. By Lemma 4(iii), that is $v(\underline{n}(h))$, and $\underline{n}(h) \leq (\delta\chi + \varepsilon)/\lambda$ at every $h > \chi$, so

it is at most $v((\delta\chi+\varepsilon)/\lambda)$. The second is the least a crossing can cost the West. A West that lets its knowledge cross spends at least one period at or below χ. In that period, the East ships $\tilde{x}(h)$ and the West earns at most

$$\bar{u}_L^W \;\equiv\; \sup_{h\le\chi}\,\max_{n\ge 0}\, u_L^W\big(n;\tilde{x}(h)\big),$$

the best flow a trapped West can obtain against those shipments, whatever it works. After that period the best the West can do is return to the high steady state, whose value under autarky is

$$V_H^W \;\equiv\; \frac{u_H^W(0)}{1-\beta}.$$

Therefore, a crossing is worth at most $\bar{u}_L^W+\beta V_H^W$ to the West, compared to V_H^W for staying above χ. The difference is $u_H^W(0)-\bar{u}_L^W$, and is incurred one period later, so the least a crossing costs is $\beta[u_H^W(0)-\bar{u}_L^W]$. The condition is as follows.

$$v\left(\frac{\delta\chi+\varepsilon}{\lambda}\right) \;<\; \beta\left[u_H^W(0)-\bar{u}_L^W\right]. \tag{C}$$

It states that the most the floor can cost in one period is less than the least a crossing costs. Like Condition (D), it involves only static objects.

Proposition 10 (Deterrence)**.** *Suppose Assumptions 3 and 4 and Condition (C) hold, and f is three times differentiable with $f'''\ge 0$.*

(i) Given N^{F}, for every $\psi\ge 0$ and every $h>\chi$, the East's value is $V^E(h;N^{\mathrm{F}})=[\Pi_H(0)-\psi g_H]/(1-\beta)$, and $X^{\mathrm{c}}(h)=0$ attains the supremum in (51); at every $h\le\chi$, $X^{\mathrm{c}}(h)=\tilde{x}(h)$ attains it, regardless of the East's security weight. Along the path generated by $(X^{\mathrm{c}},N^{\mathrm{F}})$ from h_H, the West's knowledge stays at h_H and the subsidy is zero at every date.

(ii) Given X^{c}, for every $h>\chi$ and every $x\ge 0$, $N^{\mathrm{F}}(h,x)$ attains the supremum in (52). This holds for any employment policy that agrees with N^{F} at states $h>\chi$, whatever it prescribes at or below χ.

(iii) Suppose in addition that $v\big(\underline{n}((1-\delta)\chi)\big)<\beta[u_H^W(0)-\bar{u}_L^W]$. Then for every $h\in((1-\delta)\chi,\chi]$ and every $x\ge 0$, $N^{\mathrm{F}}(h,x)=\underline{n}(h)$ attains the supremum in (52) given X^{c}. The pair $(X^{\mathrm{c}},N^{\mathrm{F}})$ is then a Markov perfect equilibrium on $((1-\delta)\chi,\infty)$. That set contains h_H, the policies do not leave it, a single departure by the West from any state above χ leaves the state in it, and a departure by the East never takes the state to χ or below.

(iv) For every $\psi>\psi^*$ and every optimal plan P of the East against the competitive West, the West's value under $(X^{\mathrm{c}},N^{\mathrm{F}})$ from h_H exceeds its value under the competitive response against P by exactly the loss $L^W(P)$ of Section 4.3.

Proof of parts (i) and (iv). (i) Fix $h>\chi$. Under N^{F} the West is in regime H at every date, whatever the East ships. Its employment is at least $\varphi_H(x_t)$, so the price $f'(\alpha_H n_t+x_t)$

is at most $q_H(x_t)$, and the East's flow payoff is $\hat{\Pi}_H(x_t, n_t) \leq \Pi_H(x_t) \leq \Pi_H(0)$, the last by Assumption 3(iii). The security loss is g_H at every date. Summing, every shipment sequence is worth at most $[\Pi_H(0) - \psi g_H]/(1-\beta)$. Shipping nothing at every date attains it, because $\hat{\Pi}_H(0, n) = \Pi_H(0)$ for every n. Hence $V^E(h; N^{\mathrm{F}})$ equals that value and $x = 0$ attains the supremum in (51). At $h \leq \chi$ the West works $\underline{n}(h)$ whatever is shipped, and the next period's state is $\chi + \varepsilon$, where the East's value is the one just computed; the East's problem at h is therefore to maximize the flow $\hat{\Pi}_L(x, \underline{n}(h)) - \psi g_L$ over x, and $\tilde{x}(h)$ does so by definition. On the path from h_H: at $x = 0$ the static best response is competitive employment, $n_H^{\mathrm{R}}(0) = \varphi_H(0)$, which leaves knowledge at h_H, above the defended level because $\varepsilon < h_H - \chi$. So $N^{\mathrm{F}}(h_H, 0) = \varphi_H(0)$, knowledge stays at h_H, and the rate that induces it is zero.

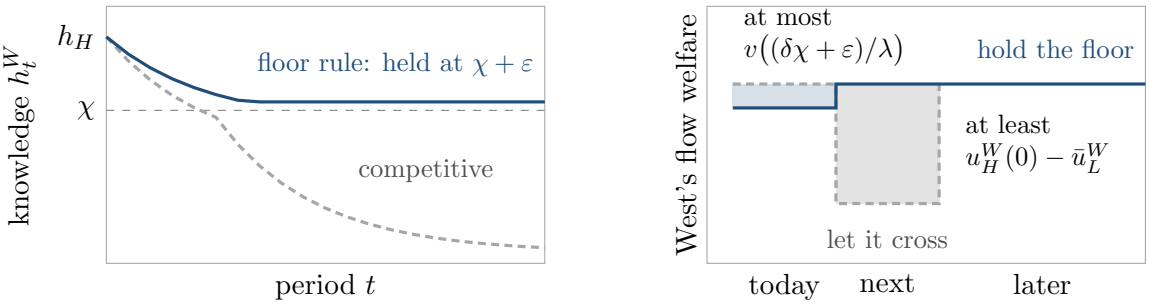
(iv) Under $(X^{\mathrm{c}}, N^{\mathrm{F}})$ the West's value from h_H is V_H^W by (i). Under the competitive response, its value against P is $V_H^W - L^W(P)$, by the definition of L^W in Section 4.3. The proofs of parts (ii) and (iii) are in Appendix A. □

The floor keeps the West's knowledge above χ against any shipment (Figure 7(a)). Because the security loss is a step, a West held at the defended level has the same capability as a West at h_H. Therefore, a flood that does not cross has no security gain for the East. It also lowers the commercial payoff of the East. Each shipped unit costs flow payoff and lowers the price the East receives. The East therefore ships nothing, and the West pays no subsidy. Condition (C) makes the West's response optimal. Holding the floor for one period costs at most $v((\delta\chi + \varepsilon)/\lambda)$, while letting knowledge cross costs at least $\beta[u_H^W(0) - \bar{u}_L^W]$ (Figure 7(b)). Part (iii) extends the equilibrium to the states just below χ from which the West returns in one period, and part (iv) says that holding the floor raises the West's value by the loss $L^W(P)$ that predation would have caused. None of this depends on ψ. The West compares two of its own flows, and the East compares shipments that gain nothing relative to autarky. Against a West that holds the floor, the East ships nothing, regardless of how large its security weight ψ is.

The West's rule departs from the competitive response to a shipment for two reasons. The first is the price. Against any shipment, the West works more than its competitive employment level, because more work lowers the price of every unit it imports. The employment chosen for this motive falls as the shipment grows. Therefore, by itself it stops only shipments below a size that Section 6 reports. The second is knowledge. When a shipment would carry knowledge below the defended level, the West chooses the floor, whatever rate it takes. The floor holds deterrence at all security weights. Part (i) only needs two facts about the West: its knowledge never reaches χ, and it works at least its competitive job. The rule that sets employment to the larger of the competitive employment and the floor satisfies both, without any import-price motive. Given the floor, the import-price motive changes how much the West works against a shipment, not whether predation succeeds. Appendix D proves the whole result for that rule, in an economy where the East's shipments are free and there is no price to move.

This result has three limits.

First, the equilibrium covers only the states near the threshold. The pair is an equilibrium



(a) A large shipment every period

(b) Hold the floor or let knowledge cross

Figure 7: The floor (Lemma 4 and Proposition 10). Panel (a): the West's knowledge when the East ships the same large quantity every period, a shipment it does not make in equilibrium, under the competitive response and under the rule N^{F}; the margin ε is drawn large for visibility. Panel (b): the West's flow welfare after a shipment at $h > \chi$ if it holds the floor and if it lets its knowledge cross, with the bounds behind Condition (C). Panel (a) is drawn at the parameters of Section 6; panel (b) is schematic.

on $((1-\delta)\chi, \infty)$, the states a single departure can reach. We do not show results on the whole state space. Far below χ, returning to the defended level in one period is costly, and at our parameters the West may prefer to climb part of the way first. This does not weaken deterrence from the high steady state. From h_H a single concession cannot bring knowledge to χ, because $(1-\delta)h_H > \chi$ at our parameters, so deterrence there rests on parts (i) and (ii). Part (ii) holds whatever the West does below χ, because the bound assumes that a West that crosses earns the best flow available there and returns immediately to the high steady state.

Second, we do not characterize equilibria in which the East pays the West to stay trapped. A West that can climb back does so (Appendix B), so an East that wanted to keep it below χ would have to ship enough, every period, so that the West's best flow below χ matches its flow at the high steady state. That payment is large, and it is profitable for the East only at a security weight well above the critical one (Section 6).

Third, the step in the security loss makes the deterrence exact. A West at the defended level is worth to the East what a West at h_H is worth, so a flood that does not reach χ gains the East nothing. With a smooth security loss, every unit of Western knowledge has some security value, and the East might gain from shipments that lower Western knowledge without trapping the West. Appendix E computes that game and finds no such flood. At every steepness and every security weight at which it finds an equilibrium, the East ships nothing, the West stays at its high steady state, and the subsidy on the path is the West's domestic correction. The step function is needed for the proof. However, the computations suggest that it is not needed for the result.

At our parameters, Condition (C) holds by a wide margin, and the condition of part (iii) holds by a narrow one, and only when ε is small. A West that holds its knowledge up against a flood sustained forever is better off than under autarky, because its own employment holds down the price it pays. Section 6 reports these numbers. Appendix D proves the same result

when the East's shipments are a gift, where the West's reason to defend itself depends on its patience.

6 A Quantitative Illustration

The results so far are qualitative. This section computes the objects discussed at one parameterization, in the order of Sections 3 to 5. The model period is assumed to be two years. The parameters are not estimated. Those that determine the threshold are chosen to give multiple steady states, and we report how far the objects move when they change.

6.1 Parameters

We adopt the canonical forms $v(n) = n^{1+\eta}/(1+\eta)$ and $f(k) = k^{\theta}$ together with the logistic α of (3). Table 1 reports parameter values and our criteria for choosing them, and lists the objects of Sections 3 to 5 that they imply.

The learning intensity λ is a normalization. It fixes the scale of the knowledge stock through $h_H^* = (\lambda/\delta)\, n_H^*$, and rescaling it rescales h, χ, and γ together without changing any observable. The parameter θ is the elasticity of f. It sets the price elasticity of the demand for manufactured goods. Assumption 1 requires only $\theta \in (0, 1)$.

The last row of the table and the vertical axis of Figure 9 read a model period as two years. The convention converts the critical duration $\bar{T}$ of Proposition 2 into years. At $\delta = 0.20$ half of the knowledge stock survives six years. Plant-level estimates of organizational forgetting are faster. Benkard (2000) finds that about 61% of a firm's production experience survives a year, a half-life of about 1.4 years. We use the slower rate because the stock here is a sector's collective knowledge rather than one plant's. The parameters δ, λ, and χ enter the steady states and the import threshold only through $\delta\chi/\lambda$ and $\gamma\lambda/\delta$. In employment units $\tilde{n} = \delta h/\lambda$ the steady-state condition reads $\tilde{n} = \phi(\lambda\tilde{n}/\delta; m)$, and α depends on $\tilde{n}$ only through these two combinations. A faster forgetting rate with the S-curve rescaled to hold them fixed leaves the steady states in employment units, the import threshold, and its penetration where they are and shortens every duration; at Benkard's rate the twenty years reported below become six.

6.2 The Mechanism

At these parameters, the hypotheses of Proposition 1 hold. The map has three steady states. Moreover, Assumption 2 holds, and the import threshold of Proposition 1(ii) is $\bar{m} = 3.67$.

Figure 8 measures the hysteresis of Proposition 2. A boom of size $m' = 4.50$, above the import threshold, has a critical duration of 25 periods. A boom of 24 periods ends 0.035 above the threshold, and the economy climbs back. A boom of 25 ends 0.28 below it, and the economy converges to h_L^*. Panel (b) shows $\bar{T}(m')$ for various boom sizes. Size and duration substitute for one another along a downward-sloping frontier.

Parameter	Value	Basis
δ	0.20	Forgetting per period. With χ and γ fixed in employment units, the steady states and the import threshold do not depend on it; durations do. Slower than the plant-level evidence; see the text
$\alpha_L,\ \alpha_H$	0.30, 3.00	Productivity plateaus. The ratio of ten is chosen to give a clear example with multiple steady states and is not estimated; Irwin and Klenow (1994) estimate a learning rate of about 20% per doubling of cumulative output in DRAMs, with no forgetting
γ	4.20	Steepness of the S-curve; not identified, chosen with χ to give multiple steady states
χ	4.17	Inflection of the S-curve; not identified, chosen with γ to give multiple steady states
λ	1.00	Units normalization; rescaling changes no observable
θ	0.65	Elasticity of f; the demand for manufactured goods has price elasticity $1/(1-\theta) \approx 2.9$
η	1.00	Frisch elasticity of labor supply $1/\eta = 1$
β	0.96	Discount factor; a two-year period at roughly a 2% annual real rate. Enters only the problems of the planners and of the West's government; the households' labor choice is static
Implied objects at the baseline flow $m_0 = 0.05$		
$h_L^*,\ h_M^*,\ h_H^*$	1.849, 4.029, 6.145	Steady states (Proposition 1); slopes $G_h = 0.80,\ 1.63,\ 0.80$
$\bar{m}$	3.67	Import threshold: above it only the low steady state remains
$\bar{T}(4.50)$	25 periods	Critical duration of a boom of 4.50 (Proposition 2); 125 periods just above the import threshold, 10 at $m' = 8.07$
Import threshold in penetration	55%	Import penetration above which the high steady state ceases to exist; independent of the period length
Trap within 20 years	75%	Penetration that traps the West within ten periods, read as two years each

Table 1: Parameters of the numerical example and the objects they imply. The S-curve parameters γ and χ and the plateau ratio α_H/α_L are chosen to give multiple steady states and are not estimated; λ is a units normalization; θ and η are chosen values; the role of δ is discussed in the text. Every implied object is computed from the model at these values, and the last row reads a period as two years.

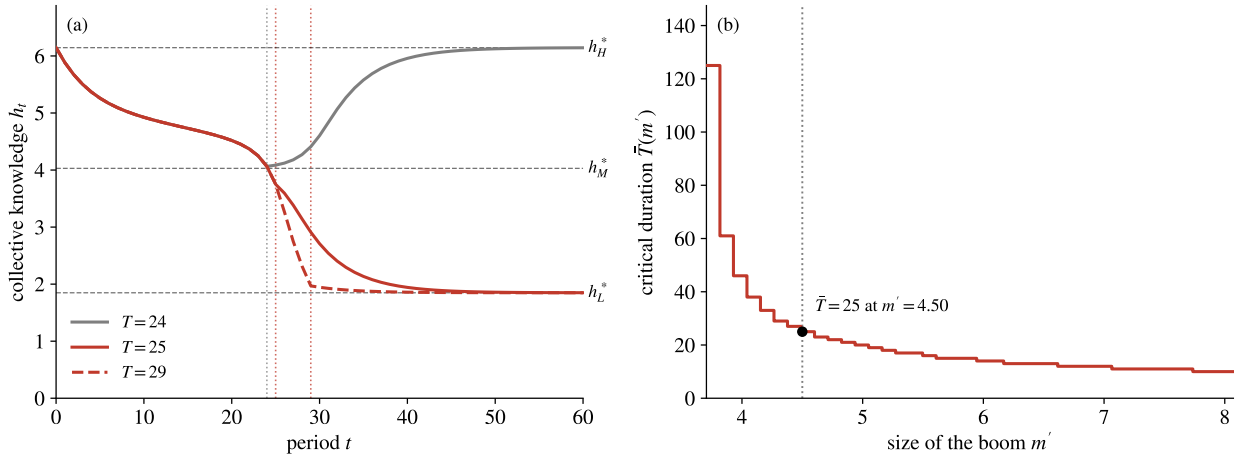


Figure 8: A temporary resource boom with a permanent consequence. Panel (a): knowledge paths when the flow of manufactured goods rises from $m_0 = 0.05$ to $m' = 4.50$ for T periods and then reverts. A boom of $T = 24$ periods (gray) ends just above the threshold h_M^* and the economy recovers to h_H^*; booms of $T = 25$ and $T = 29$ periods (red) end below it and the economy converges to h_L^*. Dotted verticals mark the end of each boom. Panel (b): the critical duration $\bar{T}(m')$ of (13) as a function of the size of the boom, which diverges as m' approaches the import threshold $\bar{m} = 3.67$ from above.

We measure the size of a surge by the West's *import penetration*, $x/(\alpha(h_H^*)\,\phi(h_H^*;x)+x)$, imports as a share of the total use of manufactured goods, and Figure 9 plots the critical duration of Proposition 2 against it in years. The import threshold sits at a penetration of about 55%. Below it, the high steady state survives a surplus of any duration. Just above it, the West takes decades to fall below the threshold, because near the import threshold the map is close to the diagonal and knowledge falls slowly. A penetration of about 75% or more traps the West within twenty years. Where the import threshold sits is a property of the S-curve in employment units and of the plateau ratio, neither of which is measured. Moving γ between 2.6 and 16 moves it between 43% and 71%, moving χ between 3.5 and 5.0 moves it between 74% and 20%, and moving α_H/α_L between 6.7 and 13.3 moves it between 12% and 73%. Keeping χ and γ fixed in knowledge units instead and moving δ across a half-life of five to ten years moves it between 12% and 91%.

6.3 Predation

Proposition 6 gives the condition under which commerce alone traps a customer. The cheapest undiscounted crossing costs $C_0 = 39.3$, while the discounted value of the East's post-crossing commercial gain is $\beta\Delta\Pi/(1-\beta) = 5.07$. Condition (D) holds by a factor of seven, so commercial predation loses money at $\psi = 0$ and the critical security weight is strictly positive. The bound of Proposition 6(iii) puts $\psi^*\Delta g$ above 1.43, close to the computed 1.66. In terms of patience, the condition holds for every discount factor below $C_0/(C_0+\Delta\Pi) = 0.99465$, and the cutoff of the proposition is $\hat{\beta} = 0.99471$. Reading a period as two years, a purely commercial East therefore traps the West only if it discounts the future at less than about 0.27%

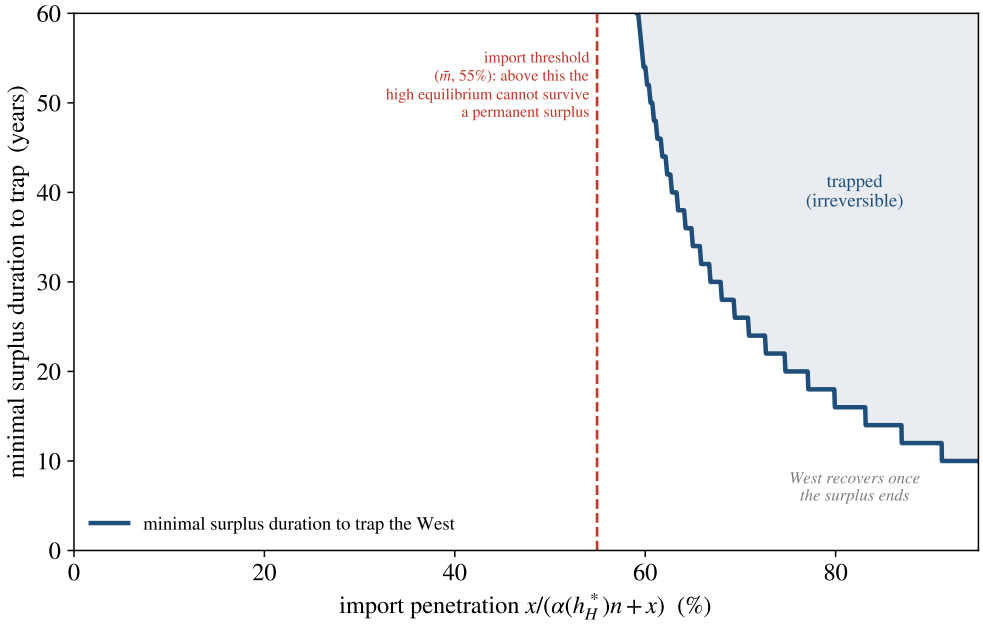


Figure 9: Critical duration of an import surge: for each level of Western import penetration (horizontal), the minimal number of years a surplus must be sustained to push the West below the threshold and into the low equilibrium, which the West does not leave on its own (vertical); the shaded region is trapped; below the curve the West recovers once the surplus ends. The import threshold (dashed) marks the penetration above which the high equilibrium ceases to exist.

a year. The two cutoffs differ by 5×10^{-5}. Both lie close to one, where the compounding within the active phase that the condition ignores is small.

The critical security weight of Proposition 5 is $\psi^* \Delta g = 1.66$, about one period of the East's autarky flow payoff $\Pi_H(0) = 1.58$. We report it as a numerical approximation because the infimum in (38) is a limit that no plan achieves. The plan that is optimal just above it is worth -16.9 at $\psi = 0$, about eleven periods of that flow. The crossing date falls as the weight rises: 22 periods at $\psi \Delta g = 1.67$, 15 at 2.0, and 10 at 4.0. The magnitudes in Table 2 are at $\psi \Delta g = 1.67$, just above the critical security weight, where the active exports rise from 0.4 to 14.3. The benchmark's import threshold is $\bar{m}_\infty = 8.85$, against $\bar{m} = 3.67$ in the smooth economy.

Figure 10 shows the optimal plan in the smooth economy, with $\mathcal{D}(z) = z^2/2$, so that the security gap between the two competitive steady states is $\Delta g = 4.45$, and with the weight at $\psi \Delta g = 1.66$. The export rises from 3.4 to a peak of 7.6 over eight periods, while the West's knowledge is still on its upper plateau. It then falls over the next six periods as the West's productivity and employment fall in the steep part of the S-curve. The West crosses the threshold in period 14, and then the export falls to its commercial level against a trapped West, about 0.9. The benchmark's profile is monotone and peaks at the crossing. The smooth economy peaks before it.

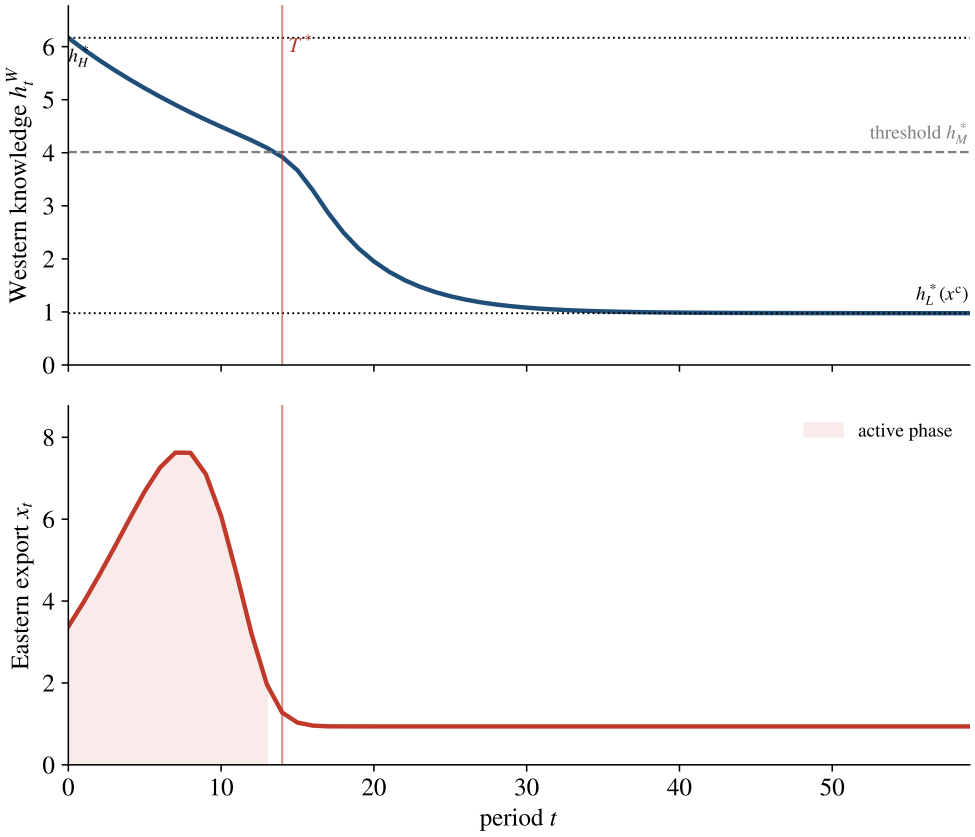


Figure 10: Optimal Eastern predation in the smooth economy, at the parameters of Section 6 with $\mathcal{D}(z) = z^2/2$ and $\psi\Delta g = 1.66$; the East is held on its upper plateau and the West is competitive. Top: Western knowledge h_t^W falls from h_H^* across the threshold h_M^* at $T^* = 14$ and converges to the low steady state under the East's commercial inflow, $h_L^*(x^c)$. Bottom: the optimal export rises to its peak of 7.6 while the West is on its plateau, falls as the West enters the steep part of the S-curve, and drops to its commercial level after the crossing.

6.4 Welfare

The West's lifetime loss is 9.8 at $\psi\Delta g = 1.67$, about six periods of its autarky flow of 1.58. The loss measured at date zero falls toward zero as the weight falls to the critical security weight, because the optimal crossing date diverges there and the trap starts later. At $\psi\Delta g = 1.6595$ the optimal plan crosses at 29 and costs the West 7.3. At 1.659091, nearer the critical security weight, which is 1.659 to three figures, it crosses at 40 and costs 4.7. Measured at the crossing date, the loss barely moves. $\beta^{-T_P} L^W(P)$ lies between 23.98 and 24.14 at every crossing date we compute, falling monotonically in the crossing date. The reason is that the loss is the perpetuity of the West's post-crossing flow gap $\Delta u = 1.21$, which is 30.3, net of the gain it makes while the shipments last, and that gain, read backward from the crossing, varies little with the horizon.

Condition (N) of Proposition 8 holds and it is not tight. Three times the West's per-

period loss in regime L is 3.64, against $\psi^*\Delta g + \Delta\Pi = 1.66 + 0.21 = 1.87$. The proposition therefore gives the negative-sum conclusion on every optimal plan for $\psi\Delta g$ up to $\hat{\psi}\Delta g = 2.25$. A direct comparison extends the interval. Comparing the East's gain with the West's loss on the plan that is optimal at each weight, the gain first exceeds the loss at $\psi\Delta g = 2.72$, and the comparison holds at every weight at which two crossing dates are both optimal, the largest of which is 2.58. Therefore, the proposition proves the conclusion up to 2.25, and the direct comparison extends it to 2.72. Table 2 presents these magnitudes together with the rest of the predation numbers.

Object	Symbol	Value	Result or interpretation
Benchmark import threshold	$\bar{m}_\infty$	8.85	Assumption 3(iv); 3.67 in the smooth economy
Cheapest undiscounted crossing; commercial perpetuity	C_0; $\beta\Delta\Pi/(1-\beta)$	39.3; 5.07	Condition (D), factor seven
Lower bound on the critical security weight	$(1-\beta)C_0/\beta - \Delta\Pi$	1.43	Proposition 6(iii)
Discount factor above which commerce alone traps the West	$\hat{\beta}$	0.99471	Proposition 6; lower bound $C_0/(C_0+\Delta\Pi) = 0.99465$; about 0.27% a year at a two-year period
Value at $\psi = 0$ of the plan optimal at $\psi\Delta g = 1.67$	$\Delta(P;0)$	-16.9	$\psi^* > 0$; about eleven periods of $\Pi_H(0) = 1.58$
Critical security weight	$\psi^*\Delta g$	1.66	Proposition 5; about one period of $\Pi_H(0)$
Crossing date; active exports	T^*; $x_0 \to x_{T^*-1}$	22; $0.4 \to 14.3$	Proposition 7, at $\psi\Delta g = 1.67$
West's autarky flow; its loss on the plan	$u_H^W(0)$; L^W	1.58; 9.8	Proposition 8, at $\psi\Delta g = 1.67$; about six periods of autarky flow
West's per-period loss in regime L, times three; East's per-period gain at ψ^*	$3\Delta u$; $\psi^*\Delta g + \Delta\Pi$	3.64; 1.87	Condition (N), factor about two; $\Delta u = 1.21$
Upper end of the negative-sum interval: proved; computed	$\psi\Delta g$	2.25; 2.72	Both intervals start at $\psi^*\Delta g$. The first is $\hat{\psi}\Delta g$ of Proposition 8, the second a direct optimal-plan envelope calculation, ties included

Table 2: Magnitudes of the benchmark at the parameters of Table 1: predation. Security gains are in units of the security gap Δg; objects that depend on the weight are reported at $\psi\Delta g = 1.67$, just above the critical security weight. The magnitudes are illustrative, as the parameters are.

6.5 Defense

The autarky price of the manufactured good is $q_H(0) = 0.41$, which is the scale against which a tariff should be read. A tariff of 0.1 raises the critical security weight from 1.66 to 2.27. The tariff that deters a security gain of 2.0 per period is 0.054 and the one that deters 2.5

is 0.14, a third of the autarky price. Under a tariff of 1 the plan that is optimal just above the critical security weight under that tariff crosses after 12 periods rather than 22.

With a margin $\varepsilon = 0.01$, the floor at the defended level is 0.84, below the competitive autarky employment of 1.23. Holding this floor costs the West at most 0.35 per period whatever the shipment. The bound 0.36 that Condition (C) uses is the same quantity taken over every state above χ rather than at the defended level alone. Against the shipments of the optimal plan, the floor does not bind, and the rate is set by the import-price motive. The West works 1.01 against a shipment of 10 and 0.93 against the peak shipment of 14.3. These are well above the floor, and the subsidy rate against 14.3 is 0.29. All of it is the import-price motive. At the defended level, the floor binds only against shipments that are above 22.1. By itself, the import-price motive keeps the West out of the trap against any constant shipment below 22.3, where the competitive West's limit is 8.85, and a West that used the subsidy for the price alone would raise the critical security weight from 1.66 to 13.5. Following N^{F} against a flood of 14.3 forever is worth 61 to the West, against 39.5 at its high steady state under autarky. The West's knowledge settles at 4.67, above the defended level.

The caps of 0.15, 0.30, 0.50, and 1.00 raise the critical security weight to 5.17, 12.1, 15.7, and 27.9. The cap of 0.15 gives the protection of a tariff of about 0.69 and the cap of 0.50 that of a tariff of about 3.1. These weights are high because under the cap the West still works more than its households would, to lower the import price. A trapped West cannot climb back under any cap below 1.63, the rate that holds a stock of χ in place in regime L in autarky.

Condition (C) holds by a factor of three, 0.36 against 1.16. The condition in part (iii) of Proposition 10 holds by 0.02, and it holds for every margin up to $\bar{\varepsilon} \approx 0.02$, which is why the margin is small. Below χ the one-period return is the West's best reply down to about 42 percent of χ, and not below. Paying a West to stay trapped would take permanent shipments of about 10.1 per period, costing the East about 4.4 per period, which pays only if $\psi\Delta g$ exceeds about 4.20, two and a half times the critical security weight.

In the smooth economy, computed in Appendix E, the pair in which the East ships nothing and the West defends its knowledge with the subsidy is an equilibrium at every steepness and every weight we compute, up to $\psi\Delta g = 4$, which is 2.4 times the benchmark's critical security weight, with the subsidy on the path equal to the West's domestic correction to within one step of the employment grid. Appendix E reports the West's response off that path, which rises steeply as its knowledge falls into the steep region, and the present value the East loses on the shipment paths we computed.

7 Discussion

7.1 Historical Precedents

Treating industrial capability as a strategic, contestable asset is not new. The decline of India's cotton-textile industry under British rule is a historical episode that motivates the analysis. India produced about a quarter of world manufacturing output in 1750 and a few

Object	Symbol	Value	Result or interpretation
Tariffs (Section 5.1)			
Autarky price	$q_H(0)$	0.41	scale of ξ
Critical security weight at $\xi = 0.1$; crossing date at $\xi = 1$	$\psi^*(0.1)\Delta g$; T^*	2.27; 12	Proposition 9(ii)
Tariff that deters $\psi\Delta g = 2.0$; 2.5	$\bar{\xi}(\psi)$	0.054; 0.14	Proposition 9(iv)
The subsidy (Section 5.2)			
Floor at the defended level; competitive autarky employment	$\delta(\chi+\varepsilon)/\lambda$; $\varphi_H(0)$	0.84; 1.23	floor rule (49)
Employment against 10; 14.3	$n_H^{\mathrm{R}}(x)$	1.01; 0.93	(46)
Rate against 14.3, all import-price motive	$-xf''/f'$	0.29	(47)
Shipment above which the floor binds at the defended level; the import-price motive's own limit		22.1; 22.3	Lemma 4(ii)
Critical security weight against a West that uses only the import-price motive	$\psi\Delta g$	13.5	Section 5.3
Critical security weight against caps 0.15; 0.50	$\psi^*_{\mathrm{cap}}(\bar{\tau})\Delta g$	5.17; 15.7	capped-rate remark; tariff equivalents 0.69; 3.1
Deterrence (Section 5.3)			
Floor bound; continuation loss	$v((\delta\chi+\varepsilon)/\lambda)$; $\beta[u_H^W(0)-\bar{u}_L^W]$	0.36; 1.16	Condition (C), factor three
One-step return; its disutility	$\underline{n}((1-\delta)\chi)$; $v(\cdot)$	1.51; 1.14	Proposition 10(iii), margin 0.02
Value under N^{F} against 14.3 forever; autarky value	V^W; V_H^W	61; 39.5	Section 5.3
Shipment that keeps a trapped West low; cost per period; break-even weight	x^b; cost; $\Pi_H(0)-\Pi_L(x^b)$	10.1; 4.4; 4.20	Section 5.3, second limit

Table 3: Magnitudes of the benchmark at the parameters of Table 1: the counterpolicy. Security gains are in units of Δg, and the margin is $\varepsilon = 0.01$. The magnitudes are illustrative, as the parameters are.

percent by 1900 (Bairoch, 1982), as mechanized British cloth displaced Indian production. The decline took place under colonial rule and alongside mechanization, so trade was not the only channel, and we do not claim that it was. The mirror strategy, guarding one's own technique, also recurs. China held a near-monopoly on silk for roughly two millennia and, by repute, forbade the export of silkworms (Liu, 2010), and Britain prohibited the export of textile machinery and the emigration of skilled artisans for more than half a century (Jeremy, 1977). The model has no instrument of this kind, and we do not analyze export controls.

7.2 Western Investment in China

A common account of China's rise gives the leading role to Western investment. Western firms built plants in China and brought their technology with them. The mechanism of the model applies to this account, although the model does not formalize it. Net capital flows and gross direct investment play different roles in this account. The net flow of capital ran from China to the West. China ran trade surpluses and accumulated claims on the United States, part of the pattern that Prasad et al. (2007) describe as capital flowing uphill, from poorer to richer countries. The East's reserve position b^E in Section 4 is a stylized counterpart of these claims. The gross flow of direct investment ran from the West to China and bears on how China industrialized, which the benchmark takes as given in Assumption 3(ii). China also made access to its market conditional on transfer of technology (Holmes et al., 2015). In the model, knowledge accumulates where production takes place, whoever owns the plant. A Western firm that moves production to China therefore moves the learning with it, and, like the households of Section 3, it does not take into account the knowledge the West loses. The move lowers Western employment and raises Eastern employment, which is what an import surge does, except that the choice is made by Western firms and not by the East's government.

7.3 Contemporary Industrial Policy

Since about 2018 the United States, the European Union, and China have run concurrent large-scale subsidies in advanced manufacturing. The model identifies different conditions for deterrence through tariffs and employment subsidies. A tariff below the prohibitive level does not deter an exporter whose security weight exceeds the critical security weight at that tariff, and the importer does not observe that weight. In the benchmark, an uncapped employment subsidy set as a rule that keeps employment at or above the floor prevents the loss of industrial knowledge at every security weight, provided Condition (C) makes the rule credible. The subsidy paid in equilibrium is zero because the exporter anticipates the rule and ships nothing. What deters is the rule's response to a flood. In the smooth economy of Appendix E, the subsidy on the path is the domestic learning correction, and the exporter ships nothing in every case we compute. A prohibitive tariff also prevents deindustrialization and has no static cost in the benchmark, so ranking the two instruments requires static gains from trade between industrialized countries.

7.4 Limitations

The benchmark gives two industrialized countries no static gain from trade. A prohibitive tariff then protects as well as the floor does, and it needs neither the credibility condition nor knowledge of where the threshold lies. Ranking the two instruments requires static gains from trade, which we leave open. The policy analysis also restricts the Western government. It bars the government from placing its knowledge in a small band just above the threshold, finances the subsidy lump sum, lets the rate condition on the realized shipment, and leaves the rate uncapped. A capped subsidy deters only security weights below a level that rises with the cap (Section 5.2). The strategic results are proved only in the sharp benchmark, and they characterize play only near the threshold, leaving open the equilibria of a West that has been trapped and is being paid to stay there. The smooth economy is treated numerically in Appendix E, where the band is not needed. Finally, the parameters of Section 6 are chosen for illustration and are not estimated. An empirical assessment would require estimates of learning and forgetting rates and evidence on sectoral employment, imports, and productive capacity, such as the joint path of Western net imports of semiconductors and Western fabrication capacity.

The monetary side of the model is a separate direction. It is spare, and a fuller treatment of the bond, its return, and the sustainability of the issuer's debt would connect the mechanism more tightly to the literature on exorbitant privilege.

8 Conclusion

When industrial knowledge accumulates through learning by doing and the feedback between employment and productivity is strong enough, the competitive economy has two stable steady states separated by an unstable one, the threshold. An inflow of manufactured goods lowers the importer's industrial employment, erodes its knowledge, and, if it is large and persistent enough, moves the importer's knowledge below the threshold, so the loss persists after the inflow ends. In a benchmark in which the return to knowledge is a step, we prove four things. A security-motivated exporter produces such an inflow once its security weight passes a critical value. An exporter with no security motive does the same if it is patient enough and not otherwise. A tariff protects the importer only if it exceeds a level that depends on a motive the importer cannot observe, although a prohibitive tariff protects against every motive and, in the benchmark, has no static cost. And a labor subsidy, set as a rule that keeps employment at or above the floor against any shipment, deters the flood without being paid, provided a condition on the primitives makes the rule credible and the rate is uncapped. The mechanism also applies to the exporter. An exporter that starts below its own threshold and runs a large enough surplus for long enough ends at its high steady state (Appendix C).

A Proofs

Proofs for Section 3

Proof of Proposition 1. By Lemma 1, $D_m(h;m) = \lambda\phi_m(h;m) < 0$, so an increase in imports shifts $D(\cdot;m)$ downward pointwise.

For every $m \in [0, m_1]$, $D(0;m) = \lambda\phi(0;m) > 0$. Moreover, since $\alpha_L \le \alpha(h) \le \alpha_H$ and f' is decreasing, for $n \ge 1$ we have $\alpha(h)f'(\alpha(h)n + m) \le \alpha_H f'(\alpha_L)$. Because $v'(n) \to \infty$, the solution $\phi(h;m)$ remains bounded as $h \to \infty$. It follows that $D(h;m) = \lambda\phi(h;m) - \delta h \to -\infty$ as $h \to \infty$.

Define the values of D at its local minimum and local maximum,

$$L(m) \equiv D(h_1(m);m) \qquad \text{and} \qquad M(m) \equiv D(h_2(m);m).$$

Since $D_h(h_i(m);m) = 0$, differentiation gives $L'(m) = D_m(h_1(m);m) < 0$ and $M'(m) = D_m(h_2(m);m) < 0$.

Because $L(0) < 0$, we have $L(m) < 0$ for every $m \in [0, m_1]$. Moreover, $M(0) > 0$ and $M(m_1) < 0$. Since M is continuous and strictly decreasing, there exists a unique $\bar{m} \in (0, m_1)$ such that $M(\bar{m}) = 0$.

For $m < \bar{m}$, the local minimum lies below zero and the local maximum lies above zero. Since $D(0;m) > 0$, $D(h;m) \to -\infty$ as $h \to \infty$, and Assumption 2 implies that D is decreasing, then increasing, and then decreasing, $D(\cdot;m)$ has exactly one zero in each of the intervals

$$(0, h_1(m)), \qquad (h_1(m), h_2(m)), \qquad (h_2(m), \infty).$$

Panel (a) of Figure 1 illustrates this configuration.

Hence, the economy has exactly three distinct steady states. At $m = \bar{m}$,

$$D(h_2(\bar{m});\bar{m}) = 0 \qquad \text{and} \qquad D_h(h_2(\bar{m});\bar{m}) = 0.$$

Because

$$D_{hh}(h_2(\bar{m});\bar{m}) < 0 \qquad \text{and} \qquad D_m(h_2(\bar{m});\bar{m}) < 0,$$

the middle and high steady states coalesce in a saddle-node bifurcation. Since $L(\bar{m}) < 0$ and $D(0;\bar{m}) > 0$, there is exactly one additional zero in $(0, h_1(\bar{m}))$. Thus, the economy has exactly two distinct steady states at $m = \bar{m}$.

For $\bar{m} < m \le m_1$, both the local minimum and the local maximum lie below zero. Since $D(0;m) > 0$, D crosses zero exactly once in $(0, h_1(m))$ and remains negative thereafter. Panel (b) of Figure 1 illustrates this configuration.

Finally, the low and high steady states are downward crossings of $D(\cdot;m)$, so $D_h < 0$ and they are locally asymptotically stable. The middle steady state is an upward crossing, so $D_h > 0$ and it is unstable. For $\bar{m} < m \le m_1$, the surviving low-level steady state is likewise a downward crossing and is locally asymptotically stable.

It remains to prove part (iv). For $m \neq \bar{m}$ each steady state is a simple zero of $D(\cdot; m)$, so $D_h(h^*(m); m) \neq 0$ and the implicit function theorem applies to the identity $D(h^*(m); m) = 0$, giving

$$\frac{dh^*}{dm} = -\frac{D_m(h^*(m); m)}{D_h(h^*(m); m)}.$$

By Lemma 1, $D_m = \lambda\phi_m < 0$, so the numerator $-D_m$ is positive at every steady state. At the low and high steady states the crossing is downward, $D_h < 0$, and therefore $dh_L^*/dm < 0$ and $dh_H^*/dm < 0$. At the middle steady state the crossing is upward, $D_h > 0$, and therefore $dh_M^*/dm > 0$. The same argument applies to the unique low-level steady state when $\bar{m} < m \leq m_1$. □

Proof of Proposition 2. (i) Since $m' > \bar{m}$, Proposition 1(iii) leaves $h_L^*(m')$ as the unique fixed point of $G(\cdot; m')$, and $D(h; m') < 0$ for every $h > h_L^*(m')$. Because $h_0 = h_H^*(m_0) > h_L^*(m')$, the sequence $h_{t+1} = G(h_t; m')$ satisfies $h_{t+1} < h_t$ as long as the flow is held at m'; since $G(\cdot; m')$ is increasing, the sequence stays above $h_L^*(m')$. It is therefore strictly decreasing on $t = 0, \ldots, T$, and if the flow were never withdrawn the sequence would be monotone and bounded below, hence convergent, with limit the unique fixed point $h_L^*(m')$. For finite T the boom path is the first $T + 1$ terms of that sequence and stops short of the limit.

(ii) By Proposition 1(iv), h_L^* is strictly decreasing in m, so $h_L^*(m') < h_L^*(m_0) < h_M^*(m_0)$. The boom path therefore falls strictly below the threshold in finite time and $\bar{T}(m')$ is well-defined. Once the dividend stops the law of motion is $G(\cdot; m_0)$, whose basins of attraction are $[0, h_M^*(m_0))$ for $h_L^*(m_0)$ and $(h_M^*(m_0), \infty)$ for $h_H^*(m_0)$. Since h_T is strictly decreasing in T by part (i) and $\bar{T}(m')$ is the first date at which $h_T < h_M^*(m_0)$, we have $h_T \geq h_M^*(m_0)$ for $T < \bar{T}(m')$, with strict inequality for $T \leq \bar{T}(m') - 2$, and $h_T < h_M^*(m_0)$ for $T \geq \bar{T}(m')$. A post-boom path that starts strictly above the threshold converges to $h_H^*(m_0)$, one that starts strictly below converges to $h_L^*(m_0)$, and one that starts at $h_M^*(m_0)$ stays there, since $G(h_M^*(m_0); m_0) = h_M^*(m_0)$. Equality can hold only at $T = \bar{T}(m') - 1$, because the path is strictly decreasing. For each T the map $m' \mapsto G^T(h_H^*(m_0); m')$ is continuous and strictly decreasing, since $G_m < 0$ and $G_h > 0$, so equality at date T holds for at most one m'; the set of boom sizes at which the exception arises is countable.

(iii) Let $m'' > m' > \bar{m}$. Then $G(h; m'') \leq G(h; m')$ for every h because $G_m < 0$, and G is increasing in h; by induction $G^T(h_0; m'') \leq G^T(h_0; m')$ for every T. The first date at which the path falls below $h_M^*(m_0)$ is therefore weakly earlier under m'', so $\bar{T}(m'') \leq \bar{T}(m')$. □

Proofs for Sections 4 and 5

Throughout, Assumption 3 holds, $h_H = \lambda\varphi_H(0)/\delta$, $\Delta\Pi = \Pi_L(x_L^c) - \Pi_H(0)$, and $\Delta g = g_H - g_L$.

Proof of Proposition 6. The discount factor enters the reduced problem (36) only through the weights β^t. Assumption 3, the flow payoffs Π_i, the cost K, the gains $\Delta\Pi$ and Δg, the set of crossing plans, and C_0 do not depend on it. For a crossing plan P with crossing date $T = T_P$ write

$$a_P(\beta) \equiv (1-\beta)\,\beta^{-T} C_P = (1-\beta) \sum_{t<T} \beta^{-(T-t)} K(x_t), \qquad A(\beta) \equiv \inf_P a_P(\beta),$$

the infimum taken over crossing plans. By (38), $\psi^*\Delta g = \max\{0, A(\beta) - \Delta\Pi\}$, and by (39), $\Delta(P;0) = \beta^T[\Delta\Pi - a_P(\beta)]/(1-\beta)$. The proof uses three facts about A.

First, A contracts as β rises. Let $\beta < \beta'$ and $r = (1-\beta')/(1-\beta) < 1$. Every term of C_P is nonnegative by Assumption 3(iii), and $\beta'^{-(T-t)} \leq \beta^{-(T-t)}$ for $t < T$, so $a_P(\beta') \leq r\, a_P(\beta)$ for every crossing plan, and taking infima, $A(\beta') \leq r\, A(\beta)$.

Second, $A(\beta) < \Delta\Pi$ for β close to one. Fix a crossing plan P; one exists by Assumption 3(iv), as in the proof of Proposition 5. Its sum $\sum_{t<T} \beta^{-(T-t)} K(x_t)$ has finitely many terms and stays bounded as $\beta \to 1$, so $A(\beta) \leq a_P(\beta) \to 0 < \Delta\Pi$.

Third, $A(\beta) \geq (1-\beta)C_0/\beta$. For $t \leq T-1$, $\beta^{-(T-t)} \geq \beta^{-1}$, so

$$\beta^{-T} C_P \;=\; \sum_{t<T} \beta^{-(T-t)} \big[\Pi_H(0) - \Pi_H(x_t)\big] \;\geq\; \beta^{-1} \sum_{t<T} \big[\Pi_H(0) - \Pi_H(x_t)\big] \;\geq\; \beta^{-1} C_0,$$

where the last inequality holds because the path $\{x_t\}_{t<T}$ carries the West's knowledge from h_H to χ or below, and C_0 in (40) is the infimum of that undiscounted sum over all such paths.

Let $S = \{\beta \in (0,1) : A(\beta) < \Delta\Pi\}$ and $\hat\beta = \inf S$. By the second fact S is nonempty, so $\hat\beta < 1$. By the first, if $\beta \in S$ and $\beta' > \beta$, then $A(\beta') \leq rA(\beta) \leq A(\beta) < \Delta\Pi$, so $\beta' \in S$. By the third, Condition (D) gives $A(\beta) \geq (1-\beta)C_0/\beta > \Delta\Pi$, so no β at which it holds lies in S; it holds exactly for $\beta < C_0/(C_0 + \Delta\Pi)$, so $\hat\beta \geq C_0/(C_0 + \Delta\Pi)$.

(i) Let $\beta > \hat\beta$. Some element of S lies below β, so $\beta \in S$, and there is a crossing plan with $a_P(\beta) < \Delta\Pi$. For that plan $\Delta(P;0) > 0$, so crossing is strictly optimal at $\psi = 0$, and $\psi^* = 0$.

(ii) Let $\beta < \hat\beta$, and choose $\beta' \in (\beta, \hat\beta)$. Then $\beta' \notin S$, so $A(\beta') \geq \Delta\Pi > 0$, and the first fact gives $A(\beta) \geq A(\beta')/r > A(\beta') \geq \Delta\Pi$. Hence $\psi^* > 0$. At $\psi = 0 < \psi^*$ crossing is strictly suboptimal by Proposition 5(i), so the optimal path never crosses; by Lemma 2(iii) it ships $x^{\mathrm{c}}_H = 0$ at every date, and under autarky the West's knowledge stays at h_H.

(iii) Under Condition (D), the third fact gives $\psi^*\Delta g \geq A(\beta) - \Delta\Pi \geq (1-\beta)C_0/\beta - \Delta\Pi > 0$. □

Proof of Proposition 7. Fix $\psi > \psi^*$ and write $\Delta(\psi) > 0$ for the East's value gain from predation, as in the proof of Proposition 5. Plans with a late crossing date are dominated: $\Delta(P;\psi) \leq \beta^{T_P}(\Delta\Pi + \psi\Delta g)/(1-\beta)$ vanishes as T_P grows while some plan gains more than $\Delta(\psi)/2$, so attention restricts to $T_P \leq \hat T$ for some finite $\hat T$. For a date T and active exports $x = (x_0, \ldots, x_{T-1})$ write

$$h_T(x) \;\equiv\; (1-\delta)^T h_H \;+\; \lambda \sum_{t<T} (1-\delta)^{T-1-t} \varphi_H(x_t),$$

the West's knowledge at T if it is in regime H at every earlier date; the function is continuous and strictly decreasing in each x_t, because φ_H is. Consider the problem of minimizing $\sum_{t<T} \beta^t K(x_t)$ over $x \geq 0$ subject to $h_T(x) \leq \chi$. Its constraint set is closed, and the objective can be restricted to a compact box, because $K(x) \to \infty$ as $x \to \infty$; so whenever the constraint set is nonempty a minimizer $x^{(T)}$ exists, with minimum $C^*(T)$. Every crossing

plan with first crossing date T is feasible for this problem, so its value (39) is at most $\Delta_T \equiv -C^*(T) + \beta^T(\Delta\Pi + \psi\Delta g)/(1-\beta)$. Let T^* maximize Δ_T over the feasible dates $T \le \hat{T}$, and let $T' \le T^*$ be the first date at which the exports $x^{(T^*)}$ carry the West's knowledge to χ or below. These exports, followed by x_L^c, form a crossing plan with first crossing date T'. Its cost is at most $C^*(T^*)$, since it drops the nonnegative terms after T', and $\beta^{T'} \ge \beta^{T^*}$, so its value is at least Δ_{T^*}; and its value is at most $\Delta_{T'} \le \Delta_{T^*}$. It is therefore an optimal plan, and an optimal plan exists. Conversely, every optimal plan minimizes cost at its own crossing date T: if some x' with $h_T(x') \le \chi$ cost less, then x' would cross at a date no later than T at a cost no larger, and its value would exceed the optimum. What follows therefore applies to every optimal plan, with T its crossing date. The passive phase is Lemma 2(ii) and (iii): from T on the regime is L and permanent, and the export is x_L^c.

For the active phase, the constraint binds, $h_T(x^{(T)}) = \chi$. Some x_t is positive, since $h_T(0) = h_H > \chi$; if the constraint were slack, that x_t could be lowered slightly, keeping $h_T \le \chi$ by continuity and lowering the cost, because $K'(x) > 0$ for $x > 0$ by Assumption 3(iii). Let $\mu \ge 0$ be the multiplier on $h_T(x) \le \chi$. The Kuhn–Tucker conditions are

$$\beta^t K'(x_t) \;-\; \mu\,\lambda\,(1-\delta)^{T-1-t}\big(-\varphi_H'(x_t)\big) \;\ge\; 0, \qquad \text{with equality if } x_t > 0, \quad t < T.$$

If $\mu = 0$ they give $K'(x_t) = 0$, hence $x_t = 0$ for all t, contradicting binding; so $\mu > 0$. At $x_t = 0$ the left side is $-\mu\lambda(1-\delta)^{T-1-t}(-\varphi_H'(0)) < 0$, which violates the condition; so $x_t > 0$ at every active date, and the conditions hold with equality. Dividing successive conditions, the ratio $r(x) \equiv K'(x)/\big(-\lambda\varphi_H'(x)\big)$ satisfies $r(x_{t+1})/r(x_t) = [\beta(1-\delta)]^{-1} > 1$, and r is strictly increasing by Assumption 3(iii), so x_t is strictly increasing and largest at $t = T-1$.

Finally, h^W falls strictly. At $t = 0$, $\lambda\varphi_H(x_0) < \lambda\varphi_H(0) = \delta h_H = \delta h_0^W$ because $x_0 > 0$, so $h_1^W < h_0^W$. For the induction step, suppose $\lambda\varphi_H(x_t) < \delta h_t^W$. Then $h_{t+1}^W < h_t^W$, and since $x_{t+1} \ge x_t$,

$$\lambda\varphi_H(x_{t+1}) \;\le\; \lambda\varphi_H(x_t) \;<\; \delta\big[(1-\delta)h_t^W + \lambda\varphi_H(x_t)\big] \;=\; \delta h_{t+1}^W,$$

where the middle inequality is $(1-\delta)\lambda\varphi_H(x_t) < \delta(1-\delta)h_t^W$, the induction hypothesis again. With $h_T^W = \chi$ this gives $h_t^W > \chi$ for $t < T$, consistent with T being the first crossing date, and the largest shipment is made in the period the West's knowledge reaches χ. □

Crossing dates and the critical security weight. Let $C^*(T)$ be the minimum of the fixed-date problem in the proof of Proposition 7, and $z(T) \equiv [(1-\beta)\beta^{-T}C^*(T) - \Delta\Pi]/\Delta g$ the break-even weight of the cheapest plan that crosses at date T.

(a) $z(T+1) < z(T)$ *at every date* T *at which a crossing is feasible.* Take the cheapest plan that crosses at T and delay it by one period: the East ships nothing in the first period and then follows the original plan. The delayed copy crosses at $T+1$ and costs $\beta C^*(T)$, so its break-even weight is $z(T)$. Now let the East ship $\epsilon > 0$ in the first period. This lowers the West's knowledge at $T+1$ by an amount proportional to ϵ, so the plan's last shipment, which is positive by the Kuhn–Tucker argument above, can be reduced by an amount proportional to ϵ while the West's knowledge still reaches χ at $T+1$. The first shipment costs $K(\epsilon)$, which is of order ϵ^2 because $K(0) = K'(0) = 0$: the two countries' autarky prices are equal, so $x = 0$ is an interior critical point of Π_H. Reducing the last shipment saves an amount of

order ϵ, because $K' > 0$ at a positive shipment. For small ϵ the modified plan costs less than $\beta C^*(T)$. Hence $C^*(T+1) < \beta C^*(T)$, which is equivalent to $z(T+1) < z(T)$.

(b) *No crossing plan attains the infimum in* (38). A crossing plan P costs at least $C^*(T_P)$, so $z_P \geq z(T_P) > z(T_P+1)$ by (a): some other plan has a lower break-even weight. When $\psi^* > 0$, this gives $z_P > \psi^*$ for every crossing plan, so every crossing plan loses money at $\psi = \psi^*$.

(c) *The optimal crossing date does not increase with ψ, and when $\psi^* > 0$ it grows without bound as ψ falls to ψ^*.* By the proof above, every optimal plan is the cheapest plan at its own crossing date, and its value is the maximum over T of Δ_T. For $\psi > \psi^*$ the optimal crossing dates are therefore the maximizers over feasible T of $f(T,\psi) \equiv \beta^T[\psi - z(T)]$, a positive multiple of Δ_T. Let $\psi < \psi'$, let T be optimal at ψ, and let T' be optimal at ψ'. Optimality gives $f(T',\psi') \geq f(T,\psi')$ and $f(T,\psi) \geq f(T',\psi)$. Adding the two inequalities and using $f(T,\psi') - f(T,\psi) = (\psi' - \psi)\beta^T$ gives $(\psi' - \psi)\beta^{T'} \geq (\psi' - \psi)\beta^T$, so $T' \leq T$. For the second claim, fix a date T. By (a), $z(s) \geq z(T)$ for every $s \leq T$, and $z(T) > \psi^*$ by (b). For $\psi \in (\psi^*, z(T))$, every plan that crosses by date T therefore loses money, and the optimal crossing date exceeds T.

The West's gain from a shipment. The proof of Proposition 8 uses a pointwise inequality between the West's gain from a shipment and the East's cost of making it, $3\big[u_H^W(x) - u_H^W(0)\big] \leq K(x)$, which holds when v is quadratic and $f''' \geq 0$. Write $q_H(x) = f'\big(\alpha_H\varphi_H(x) + x\big)$ for the price a regime-H West pays and $p(x) = f'\big(\alpha_H n^E(x) - x\big)$ for the East's own marginal product of the manufactured good when it ships x. The West's flow welfare $u_H^W(x)$ is the value of $\max_n f(\alpha_H n + x) - v(n)$ less the bill $q_H(x)x$, and by the envelope theorem that value has derivative $q_H(x)$ in x; the East's flow payoff $\Pi_H(x)$ is the value of $\max_n f(\alpha_H n - x) - v(n)$, whose derivative is $-p(x)$, plus the revenue $q_H(x)x$. Hence

$$u_H^W(x) - u_H^W(0) = \int_0^x \big[q_H(s) - q_H(x)\big]\,ds, \qquad K(x) = \int_0^x \big[p(s) - q_H(x)\big]\,ds. \tag{53}$$

Differentiating the two employment optimality conditions $v'(n) = \alpha_H f'(\alpha_H n \pm x)$ gives

$$q_H'(x) = -Q\big(k^W(x)\big), \qquad p'(x) = Q\big(k^E(x)\big), \qquad Q(k) \equiv \frac{-v''f''(k)}{v'' - \alpha_H^2 f''(k)} > 0,$$

with $k^W(x) = \alpha_H\varphi_H(x) + x$ increasing and $k^E(x) = \alpha_H n^E(x) - x$ decreasing. With v quadratic, v'' is a positive constant, and $f''' \geq 0$ makes $-f''$ nonincreasing, so Q is nonincreasing. Put $D(x) = q_H(0) - q_H(x)$ and $E(x) = p(x) - q_H(0)$, both zero at the origin. Then D is concave and E is convex, and $k^E(x) \leq k^W(x)$ with Q nonincreasing gives $E' \geq D'$ and so $E \geq D$. The autarky prices of two industrialized countries are equal, $q_H(0) = p(0)$, so (53) reads

$$u_H^W(x) - u_H^W(0) = xD(x) - \int_0^x D, \qquad K(x) = xD(x) + \int_0^x E.$$

The inequality $3\big[u_H^W(x) - u_H^W(0)\big] \leq K(x)$ is therefore $2xD(x) \leq 3\int_0^x D + \int_0^x E$, and $E \geq D$ makes the right side at least $4\int_0^x D$, while a concave D vanishing at the origin satisfies

$D(s) \geq (s/x)D(x)$ on $[0, x]$ and hence $\int_0^x D \geq xD(x)/2$. Both inequalities bind as $x \to 0$, so the constant one third cannot be improved.

Proof of Proposition 8. Condition (N) is $\hat{\psi} > \psi^*$ rearranged. Fix ψ with $\psi^* < \psi < \hat{\psi}$ and $P \in \mathcal{P}^*(\psi)$, with crossing date $T = T_P$ and break-even weight z_P. Because $\Delta(P; \cdot)$ is affine with slope $\beta^T \Delta g/(1-\beta)$ and vanishes at z_P,

$$\Delta W^E(\psi) \;=\; \Delta(P;\psi) \;=\; \frac{\beta^T \Delta g}{1-\beta}\,\big(\psi - z_P\big),$$

which is positive because P gains. Before date T the West is in regime H. From T on it is in regime L and, by Lemma 2(iii), imports x_L^{c} in every period. Its loss is therefore

$$L^W(P) \;=\; \frac{\beta^T \Delta u}{1-\beta} \;-\; \sum_{t<T} \beta^t \big[u_H^W(x_t) - u_H^W(0)\big].$$

By the inequality proved above, each term of the sum is at most $K(x_t)/3$, so the sum is at most $C_P/3 = \beta^T(z_P \Delta g + \Delta\Pi)/\big(3(1-\beta)\big)$, the equality by the definition of z_P. Hence

$$L^W(P) - \Delta W^E(\psi) \;\geq\; \frac{\beta^T}{1-\beta}\Big[\Delta u - \tfrac{1}{3}\,\Delta\Pi + \tfrac{2}{3}\,z_P \Delta g - \psi \Delta g\Big].$$

No finite crossing plan attains the infimum in (38), by part (b) of the argument on crossing dates above, so $z_P > \psi^*$, and the bracket exceeds $\Delta u - \frac{1}{3}\Delta\Pi + \frac{2}{3}\psi^*\Delta g - \psi\Delta g = (\hat{\psi} - \psi)\Delta g > 0$. World welfare, the sum of the two countries' objectives, changes by $\Delta W^E(\psi) - L^W(P) < 0$ on every optimal plan. The bound holds at every weight in the interval and at every optimal plan, so no separate treatment of the weights at which two crossing dates tie is needed. Finally, the security parameters enter a plan's value (39) only through the product $\psi\Delta g$, and $\psi^*\Delta g$, Δu, and $\Delta\Pi$ do not involve Δg, so the interval $\psi^*\Delta g < \psi\Delta g < \hat{\psi}\Delta g$ is the same for every Δg. □

Proof of Proposition 9. (i) The tariff enters the East's problem only through the revenue term $q_t x_t$, which becomes $(q_t - \xi)x_t$. The East's employment optimality condition (30) does not involve it. The argument of Lemma 2(i) therefore applies unchanged: $\mu^E \equiv 0$ on the plateau, the East's employment is the static $n^E(x)$, and its knowledge stays above χ. Substituting the budget gives the flow payoff $\Pi_{i_t}(x_t) - \xi x_t - \psi g_{i_t}$, with the West's law of motion as the only link between dates. The West is competitive, so part (ii) of the lemma holds word for word. For part (iii): in regime H, $\Pi_H(x) - \xi x \leq \Pi_H(0)$ for every $x \geq 0$, by Assumption 3(iii) and $\xi x \geq 0$, so autarky is the best never-crossing plan. In regime L the payoff involves no state. It is maximized date by date at $x_L^{\mathrm{c}}(\xi)$, which exists because $\Pi_L(x) \to -\infty$ as $x \to \infty$.

(ii) Fix ξ. Never crossing is worth $[\Pi_H(0) - \psi g_H]/(1-\beta)$. A crossing plan P is worth

$$\sum_{t<T_P} \beta^t \big[\Pi_H(x_t) - \xi x_t - \psi g_H\big] + \beta^{T_P}\,\frac{\Pi_H(0) + \Delta\Pi(\xi) - \psi g_L}{1-\beta},$$

using (i) for the continuation. Its advantage over staying is

$$\Delta(P;\psi,\xi) \;=\; -\,C_P(\xi) + \beta^{T_P}\,\frac{\Delta\Pi(\xi)+\psi\,\Delta g}{1-\beta}, \tag{54}$$

which is (39) with the tariffed cost and gain. For fixed ξ the plan-by-plan argument of Proposition 5(i) applies. Each $\Delta(P;\cdot,\xi)$ is affine and strictly increasing in ψ with zero $z_P(\xi)$, the expression inside the infimum in (42); crossing plans exist by Assumption 3(iv); above $\inf_P z_P(\xi)$ some plan gains and below it every plan loses, which is the first claim.

For the dependence on ξ: every crossing plan ships more than $\bar m_\infty$ at some active date. Otherwise $\lambda\varphi_H(x_t) \geq \delta\chi$ at every $t < T_P$, and then $h_t^W > \chi$ implies $h_{t+1}^W > (1-\delta)\chi+\delta\chi = \chi$; by induction from $h_0^W = h_H > \chi$, the plan never crosses. So there is $t < T_P$ with $x_t > \bar m_\infty$, and

$$\beta^{-T_P}\sum_{s<T_P}\beta^s x_s \;\geq\; \beta^{t-T_P}x_t \;>\; \bar m_\infty.$$

Now $C_P(\xi') - C_P(\xi) = (\xi'-\xi)\sum_{s<T_P}\beta^s x_s$, and $\Delta\Pi$ is nonincreasing, being a maximum of functions that fall in ξ. Hence

$$z_P(\xi') - z_P(\xi) \;\geq\; \frac{(1-\beta)\,\bar m_\infty}{\Delta g}\,(\xi'-\xi) \qquad \text{for every } P \text{ and } \xi' > \xi.$$

Taking the infimum over P on both sides gives the same inequality for $\inf_P z_P$. This proves the slope bound wherever $\psi^*(\xi) = \inf_P z_P(\xi) > 0$, the monotonicity of ψ^*, and the divergence. For continuity, $\inf_P z_P(\xi) = \inf_P[a_P + b_P\xi] - \Delta\Pi(\xi)/\Delta g$ with $a_P, b_P \geq 0$. An infimum of affine functions that is finite on $[0,\infty)$ is concave, hence continuous on $(0,\infty)$. At $\xi = 0$, monotonicity gives $\liminf_{\xi\downarrow 0} \geq \inf_P a_P$, and $\inf_P[a_P + b_P\xi] \leq a_P + b_P\xi$ for each P gives $\limsup_{\xi\downarrow 0} \leq \inf_P a_P$. And $\Delta\Pi$ is continuous, as the maximum of a continuous function of (x,ξ) over a set that can be taken compact, since $\Pi_L(x) - \xi x \to -\infty$ as $x \to \infty$ uniformly for ξ in a bounded set. The truncation at zero preserves continuity.

(iii) For $\psi > \psi^*(\xi)$, crossing is strictly optimal by (ii). An optimal plan exists by the existence argument in the proof of Proposition 7, which uses only that the capture set $h_T \leq \chi$ is closed, that $\Pi_H(x) - \xi x \to -\infty$ as $x \to \infty$, and that late crossings are dominated. On that plan the West is in regime L from the crossing date on, by (i). The West's flow welfare at date t is $u_{i_t}^W(x_t) + \xi x_t$: households pay the gross price $q_{i_t}(x_t)$ on x_t units, the treasury returns ξx_t to them, and the East receives the rest. For $\psi < \psi^*(\xi)$, staying is strictly optimal, the best never-crossing plan is autarky by (i), and under autarky the West's knowledge stays at h_H.

(iv) Fix $\psi > \psi^*(0)$ and let $\bar\xi(\psi) = \inf\{\xi \geq 0 : \psi^*(\xi) \geq \psi\}$. The set is nonempty by the divergence in (ii) and does not contain 0, so $\bar\xi(\psi) > 0$; by continuity $\psi^*(\bar\xi(\psi)) = \psi$. For $\xi < \bar\xi(\psi)$, $\psi^*(\xi) < \psi$ by the definition of the infimum, and the West is trapped by (iii). For $\xi > \bar\xi(\psi)$, $\psi^*(\xi) > \psi$ by the slope bound, since $\psi^*(\bar\xi(\psi)) = \psi > 0$, and the West stays at h_H by (iii). Uniqueness follows from the same strict increase. The set in the definition of $\bar\xi$ shrinks as ψ rises, so $\bar\xi$ is nondecreasing. □

Proof of Lemma 4. (i) The derivative of the flow (45) in n is $\alpha_H f'(k) - v'(n) - \alpha_H x\, f''(k)$, with $k = \alpha_H n + x$. For $n < \varphi_H(x)$ the first two terms sum to a positive number, because $f(\alpha_H n + x) - v(n)$ is strictly concave with its maximum at $\varphi_H(x)$, and the last term is nonnegative; so the flow is strictly increasing on $[0, \varphi_H(x)]$. Its second derivative is $\alpha_H^2[f''(k) - xf'''(k)] - v''(n) < 0$ when $f''' \geq 0$, so it is strictly concave, and its unique maximizer $n_H^{\mathrm{R}}(x)$ solves (46) and is at least $\varphi_H(x)$. Differentiating (46) gives $dn_H^{\mathrm{R}}/dx = \alpha_H x f'''(k) / \big(\alpha_H^2[f''(k) - xf'''(k)] - v''(n)\big) \leq 0$, and at $x = 0$ (46) is the household condition, so $n_H^{\mathrm{R}}(0) = \varphi_H(0)$; the same holds in regime L. The levels that place next period's knowledge at or above $\chi + \varepsilon$ are those with $n \geq \underline{n}(h)$. The maximizer of a strictly concave function over such a set is its unconstrained maximizer if that lies in the set and the endpoint otherwise, which is (49). It is at least $n_H^{\mathrm{R}}(x) \geq \varphi_H(x)$.

(ii) At $h = \chi + \varepsilon$, $\underline{n}(\chi + \varepsilon) = \delta(\chi + \varepsilon)/\lambda$, and by (i) the floor binds exactly when $n_H^{\mathrm{R}}(x) < \underline{n}(\chi + \varepsilon)$. Where it binds, the rate is $v'(\underline{n})/(\alpha_H f'(\alpha_H \underline{n} + x)) - 1$ with $\underline{n} = \underline{n}(\chi + \varepsilon)$ fixed, which is strictly increasing in x because f' is strictly decreasing.

(iii) Let $n' < n$, and write $k' = \alpha_i n' + x < k = \alpha_i n + x$. Then

$$u_i^W(n'; x) - u_i^W(n; x) \;=\; \big[f(k') - f(k)\big] - x\big[f'(k') - f'(k)\big] + v(n) - v(n').$$

The first bracket is negative because f is strictly increasing, and the second term is nonpositive because f' is decreasing, so the difference is less than $v(n) - v(n')$. □

Proof of Proposition 10, parts (ii) and (iii). (ii) Two bounds on the West's value function under X^{c} come first. From any $h' > \chi$, the East ships nothing, and any plan earns at most $\max_n u_H^W(n; 0) = u_H^W(0)$ in each period it spends above χ and at most $\bar{u}_L^W$ in each period it spends at or below, since the East ships $\tilde{x}(h)$ there and $\bar{u}_L^W$ bounds the West's flow against every such shipment at every employment. Since $\bar{u}_L^W < u_H^W(0)$ by (C), the best plan stays above χ and works $\varphi_H(0)$ at every date, which is allowed because $(1 - \delta)h' + \lambda\varphi_H(0) > (1 - \delta)\chi + \delta h_H > \chi + \varepsilon$ for $h' > \chi$, by the bound on the margin in Assumption 4; so $V^W(h', 0; X^{\mathrm{c}}) = V_H^W$. Following N^{F} attains it, because $\underline{n}(h') < (\delta\chi + \varepsilon)/\lambda < \delta h_H/\lambda = \varphi_H(0)$ for $h' > \chi$, so $N^{\mathrm{F}}(h', 0) = \varphi_H(0)$. From any $h' \leq \chi$, the same bounds show that the value is at most $\bar{u}_L^W + \beta V_H^W$: at least one period is spent at or below χ, and the bound is largest when exactly one is.

Now fix $h > \chi$, a shipment x, and an employment policy that agrees with N^{F} above χ. The levels in $\mathcal{N}(h)$ split in two. Those that place next period's knowledge at or above $\chi + \varepsilon$ all leave the continuation at V_H^W by the first bound, so among them $N^{\mathrm{F}}(h, x)$ is best, by Lemma 4(i). Those that place it at or below χ leave the continuation at most at $\bar{u}_L^W + \beta V_H^W$, by the second bound, so each loses at least $\beta V_H^W - \beta[\bar{u}_L^W + \beta V_H^W] = \beta[u_H^W(0) - \bar{u}_L^W]$ in continuation. Take such a level n'. If $N^{\mathrm{F}}(h, x) = n_H^{\mathrm{R}}(x)$, the unconstrained maximizer of the flow, then n' has no higher flow either, and it does not gain. Otherwise $N^{\mathrm{F}}(h, x) = \underline{n}(h) > n'$, and by Lemma 4(iii) the gain of n' in flow is less than $v(\underline{n}(h)) \leq v((\delta\chi + \varepsilon)/\lambda)$, because $\underline{n}(h) \leq (\delta\chi + \varepsilon)/\lambda$ for $h > \chi$. By (C) the loss exceeds the gain.

(iii) Fix $h \in ((1 - \delta)\chi, \chi]$ and a shipment x. The levels that return the West to the high regime are those with $n \geq \underline{n}(h)$, and they all leave the continuation at V_H^W by the first bound in (ii). Among them the flow $u_L^W(\cdot; x)$ is highest at $\underline{n}(h)$, because it is strictly concave

with maximizer $n_L^{\mathrm{R}}(x) \le n_L^{\mathrm{R}}(0) = \varphi_L(0) < \delta\chi/\lambda < \underline{n}(h)$, by the proof of Lemma 4(i) and Assumption 3(i). Returning to the defended level means working $\underline{n}(h) < \underline{n}((1-\delta)\chi)$. The other levels in $\mathcal{N}(h)$ leave the West at or below χ for another period, so their continuation is at most $\bar{u}_L^W + \beta V_H^W$ by the second bound in (ii), and each lies below $\underline{n}(h)$. By Lemma 4(iii), the gain in flow of each over $\underline{n}(h)$ is less than $v(\underline{n}(h))$. Its loss in continuation is at least $\beta[u_H^W(0) - \bar{u}_L^W]$. The stated condition makes the loss exceed the gain for every such h. Neither bound involves x, so the return is optimal whatever the East ships. Finally, from h_H the policies visit only h_H. From any $h \in ((1-\delta)\chi, \chi]$ the policies move the state to $\chi + \varepsilon$, and from $h > \chi$ they keep it above χ, so the policies do not leave $((1-\delta)\chi, \infty)$. A single departure by the West from a state $h > \chi$ reaches $h' \ge (1-\delta)h > (1-\delta)\chi$, and a single departure by the East is followed by N^{F} and leaves the state above χ. On $((1-\delta)\chi, \infty)$ condition (i) of Definition 1 holds by (i), condition (ii) by (ii) and the argument just given, and condition (iii) because both payoffs are defined at the price (50). □

B Efficient Basins of Attraction

The multiplicity of Section 3 is a property of the competitive equilibrium. This appendix asks when it is also a property of the efficient allocation. The setting is a single closed economy in the sharp-threshold benchmark of Section 4.3, run by a planner:

$$\max_{\{n_t \ge 0\}} \sum_{t=0}^{\infty} \beta^t \big[f\big(\alpha(h_t)\, n_t\big) - v(n_t) \big] \qquad \text{subject to} \quad h_{t+1} = (1-\delta)h_t + \lambda n_t, \quad h_0 \text{ given,} \tag{55}$$

with $\alpha(h) = \alpha_L$ for $h \le \chi$ and $\alpha(h) = \alpha_H$ for $h > \chi$, the convention of Assumption 3. The problem is not concave, so we argue from plans, not from first-order conditions.

Write $u_i(n) \equiv f(\alpha_i n) - v(n)$ for the regime-i flow, n_i for its maximizer, the solution of $v'(n) = \alpha_i f'(\alpha_i n)$, and $u_i^* \equiv u_i(n_i)$. We use two facts. The flow depends on the knowledge stock only through the regime. And the flow maximizer sustains its own steady state: n_i is also the employment that holds h constant at $h_i = \lambda n_i/\delta$. We impose the autarky part of Assumption 3(i), $\lambda n_L < \delta\chi < \lambda n_H$, so that $h_L < \chi < h_H$; note that $n_i = \varphi_i(0)$.

The results depend on three numbers. The gain from crossing is the flow gap $\Delta u \equiv u_H^* - u_L^*$, positive because $u_H(n) > u_L(n)$ for every $n > 0$. The floor is

$$s^* \;\equiv\; u_L^* - u_L(\delta\chi/\lambda) \;>\; 0 :$$

a plan that crosses χ must, in its crossing period, carry knowledge from at most χ to above it, so it employs more than $\delta\chi/\lambda$, which exceeds n_L by assumption, and u_L is strictly concave, so the crossing period sacrifices at least s^* of flow. The climb cost is

$$C_0^{\mathrm{up}}(h_0) \;\equiv\; \inf\Big\{ \textstyle\sum_{t=0}^{T-1} \big[u_L^* - u_L(n_t)\big] \;:\; h_{t+1} = (1-\delta)h_t + \lambda n_t,\; h_T > \chi \Big\},$$

the undiscounted counterpart of (40) for a crossing from below; it is nonincreasing in h_0 and at least s^*.

Proposition 11 (Basins of the efficient allocation)**.** *Let $\lambda n_L < \delta\chi < \lambda n_H$.*

(i) From every $h_0 > \chi$ the unique optimal plan sets $n_t = n_H$ at every date, and $h_t \to h_H$.

(ii) If $\beta\,\Delta u < (1-\beta)\,s^*$, then from every $h_0 \le \chi$ the unique optimal plan sets $n_t = n_L$ at every date, and $h_t \to h_L$. The efficient allocation then has two basins of attraction, separated at χ, and two efficient steady states.

(iii) If $(1-\beta)\,s^* < \beta\,\Delta u < (1-\beta)\,C_0^{\mathrm{up}}(h_L)$, the efficient allocation still has two basins: there is a threshold $\hat{h} \in (h_L, \chi)$ such that from every $h_0 < \hat{h}$ the plan $n_t = n_L$ at every date is optimal and knowledge converges to h_L, while from every $h_0 \in (\hat{h}, \chi]$ every plan that never crosses χ is strictly suboptimal.

Proof. (i) From $h_0 > \chi$, the plan $n_t = n_H$ keeps $h_{t+1} > (1-\delta)\chi + \lambda n_H > \chi$, so its flow is u_H^* at every date and its value is $u_H^*/(1-\beta)$. No plan does better, since every flow is at most $\max\{u_L^*, u_H^*\} = u_H^*$. A plan attaining the bound must earn u_H^* at every date, which forces regime H and $n_t = n_H$ throughout, by strict concavity of u_H. The law of motion then converges to h_H.

(ii) From $h_0 \le \chi$, the plan $n_t = n_L$ stays in regime L, because $h_{t+1} \le (1-\delta)\chi + \lambda n_L < \chi$; its value is $u_L^*/(1-\beta)$ and it converges to h_L. Every plan that never crosses has flows at most u_L^*, so none does better, and one that ties must set $n_t = n_L$ throughout, by strict concavity of u_L. Now take a plan that first crosses at date T. Its pre-crossing flows are $u_L^* - s_t$ with $s_t \ge 0$ and $s_{T-1} \ge s^*$, and its flows from T on are at most $u_L^* + \Delta u$. Its value therefore exceeds $u_L^*/(1-\beta)$ by at most

$$-\beta^{T-1} s^* + \beta^T \frac{\Delta u}{1-\beta} \;=\; \beta^{T-1}\Big[\frac{\beta\,\Delta u}{1-\beta} - s^*\Big] \;<\; 0.$$

(iii) Three steps. First, the gain from crossing is monotone in the initial stock. Let $V^{\mathrm{x}}(h_0)$ be the supremum of values over crossing plans; the best never-crossing value is $u_L^*/(1-\beta)$ from every $h_0 \le \chi$, as in (ii). For $h_0' < h_0'' \le \chi$, any crossing plan from h_0', run from h_0'' with the same employment path, has a pointwise higher knowledge path, crosses no later, and earns weakly higher flows at every date, so V^{x} is nondecreasing, and the set of stocks from which crossing dominates is an interval reaching to χ. Let $\hat{h}$ be its infimum. Second, crossing dominates near χ. From $h_0 \le \chi$ the plan with $n_0 = (\chi' - (1-\delta)h_0)/\lambda$ for some $\chi' > \chi$ and $n_t = n_H$ thereafter crosses at once; its gain over staying is $\beta\Delta u/(1-\beta) - \big[u_L^* - u_L(n_0)\big]$, and as $h_0 \uparrow \chi$ and $\chi' \downarrow \chi$ the bracket falls to s^*, which lies below $\beta\Delta u/(1-\beta)$ by the left inequality. Third, staying dominates near h_L. The bound in (ii), run with the full climb cost in place of the single-period floor, caps the gain of any plan crossing at T from h_0 by $\beta^{T-1}\big[\beta\Delta u/(1-\beta) - C_0^{\mathrm{up}}(h_0)\big]$, which is negative at $h_0 = h_L$ by the right inequality. And C_0^{up} is continuous at h_L from above: it is nonincreasing, and concatenating the one-period step from h_L to h_0, whose sacrifice $u_L^* - u_L\big((h_0 - (1-\delta)h_L)/\lambda\big)$ vanishes as $h_0 \downarrow h_L$, with the cheapest climb from h_0 gives $C_0^{\mathrm{up}}(h_L) \le u_L^* - u_L\big((h_0 - (1-\delta)h_L)/\lambda\big) + C_0^{\mathrm{up}}(h_0)$. So staying dominates on an interval above h_L, and the threshold $\hat{h}$ lies strictly between h_L and χ. □

This is the history dependence of Skiba (1978) and Dechert and Nishimura (1983): in a planner problem with a convex-concave technology, optimal paths converge to different steady states depending on the initial stock, and $\hat{h}$ plays the part of the Skiba point: it separates the two basins. We have not shown that the planner is indifferent at $\hat{h}$ between staying and an attained crossing plan, which is how the point is usually defined; that would require the crossing value to be continuous and attained at $\hat{h}$, and part (iii) does not need it. Under the convention that places χ itself in the low regime, the set of plans that cross is open, so from $h_0 \in (\hat{h}, \chi]$ the supremum over crossing plans need not be attained; the three parts of the proposition are proved by bounds and do not need it to be. The benchmark adds explicit boundaries. At the parameters of Section 6, $u_L^* = 0.17$, $u_H^* = 1.58$, $s^* = 0.11$, and $C_0^{\text{up}}(h_L) = 1.01$, so part (ii) applies for $\beta < 0.07$ and part (iii) for β between 0.07 and 0.42. At $\beta = 0.96$ neither applies: the climb from h_L gains 31.2 in present value over staying, the climb from $h_0 = 0$ gains 28.4, and by the monotonicity in the proof of (iii) the planner's low basin is empty. This transition requires the planner's employment allocation: a government whose subsidy rate is capped below the rate the floor employment requires in regime L, which Section 6 reports, cannot implement it. The competitive economy of Section 3 has two basins at every discount factor, because no household takes the knowledge externality into account. The planner's two-basin region requires heavy discounting, and at the discount factor of the numerical example it is empty, so at that discount factor every competitive path that converges to the low steady state is inefficient.

C Predation from a Pre-industrial Position

Assumption 3(ii) starts both countries industrialized. This appendix examines what that leaves out: a West that begins in regime H facing an East that begins at or below χ. With the East's own regime free to change, its knowledge is a second state, the reduction of Lemma 2 fails, and we do not solve the problem. One structural fact can still be proved: under a condition on the primitives, no plan traps the West while the East remains pre-industrial. A flood that traps the West requires employment, and at low productivity that employment raises the exporter's knowledge above χ. The condition is

$$\lambda \bar{m}_\infty \;\geq\; \alpha_L \chi, \tag{P}$$

which says that producing a flow of $\bar{m}_\infty$ at low productivity, employment of $\bar{m}_\infty/\alpha_L$, generates at least the threshold stock of knowledge in a single period. At the parameters of Section 6 the left side is 8.85 against $\alpha_L \chi = 1.25$: the condition holds by a factor of seven.

Proposition 12 (Predation from below industrializes the exporter)**.** *Let the West begin in regime H, and let part (iv) of Assumption 3 and Condition (P) hold. Consider any export plan that carries the West's knowledge to χ or below, and let T be the first date at which it is there. Then the East is in regime H at date $T-1$ or at date T. In particular, no plan traps the West while the East remains pre-industrial throughout.*

Proof. The West is in regime H at every date before T, so $h_{T-1}^W > \chi$ and

$$\lambda \varphi_H(x_{T-1}) \;=\; h_T^W - (1-\delta) h_{T-1}^W \;\leq\; \chi - (1-\delta) h_{T-1}^W \;<\; \chi - (1-\delta)\chi \;=\; \delta\chi.$$

Since φ_H is decreasing and $\lambda\varphi_H(\bar{m}_\infty) = \delta\chi$, the crossing shipment satisfies $x_{T-1} > \bar{m}_\infty$. If the East is in regime H at date $T-1$, the claim holds at that date. If it is in regime L, feasibility $x_{T-1} \le \alpha_L n_{T-1}$ gives $n_{T-1} > \bar{m}_\infty/\alpha_L$, so

$$h_T^E \;=\; (1-\delta)\, h_{T-1}^E + \lambda\, n_{T-1} \;>\; \lambda\, \bar{m}_\infty/\alpha_L \;\ge\; \chi$$

by Condition (P), so the East is in regime H at date T. □

The proposition leaves two cases, and both end at the configuration the text assumes. If the East industrializes before the final flood, then from that date it is an industrialized country facing a still-industrialized West, deciding whether to finish the crossing: the position of Assumption 3(ii). If instead it is still pre-industrial in the crossing period, its own shipments complete its industrialization in the same period the West falls, and the two countries exchange positions at once. This exchange is the role reversal announced in the introduction. The exporter need not start in the high regime. A country at or below χ that ships a sustained surplus raises its own employment and knowledge while depressing its customer's. The feasibility argument of the proof applies to any shipment: in regime L a shipment x requires employment of at least x/α_L, so a shipment above $\alpha_L\chi/\lambda$ carries the exporter across its own threshold in one period. An East that begins at the low steady state and runs a large enough surplus for long enough ends at the high one, and the West it ships to ends at the low one. Whether such a surplus is commerce is a different question, and we answer only part of it. The next proposition, which holds in the smooth model and in the benchmark alike, says that while the East lags, shipping has no commercial value.

Proposition 13 (A lagging East sells nothing)**.** *Set $\psi = 0$. Call a plan of the East* lagging *if*

$$\alpha(h_t^E)\, n_t^E \;\le\; \alpha(h_t^W)\, n_t^W + x_t \qquad \textit{for all } t:$$

keeping every shipped unit at home, the East would still have a smaller supply of manufactured goods than the West holds with them. Removing every shipment from a lagging plan weakly raises its value, and strictly so if $x_t > 0$ at some date. At $\psi = 0$, therefore, no optimal plan is both lagging at every date and exports a positive amount at some date: an optimal plan that keeps the East lagging throughout ships nothing.

Proof. Fix a lagging plan $\{n_t, x_t\}$ and let the alternative plan keep the same employment and ship nothing. The East's knowledge path depends only on employment, so it is unchanged, and so is the disutility of labor. The value difference between the alternative and the original is

$$\sum_{t=0}^{\infty} \beta^t \Big[f\big(\alpha(h_t^E) n_t\big) - f\big(\alpha(h_t^E) n_t - x_t\big) - q_t\, x_t \Big].$$

By strict concavity of f, $f(\alpha n_t) - f(\alpha n_t - x_t) \ge x_t\, f'(\alpha n_t)$, strictly if $x_t > 0$. Because the plan is lagging and $f'' < 0$, $f'(\alpha(h_t^E) n_t) \ge f'\big(\alpha(h_t^W) n_t^W + x_t\big) = q_t$. Every term is therefore nonnegative, and positive at any date with $x_t > 0$: removing a shipment loses the revenue $q_t x_t$ but returns goods worth more at home. □

When the East is at its low steady state and the West at its high one, the East's output of manufactured goods is a fraction of the West's supply, 0.12 against 3.7 at the parameters of Section 6, and a shipment only widens the gap, since it raises the West's supply (Lemma 1) and leaves the East's output unchanged. A plan that keeps employment near its steady-state level therefore lags at every date: the East lags because its knowledge is low. The proposition says that such an East gains nothing from shipping: whatever revenue a shipment earns, the goods are worth more at home. The class does not contain every plan an East with less knowledge can run. A plan that raises employment enough to industrialize the East leaves it once the East's productivity is high, and from that date the East faces the export decision of Section 4, including the commercial gain $\Delta\Pi$ from a West in regime L. A role-reversal surplus is therefore not covered by the proposition, and whether a pre-industrial East would choose one requires comparing the cost of the transition with its own industrialization gain, the subject of Appendix B, and the gain from trapping the West; we do not characterize that comparison. The two propositions establish that predation cannot be completed from below, that carrying it out puts the East in regime H, the configuration of Assumption 3(ii), and that at $\psi = 0$ no optimal plan both keeps the East lagging at every date and ships a positive amount.

D Predation and Deterrence with Free Shipments

Section 4.3 lets the East sell its shipments at the West's marginal product. This appendix removes the payment. The East ships manufactured goods and receives nothing for them, so every unit is a gift. We show that the security motive alone produces the flood and the trap of Proposition 7 above the critical security weight of Proposition 4, which Proposition 14 shows is at least ψ^*, and the plan ends differently from the paid one: once the West is trapped the East ships nothing at all.

Environment. Everything is as in Section 4.3 except that the East receives nothing for its shipments. At a price of zero the West's import margin (19) has no finite solution, so the West does not choose the quantity; the East does, and the West's households receive it as the inflow of Section 3 with $m_t = x_t$. Accepting it is individually rational, since a free unit of the manufactured good raises a household's output, and only the government could refuse or destroy it; we take the delivery as given and return to refusal below. The West's labor margin is unchanged, its employment in regime i is $\varphi_i(x)$, and its flow welfare is $u_i^{W,0}(x) \equiv f(\alpha_i\varphi_i(x) + x) - v(\varphi_i(x))$, which exceeds the autarky flow $u_i^W(0)$ for every $x > 0$. The East's budget is $c_t^E = f(\alpha_H n_t^E - x_t)$ and its objective is (22). Its employment at a given shipment is the same $n^E(x)$ as before, because the price never entered the East's employment optimality condition (30), and its flow payoff from shipping x is

$$\Pi^0(x) \;\equiv\; f\big(\alpha_H n^E(x) - x\big) - v\big(n^E(x)\big), \tag{56}$$

the gift payoff of Section 4.3, the payoff (34) without its revenue term. It does not depend on the West's regime. By the envelope theorem, $\Pi^{0\prime}(x) = -f'(\alpha_H n^E(x) - x) < 0$: every

unit shipped costs the East its marginal product at home, so the commercial export is zero in both regimes. The flow cost K^0 and the plan cost C_P^0 are as defined there.

Assumption 5 (Gift benchmark)**.** *The step* (35) *and parts (i), (ii), and (iv) of Assumption 3 hold, and the ratio* $-\Pi^{0\prime}(x)\big/\big(-\lambda\varphi_H'(x)\big)$ *is strictly increasing on* $[0,\infty)$*.*

The ratio condition replaces part (iii) of Assumption 3. The first half of that part is not needed, since Π^0 is decreasing and so is maximized at zero. The second half is the same shape restriction as before, on the cost of a gift rather than of a sale. It holds at the parameters of Section 6.

Lemma 2 holds without change, because its proof does not use the price: the East stays industrialized, regime L is absorbing, the best plan that never crosses ships nothing, and the best continuation after a crossing ships nothing, since Π^0 is decreasing and g_L is fixed. The reduced problem is (36) with Π^0 in place of Π_{i_t}, and Proposition 4 gives the critical security weight ψ_0^* and, through (37), its ordering against ψ^*. Proposition 14 adds two results.

Proposition 14 (Predation with free shipments)**.** *Under Assumption 5:*

(i) The critical security weight with gifts is strictly positive,

$$\psi_0^* \;\geq\; \frac{1-\beta}{\beta}\,\frac{r_0\,(h_H-\chi)}{\Delta g}, \qquad r_0 \;\equiv\; \frac{f'\big(\alpha_H n^E(0)\big)}{-\lambda\varphi_H'(0)} \;>\; 0,$$

and if Assumption 3(iii) also holds, then $\psi_0^* \geq \psi^*$*: a gift requires at least the weight a sale requires.*

(ii) For $\psi > \psi_0^*$ *an optimal plan exists, and every optimal plan has two phases. In the active phase,* $0 \leq t < T^*$ *with* T^* *finite, the shipment is strictly positive, strictly increasing, and largest in the period the West's knowledge reaches* χ*, and the West's knowledge falls strictly. In the passive phase,* $t \geq T^*$*, the West's regime is L and permanent, and the East ships nothing.*

Proof. (i) For the bound, write $K^0(x) = \int_0^x r(y)\big(-\lambda\varphi_H'(y)\big)\,dy$ with r the ratio of Assumption 5, so that $K^0(x) \geq r_0\lambda[\varphi_H(0) - \varphi_H(x)]$, because $r \geq r(0) = r_0$. A plan that crosses at T_P has $h_{T_P}^W \leq \chi$, and since $h_H = (1-\delta)^{T_P}h_H + \lambda\sum_{t<T_P}(1-\delta)^{T_P-1-t}\varphi_H(0)$,

$$\lambda\sum_{t<T_P}\big[\varphi_H(0)-\varphi_H(x_t)\big] \;\geq\; \lambda\sum_{t<T_P}(1-\delta)^{T_P-1-t}\big[\varphi_H(0)-\varphi_H(x_t)\big] \;=\; h_H - h_{T_P}^W \;\geq\; h_H-\chi.$$

Hence $C_P^0 \geq \beta^{T_P-1}\sum_{t<T_P}K^0(x_t) \geq \beta^{T_P-1}r_0(h_H-\chi)$, so $\beta^{-T_P}C_P^0 \geq r_0(h_H-\chi)/\beta$ for every crossing plan, and the infimum obeys the same bound. For the ordering, both bracketed terms in (37) are nonnegative, the revenue because $q_H(x)x \geq 0$ and $\Delta\Pi$ because $\Pi_L(x_L^{\mathrm{c}}) \geq \Pi_L(0) = \Pi_H(0)$ under Assumption 3(iii); so each plan's zero is at least as large under the gift as under the sale, and so is the infimum in (38).

(ii) The proof of Proposition 7 applies with K^0 in place of K and $\Delta\Pi = 0$. Existence uses the closed capture set and $\Pi^0(x) \leq \Pi_H(x) \to -\infty$ as $x \to \infty$. The constraint binds

because $K^{0\prime} > 0$, the multiplier is positive, the Kuhn–Tucker conditions hold with equality at every active date, the ratio argument runs under Assumption 5, and the induction for h^W is unchanged. The passive phase is Lemma 2(ii) together with the fact that, once the regime is L, the flow $\Pi^0(x) - \psi g_L$ is maximized at $x = 0$ at every date. □

What the gift changes. Relative to Section 4.3, the gift removes the commercial motive, so there is no Condition (D) to check and no trade with a West in regime L afterward. It removes the revenue that covers part of the cost of the shipments, so the critical security weight is higher. And it adds a transfer to the West. Because the goods are free, every active period raises the West's flow welfare above its autarky level, so the West's lifetime loss from predation is the discounted loss after the crossing net of the gain from free imports before it, and it can have either sign: an impatient West can be better off accepting the flood, a patient one is not, because the trap is permanent and the gift is not. This is the qualification under which the deterrence results of Section 5 carry over. The gift also changes the border instrument. A tariff acts on the West's import margin, which a delivered quantity bypasses, so under a gift the border instruments are refusal and a ban. Both are costless here, as the prohibitive tariff of Section 5.1 is costless in the paid benchmark because the commercial export between industrialized countries is zero, and neither is formalized in this paper.

Deterrence with free shipments. The game of Section 5.3 simplifies in the same way. The East's flow payoff from a shipment is $\Pi^0(x)$ whatever the West works, so the West's employment reaches the East only through the state, and the price condition of Definition 1 is empty. The West's flow welfare when it works n and receives x in regime i is $u_i^{W,0}(n; x) = f(\alpha_i n + x) - v(n)$, which is strictly concave in n with maximizer $\varphi_i(x)$. There is no price to move, so the West's static best response is its competitive employment, and Assumption 4, with the same defended level $\chi + \varepsilon$, is the only restriction on its choice. Export and employment policies, the rule N^{F}, and the Bellman equations are as before with these payoffs; above χ the rule is $N^{\mathrm{F}}(h, x) = \max\{\varphi_H(x), \underline{n}(h)\}$. The East's export policy X^{c} ships nothing at every state: above χ as in Section 5.3, and at or below χ because the maximizer of Π^0 is zero. The best flow a trapped West can earn is therefore its autarky flow $u_L^W(0)$, and Condition (C) becomes a comparison of autarky objects:

$$v\left(\frac{\delta\chi + \varepsilon}{\lambda}\right) \;<\; \beta\left[u_H^W(0) - u_L^W(0)\right]. \tag{C^0}$$

Its right side rises with β, so once it holds it holds for every more patient West.

Proposition 15 (Deterrence with free shipments)**.** *Suppose Assumptions 4 and 5 and Condition* (C^0) *hold.*

(i) Given N^{F}, for every $\psi \geq 0$ and every $h \geq 0$, shipping nothing is the unique maximizer in the East's Bellman equation, and at every $h > \chi$, $V^E(h; N^{\mathrm{F}}) = [\Pi^0(0) - \psi g_H]/(1-\beta)$.

(ii) Given X^{c}, for every $h > \chi$ and every $x \geq 0$, $N^{\mathrm{F}}(h, x)$ attains the supremum in the West's Bellman equation, whatever the policy prescribes at or below χ.

(iii) If in addition $v(\underline{n}((1-\delta)\chi)) < \beta[u_H^W(0) - u_L^W(0)]$*, then* $(X^{\mathrm{c}}, N^{\mathrm{F}})$ *is a Markov perfect equilibrium on* $((1-\delta)\chi, \infty)$*.*

(iv) Let P *be a finite flood: a plan that ships* x_t *for* $t < T_P$*, carries a West using the competitive response to* χ *or below at* T_P*, and ships nothing afterward, as the optimal plan of Proposition 14(ii) does. The West's value under* $(X^{\mathrm{c}}, N^{\mathrm{F}})$ *from* h_H *exceeds its value under the competitive response against* P *by*

$$\sum_{t<T_P} \beta^t \left[u_H^W(0) - u_H^{W,0}(x_t)\right] \;+\; \beta^{T_P} \frac{u_H^W(0) - u_L^W(0)}{1-\beta}.$$

The first term is the gift; it is negative and bounded. The second is the trap; it is positive and grows without bound as $\beta \to 1$*. For every such plan there is therefore a discount factor below one above which the West prefers the floor to accepting the gift.*

Proof. (i) From $h > \chi$, under N^{F} the West's regime is H at every date whatever is shipped, so a shipment sequence is worth $\sum_t \beta^t[\Pi^0(x_t) - \psi g_H]$. Since Π^0 is strictly decreasing, this is at most $[\Pi^0(0) - \psi g_H]/(1-\beta)$, with equality only if every shipment is zero. From $h \le \chi$ the West returns to the defended level whatever is shipped, so the East's continuation is fixed and it maximizes the flow $\Pi^0(x) - \psi g_L$, shipping nothing.

(ii) Fix $h > \chi$ and x. The levels that keep knowledge at or above $\chi + \varepsilon$ all leave the continuation at V_H^W, by the first bound in the proof of Proposition 10(ii), and among them $N^{\mathrm{F}}(h, x)$ has the highest flow, by strict concavity. Take a level n' that sends knowledge to χ or below. If $N^{\mathrm{F}}(h, x) = \varphi_H(x)$, the unconstrained maximizer, n' has no higher flow. Otherwise $N^{\mathrm{F}}(h, x) = \underline{n}(h) > n'$, and the gain in flow of n' is $u_H^{W,0}(n'; x) - u_H^{W,0}(\underline{n}(h); x) < v(\underline{n}(h)) \le v((\delta\chi + \varepsilon)/\lambda)$, because $f(\alpha_H \underline{n}(h) + x) > f(\alpha_H n' + x)$ and $\underline{n}(h) \le (\delta\chi + \varepsilon)/\lambda$ for $h > \chi$. At or below χ the East ships nothing, so the West's flow there is at most $\max_n u_L^{W,0}(n; 0) = u_L^W(0)$, and the best it can do afterward is return to $V_H^W = u_H^W(0)/(1-\beta)$; its loss in continuation is at least $\beta[u_H^W(0) - u_L^W(0)]$, which exceeds the gain by (C^0).

(iii) For the West at $h \in ((1-\delta)\chi, \chi]$, the best return is to the defended level, because $\varphi_L(x) \le \varphi_L(0) < \delta\chi/\lambda < \underline{n}(h)$. Relative to any level that stays at or below χ, it costs less than $v(\underline{n}((1-\delta)\chi))$ in flow, by the argument in (ii), and it secures V_H^W, while staying at or below χ costs at least $\beta[u_H^W(0) - u_L^W(0)]$ by the bound in (ii); the stated condition orders the two, whatever the East ships. The policies do not leave $((1-\delta)\chi, \infty)$, from h_H they visit only h_H, and a single departure by either player leaves the state in that set, as in the proof of Proposition 10(iii).

(iv) Under $(X^{\mathrm{c}}, N^{\mathrm{F}})$ the West's value from h_H is V_H^W by (i). Under the competitive response against P, its knowledge crosses at T_P and it receives nothing afterward, so its value is $\sum_{t<T_P} \beta^t u_H^{W,0}(x_t) + \beta^{T_P} u_L^W(0)/(1-\beta)$. Subtracting gives the display. The first sum has finitely many terms, each between $u_H^W(0) - u_H^{W,0}(x_t)$ and zero, so it is bounded uniformly in β; the second term grows without bound as $\beta \to 1$ because $u_H^W(0) > u_L^W(0)$, which holds since $\alpha_H > \alpha_L$ and f is increasing. □

The gift case separates the two reasons deterrence works in Section 5.3. The floor deters because a West that holds the floor leaves the East nothing to gain from a flood, and the

floor is credible because holding it for a period costs less than the discounted loss from a period in the low regime (Condition (C^0)). The gift gives the West one reason not to defend itself: during the active phase it receives goods free, and whether their value exceeds the loss from the trap depends on its discount factor. The ratio condition of Assumption 5, the ordering of the two critical security weights, and the sign of the West's lifetime loss hold at the parameters of Section 6, where we compute them.

E The Counterpolicy in the Smooth Economy

Section 5 proves its results in the sharp benchmark. This appendix computes the same game with the logistic productivity schedule (3), at the parameters of Section 6 and at larger values of the steepness γ, to see which results hold without the step. The West that uses the corrective subsidy is not trapped at any weight at which we find an equilibrium, the East ships nothing on the equilibrium path, and predation against it is unprofitable. Two objects differ from the benchmark: the subsidy on the path, zero in the benchmark, becomes the West's domestic correction, and the floor at χ becomes a level inside the steep region.

The game. The state is the West's knowledge h. Productivity is $\alpha(h)$, and the security loss is $g(h) = \mathcal{D}(\alpha(h))$ with $\mathcal{D}(z) = z^2/2$. The gap between the two competitive steady states is then $\Delta g = g(h_H^*) - g(h_L^*) = 4.45$, and we report the security weight as $\psi \Delta g$, as in Section 4.3. We hold the East on its upper plateau, where $\alpha'(h_H^*) = 0.003$, so its employment is the static $n^E(x)$ of Section 4.3 and its knowledge is not a state. Timing and payoffs are those of Definition 1 with $\alpha(h)$ in place of α_i: each period the East ships $x = X(h)$, the West observes x and works $n = N(h, x)$, and the price is $f'(\alpha(h)n + x)$. The West chooses its employment at realized welfare (45), which is the criterion in its Bellman equation (52), and it chooses from the whole employment grid. The equilibrium concept is Definition 1 with the feasible set $\mathcal{N}(h)$ equal to that grid. This is the game of Section 5.3 with $\alpha(h)$ in place of α_i, and it needs no margin: productivity rises with knowledge, so the West has a reason to prefer one level of knowledge to another. We call it the *unrestricted* smooth game. As in Section 5.2, the West's choice reflects two motives, its knowledge and the price it pays for imports. On the equilibrium path nothing is shipped, so the import-price motive is absent and the subsidy there is the domestic learning correction. Off the path both are at work, and we separate them when we report the West's response to a flood.

Method. We solve each player's problem by value-function iteration on grids: 240 points of knowledge on $[0.3, 8]$; shipments in steps of 0.01 on $[0, 1]$ and 0.2 beyond, to 16, or to 30 in the runs with large weights; employment in steps of 0.0125 on $[0, 3]$. Continuation values are interpolated linearly between knowledge points. The object we compute is therefore an equilibrium of the discretized game, in which each player's actions are restricted to the grids, and not of the continuous game. We iterate best responses. Given the East's export policy, the West's employment policy solves its dynamic problem at every knowledge point and every shipment on the grid; given the West's policy, the East's export policy solves its

problem at every knowledge point. The iteration stops when neither policy changes at any grid point, and we run it from two starting policies for the West, the competitive response and a uniformly high employment. A no-defense benchmark fixes the West at its competitive response and lets the East optimize against it.

Equilibrium checks. A pair at which the iteration stops is checked directly. We evaluate each player's value from following its policy against the other's and compute the largest gain from a one-stage deviation over the whole grid: for the East over every shipment at every knowledge point, for the West over every employment at every pair of knowledge and shipment, both at realized welfare, as condition (ii) of Definition 1 requires. A policy that no one-stage deviation improves is optimal in a discounted dynamic program, so a gain of zero at every grid state rules out every deviation plan on the grids. We call the pair an equilibrium only when neither gain exceeds 10^{-8}; in every cell of Table 4 the East's largest gain is below 10^{-9} and the West's is zero. Where the iteration does not stop we say so, describe the iterates, and claim no equilibrium. No policy or best response sits at the top of an action grid, and no path reaches the top of the knowledge grid.

Results. Table 4 reports the outcomes. Against the passive West, the East ships at most 0.03 at $\psi\Delta g = 0.3$ and floods at 0.6, so the critical security weight in the smooth economy lies between the two, below the benchmark's 1.66. The flood that traps the passive West peaks at 8.0 at $\gamma = 4.2$ and at 13.4 at $\gamma = 32$, toward the benchmark's 14.3. These peaks are on this appendix's shipment grid, whose step is 0.2 above one unit; on the finer grid of Figure 10 the peak at $\gamma = 4.2$ is 7.6. Against the West that uses the corrective subsidy, the outcome is the same in every cell of the table. At every steepness and every weight we compute, up to $\psi\Delta g = 4$, which is 2.4 times the benchmark's critical security weight, the iteration stops from both starting policies at the same pair, and the pair passes the checks above. In it the East ships nothing, as in the benchmark, the West's knowledge sits at the planner's high steady state, 6.19, and the subsidy on the path equals the domestic correction of Section 3.3 at that state to the precision of the employment grid: the rate is 0.004 at every steepness, and the correction is 0.004 at $\gamma = 4.2$ and below 10^{-4} beyond, the two agreeing within one step of that grid. The West is never trapped.

Grid sensitivity. On a grid with twice the knowledge points and half the action steps the passive West's peaks move at the first decimal, to 7.5 at $\gamma = 4.2$ and 13.5 at $\gamma = 32$, the trapped West's knowledge settles at 0.97 rather than 0.94, the East ships nothing at $\psi\Delta g = 0.3$, and the bracket on the critical security weight is unchanged. The equilibrium at $\psi\Delta g = 4$ is found again on that grid, at $\gamma = 4.2$, 8, and 32, from the competitive start, and at $\gamma = 4.2$ also from the base-grid policy mapped onto it: the East ships nothing, knowledge settles at 6.19 at $\gamma = 4.2$ and 6.16 at the two larger steepnesses, and the subsidy on the path is within one grid step of the domestic correction.

The learning motive alone. The unrestricted game lets the West act on the price it pays as well as on its own knowledge, so it shows whether the no-trap outcome holds with

	Passive West		Unrestricted smooth-game equilibrium			Price-taking West		
	$\psi\Delta g = 1.66$		all three weights			deviation gain at $\psi\Delta g =$		
γ	peak x	h	x	τ	h	1.66	2.5	4
4.2	8.0	0.9	0	0.004	6.19	0.036	0.042	none found
8	9.6	0.9	0	0.004	6.19	0.041	0.040	0.044
16	11.8	0.9	0	0.004	6.19	0.044	0.044	0.047
32	13.4	0.9	0	0.004	6.19	0.046	0.046	0.051

Table 4: The counterpolicy game with a smooth productivity schedule, at the parameters of Section 6 and larger steepness. Left: the East's optimal plan against a West that does nothing. Middle: the equilibrium of the unrestricted smooth game, in which the West chooses employment from the whole grid at realized welfare, which is the same at all three security weights, $\psi\Delta g \in \{1.66, 2.5, 4\}$; x is the largest shipment on the path, τ is the West's subsidy rate on the path, and h is knowledge after 300 periods. The rate equals the domestic correction of Section 3.3 at the high steady state, 0.004 at $\gamma = 4.2$ and below 10^{-4} beyond, to within one step of the employment grid, which is 0.0125 in employment and about 0.01 in the rate. Right: the largest gain in realized welfare the West could obtain by changing its employment for one period, at the pair reached when it is restricted to choose its employment taking the import price as given; this gain is the cost of the restriction to the West, and because it is positive the pair is not an equilibrium of Definition 1. At $\gamma = 4.2$ and $\psi\Delta g = 4$ the restricted iteration does not stop and no such pair was found. The threshold is 4.01 at $\gamma = 4.2$.

a smooth schedule; it does not show whether the learning correction alone produces it. To isolate the second question we restrict the government to choose employment taking the import price as given, which removes the import-price motive. That choice is not optimal at realized welfare, so the object is no longer an equilibrium of Definition 1 with $\mathcal{N}$ the grid; we call it a price-taking policy fixed point and report what it looks like. The iteration stops at eleven of the twelve cells, at pairs with the same deterrence outcome: the East ships nothing, except 0.07 every fifth period at $\gamma = 4.2$ and $\psi\Delta g = 2.5$, and the West's knowledge stays at the high steady state with the same subsidy on the path. So in the cells where the restricted fixed point exists, the learning correction alone deters. Those pairs are not equilibria of Definition 1: at each of them the West could raise its value by between 0.036 and 0.051 by changing its employment for one period at some state, against a value of about 39 at the high steady state. Under the restriction the iteration also fails to converge in several cases. At $\gamma = 4.2$ and $\psi\Delta g = 4$ the restricted iteration does not stop from either starting policy, nor at the intermediate weights from 2.75 to 3.75 when each is started from the last pair found, and on the finer grid it does not stop at $\gamma = 4.2$ at the weight we ran there, while the unrestricted iteration stops at every cell we ran. A government that chooses under one flow and is evaluated under another does not solve a contraction, and its inner iteration need not converge; a government that both chooses and is evaluated at realized welfare does.

The response off the path. Deterrence comes from the West's response to a flood off the equilibrium path, which Figure 11 shows, from the equilibrium at $\gamma = 4.2$ and $\psi\Delta g = 4$.

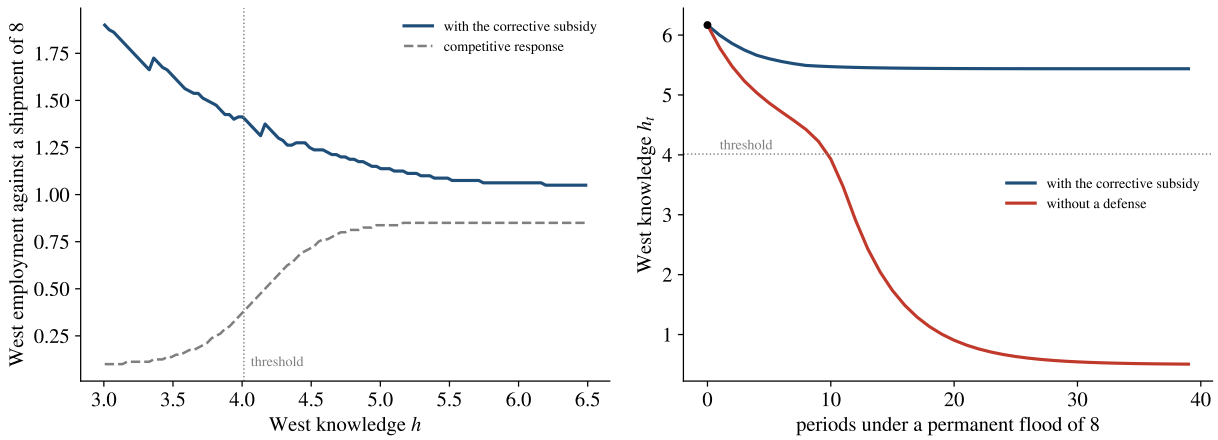


Figure 11: The West's defense in the smooth economy, at the parameters of Section 6 and $\psi\Delta g = 4$, from the equilibrium of Table 4. Left: the West's employment against a shipment of 8 as a function of its knowledge, with the corrective subsidy (solid) and under the competitive response (dashed); the gap widens as knowledge falls into the steep region of the S-curve and below the threshold. Right: the West's knowledge under a permanent flood of 8, with the corrective subsidy (blue) and without a defense (red); the dotted line is the threshold.

Against a shipment of 8, the West's wedge is 0.27 at the high steady state, 0.40 at $h = 5.0$, 1.7 at the inflection $h = 4.2$, 3.0 at the threshold, and 11.0 at $h = 3.5$. The shares of the two motives in the wedge vary with the knowledge stock. At the high steady state the shadow value of knowledge is nil, so the 0.27 is entirely the price term: a West facing a shipment of 8 works more than its competitive level to lower what it pays for it. Deep in the steep region the price term is a small part of a much larger wedge, and the response is driven by the shadow value of knowledge, which grows steeply below the threshold, where without the subsidy knowledge would slide to the low steady state. The response is not proportional to $\alpha'(h)$ at the current state, which is smaller at $h = 3.5$ than at the threshold. On the equilibrium path, where nothing is shipped, the price term is zero and the wedge is the domestic correction alone. Under a permanent flood of 8 the West's knowledge falls from 6.17 to 5.44 and stays there, well above the threshold, while the competitive West falls below it and settles at 0.5. Every flood we computed yields the East less than autarky. A shipment of 16 sustained for 40 periods costs the East 240 in present value and never carries the West's knowledge below 5.1. A shipment of 30 sustained for 60 periods costs 1,042 and never carries it below 4.95. As in Proposition 10, the West's cost of holding a level is bounded, the East's cost of lowering it is not, and shipments that do not bring knowledge into the steep region gain the East little, because g is nearly flat on the plateau.

Larger weights and tariffs. At weights far above the critical security weight, $\psi\Delta g = 10$ and 20, the iteration does not stop from either starting policy, with or without the price-taking restriction, and no equilibrium was found; we draw no conclusion at those weights.

Tariffs behave as in Proposition 9. Against the passive West at $\gamma = 4.2$, a tariff of 0.1 is absorbed at every weight tested, and a tariff of 0.3 deters at $\psi\Delta g = 1.66$ and is absorbed at 2.5 and 4.

What the step changes. On the equilibrium path the subsidy is the domestic learning correction rather than zero; the correction vanishes as the plateau flattens and falls below 10^{-4} at $\gamma \geq 8$, which the employment grid cannot resolve. The defended level becomes a level between the threshold and the high steady state, 5.4 against a flood of 8 here, which the West holds by an employment response that rises with its shadow value of knowledge. Otherwise the computations reproduce the results of Section 5 at every steepness and weight we compute: the West that uses the corrective subsidy is not trapped, the East ships nothing on the equilibrium path, every flood we computed against it yields the East less than autarky, and a tariff below a level that depends on the East's motive is absorbed. These are computations at selected parameter values in a discretized game. They do not establish deterrence at every security weight or characterize the continuous game, and at the larger weights at which the iteration does not stop, the existence of an equilibrium is open.

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